

# popgenVCF quickstart example: chromosome 22 subset (160 samples, 8 populations)

popgenVCF

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## Contents

|    |                                                                                                                                                                                                                                                                                                                               |   |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1  | Executive summary                                                                                                                                                                                                                                                                                                             | 2 |
| 2  | Pipeline notices                                                                                                                                                                                                                                                                                                              | 3 |
| 3  | Quality control                                                                                                                                                                                                                                                                                                               | 3 |
| 4  | Genetic diversity                                                                                                                                                                                                                                                                                                             | 3 |
| 5  | Principal component analysis                                                                                                                                                                                                                                                                                                  | 4 |
| 6  | PCA SNP loadings (top 20 by  loading ; see 31_PCA_loadings.tsv for the complete top-N list)                                                                                                                                                                                                                                   | 4 |
| 7  | Population differentiation                                                                                                                                                                                                                                                                                                    | 5 |
| 8  | Nei's standard genetic distance between populations (see 46_population_genetic_distance.tsv; a neighbour-joining tree with bootstrap support at each node, alongside the individual-level IBS tree, is written and rendered as a figure when at least three populations are present – see 52_IBS_tree and 53_population_tree) | 5 |
| 9  | Genome scan FST outliers (top 20 windows by FST; see 39_genome_scan_fst.tsv for the complete list)                                                                                                                                                                                                                            | 6 |
| 10 | pcadapt outlier scan (top 20 significant loci by q-value; see 59_pcadapt_outliers.tsv for the complete results and 59b_pcadapt_significant_outliers.tsv for the complete significant list)                                                                                                                                    | 7 |
| 11 | Linkage disequilibrium decay (mean r-squared by physical distance bin; see 43_LD_decay.tsv for the complete list)                                                                                                                                                                                                             | 7 |
| 12 | LD-based effective population size (per population; see 45_Ne_LD.tsv for full diagnostics)                                                                                                                                                                                                                                    | 8 |
| 13 | Close relatives (up to 20 shown; see 36_close_relatives.tsv for the complete list)                                                                                                                                                                                                                                            | 9 |
| 14 | Sex check (reported vs inferred sex from X-chromosome heterozygosity and Y-chromosome call rate; up to 20 shown, discordant and ambiguous calls first; see 42_sex_check.tsv for the complete list)                                                                                                                            | 9 |

|      |                                                                                                                         |     |
|------|-------------------------------------------------------------------------------------------------------------------------|-----|
| 15   | Runs of homozygosity (highest FROH samples, up to 20 shown; see 38_ROH_sample_summary.tsv for the complete list)        | 10  |
| 16   | Discriminant analysis of principal components                                                                           | 10  |
| 17   | DAPC SNP loadings (top 20 by contribution across all K; see 22f_DAPC_loadings_K.tsv for the complete per-K top-N lists) | 10  |
| 18   | Analysis of molecular variance                                                                                          | 11  |
| 19   | Multilocus genotype (MLG) and clonal diversity                                                                          | 11  |
| 20   | Sex-biased dispersal test                                                                                               | 11  |
| 21   | Mantel test and isolation by distance                                                                                   | 11  |
| 22   | Admixture-model cross-validation                                                                                        | 11  |
| 23   | Chromosome-specific results                                                                                             | 12  |
| 24   | Figure gallery                                                                                                          | 12  |
| 25   | Reproducibility                                                                                                         | 158 |
| 25.1 | Population-structure reproducibility . . . . .                                                                          | 158 |
| 25.2 | fastStructure . . . . .                                                                                                 | 158 |
| 25.3 | Sparse non-negative matrix factorization . . . . .                                                                      | 158 |

## 1 Executive summary

This PDF file is a portable, human-readable analysis report. Figures are embedded directly in the file. The HTML version can be opened in a standard web browser, while the PDF version is optimized for compact exchange, printing, and email. The accompanying `analysis_results.rds` file is a reproducibility archive for R users; it is not required to read this report.

Generated with **popgenVCF v1.0.7.9000**.

The analysis retained **160 samples** from **8 populations**, with **1969 QC-passing SNPs** and **357 LD-pruned SNPs**.

## 2 Pipeline notices

| stage                 | level   | message                                                                                                                                                 |
|-----------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| VCF preparation       | INFO    | 98,922 of 98,922 VCF record(s) are biallelic SNPs (0 dropped as indels, multiallelic sites, MNPs, or other variant types); see 00_vcf_variant_types.tsv |
| pca                   | INFO    | requested 10 bounded component(s); eigensystem source: SNPRelate                                                                                        |
| population_assignment | WARNING | 54 sample(s) assign to a population other than their recorded label                                                                                     |
| pcadapt               | WARNING | 708 of 1969 tested loci are significant outliers at $q < 0.05$ (Benjamini-Hochberg FDR)                                                                 |
| ne_ld                 | WARNING | LD-based $N_e$ estimation requires unlinked (cross-chromosome) marker pairs; the retained autosomal marker set spans only one chromosome                |
| admixture             | INFO    | PLINK input generated with 160 samples and 357 SNPs                                                                                                     |
| faststructure         | INFO    | PLINK input generated with 160 samples and 357 SNPs                                                                                                     |
| snmf                  | INFO    | sNMF input generated with 160 samples and 357 SNPs                                                                                                      |
| bottleneck            | WARNING | 8 population(s) show a mode-shifted site frequency spectrum, a possible recent-bottleneck signature                                                     |

The global Weir-Cockerham  $F_{ST}$  estimate was **0.0915**.

## 3 Quality control

| step              | variants | removed_at_step | retained_percent |
|-------------------|----------|-----------------|------------------|
| Input biallelic   | 98922    | 0               | 100.00           |
| After MAF         | 63585    | 35337           | 64.28            |
| After missingness | 63585    | 0               | 64.28            |
| After LD pruning  | 357      | 63228           | 0.36             |

## 4 Genetic diversity

| population | n_samples | n_loci | polymorphic_loci | polymorphic_fraction | observed_heterozygosity | expected_heterozygosity | inbreeding_coefficient | mean_minor_allele_frequency | mean_locus_call_rate | hws_tested_loci | hws_significant_loci | hws_significant_loci_fdr | private_allele_loci | mean_allele_richness | mean_effective_allele |
|------------|-----------|--------|------------------|----------------------|-------------------------|-------------------------|------------------------|-----------------------------|----------------------|-----------------|----------------------|--------------------------|---------------------|----------------------|-----------------------|
| CHI        | 20        | 1969   | 1653             | 0.8395               | 0.2683                  | 0.2855                  | 0.0003                 | 0.2133                      | 1                    | 1653            | 149                  | 55                       | 0                   | 1.8395               | 1.5051                |
| GHR        | 20        | 1969   | 1756             | 0.8817               | 0.2761                  | 0.2927                  | 0.0026                 | 0.2044                      | 1                    | 1756            | 99                   | 31                       | 0                   | 1.8817               | 1.4800                |
| ITU        | 20        | 1969   | 1829             | 0.9289               | 0.2736                  | 0.2919                  | 0.0026                 | 0.2059                      | 1                    | 1829            | 100                  | 43                       | 0                   | 1.9289               | 1.4893                |
| LWK        | 20        | 1969   | 1622             | 0.8228               | 0.2636                  | 0.2727                  | 0.0026                 | 0.1963                      | 1                    | 1622            | 66                   | 14                       | 0                   | 1.8228               | 1.4688                |
| PEL        | 20        | 1969   | 1800             | 0.9142               | 0.2330                  | 0.2579                  | 0.0068                 | 0.1851                      | 1                    | 1800            | 147                  | 45                       | 0                   | 1.9142               | 1.4562                |
| PUR        | 20        | 1969   | 1844             | 0.9365               | 0.2748                  | 0.2730                  | -0.0007                | 0.1871                      | 1                    | 1844            | 81                   | 21                       | 0                   | 1.9365               | 1.4487                |
| STU        | 20        | 1969   | 1788             | 0.9081               | 0.2561                  | 0.2863                  | 0.1056                 | 0.2013                      | 1                    | 1788            | 131                  | 33                       | 0                   | 1.9081               | 1.4607                |
| YRI        | 20        | 1969   | 1636             | 0.8309               | 0.2731                  | 0.2705                  | -0.0006                | 0.1944                      | 1                    | 1636            | 92                   | 18                       | 0                   | 1.8309               | 1.4637                |

## 5 Principal component analysis

| sample  | vcf_sample | PC1     | PC2     | PC3     | PC4     | PC5     | PC6     | PC7     | PC8     | PC9     | PC10    | population |
|---------|------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|------------|
| NA18536 | NA18536    | -0.0575 | 0.0756  | -0.1958 | -0.0176 | 0.0470  | 0.0646  | -0.0320 | -0.0509 | -0.0829 | 0.0169  | CHB        |
| NA18538 | NA18538    | -0.0665 | 0.1310  | -0.0958 | -0.0006 | 0.0158  | -0.0224 | 0.1012  | 0.0074  | -0.2047 | -0.0051 | CHB        |
| NA18541 | NA18541    | -0.0654 | 0.0991  | -0.0443 | -0.0362 | 0.0684  | 0.0917  | 0.0875  | -0.0995 | -0.0353 | 0.0002  | CHB        |
| NA18552 | NA18552    | -0.0714 | 0.1650  | -0.1811 | -0.1448 | 0.0223  | 0.0384  | 0.0671  | 0.0537  | 0.0328  | -0.0593 | CHB        |
| NA18553 | NA18553    | -0.0616 | -0.0660 | -0.1590 | 0.0237  | -0.2193 | -0.0004 | -0.1910 | -0.0064 | -0.1380 | 0.0197  | CHB        |
| NA18555 | NA18555    | -0.0312 | 0.0930  | -0.1309 | -0.1100 | 0.0308  | 0.0717  | -0.0562 | 0.0124  | 0.0901  | -0.1218 | CHB        |
| NA18557 | NA18557    | -0.0893 | 0.2057  | -0.1264 | -0.0737 | 0.0509  | 0.0121  | 0.0892  | 0.0377  | -0.0913 | 0.0984  | CHB        |
| NA18562 | NA18562    | -0.0381 | 0.0209  | -0.0813 | 0.0459  | 0.0625  | -0.0312 | 0.1107  | -0.1086 | 0.0245  | -0.1299 | CHB        |
| NA18564 | NA18564    | -0.0305 | 0.0841  | -0.1174 | -0.1036 | -0.0883 | -0.0264 | -0.1778 | 0.0857  | 0.2782  | -0.1910 | CHB        |
| NA18573 | NA18573    | -0.0685 | 0.0633  | 0.0378  | 0.0394  | -0.0765 | 0.0588  | -0.0318 | -0.1708 | 0.0201  | -0.0102 | CHB        |
| NA18574 | NA18574    | -0.0873 | 0.1260  | -0.1222 | -0.0647 | 0.0656  | 0.0098  | 0.0426  | -0.0156 | -0.0750 | -0.0113 | CHB        |
| NA18606 | NA18606    | -0.0584 | 0.0752  | -0.1100 | 0.0288  | 0.0297  | -0.0415 | 0.1097  | 0.1042  | -0.0327 | -0.0739 | CHB        |
| NA18611 | NA18611    | -0.0717 | 0.1273  | -0.0419 | -0.0468 | -0.0176 | 0.1518  | -0.0208 | 0.0271  | -0.1508 | -0.0482 | CHB        |
| NA18612 | NA18612    | -0.0504 | -0.0424 | -0.2118 | 0.1398  | -0.1597 | -0.0342 | -0.1189 | -0.0957 | -0.0276 | 0.1465  | CHB        |
| NA18613 | NA18613    | -0.0583 | 0.1108  | -0.0705 | -0.0469 | 0.0092  | 0.0932  | 0.0875  | 0.0691  | -0.1390 | 0.1136  | CHB        |
| NA18616 | NA18616    | -0.0173 | 0.0510  | -0.1403 | -0.1915 | 0.0842  | 0.1383  | -0.0259 | -0.1606 | 0.0728  | -0.0742 | CHB        |
| NA18617 | NA18617    | -0.0307 | 0.1223  | -0.1382 | -0.1085 | -0.0277 | 0.0449  | -0.0945 | 0.0113  | 0.0748  | -0.1828 | CHB        |
| NA18638 | NA18638    | -0.0722 | 0.1224  | -0.0679 | -0.0460 | -0.0106 | 0.1791  | -0.0069 | -0.1120 | -0.1554 | 0.0068  | CHB        |
| NA18643 | NA18643    | -0.0437 | 0.0467  | -0.1922 | 0.0077  | 0.0101  | -0.0009 | 0.0329  | -0.0462 | 0.0704  | -0.0114 | CHB        |
| NA18647 | NA18647    | -0.0558 | 0.1011  | -0.1989 | -0.0603 | 0.1633  | 0.1660  | 0.0094  | -0.0507 | 0.0072  | 0.0560  | CHB        |

## 6 PCA SNP loadings (top 20 by |loading|; see 31\_PCA\_loadings.tsv for the complete top-N list)

| axis | snp_id | chromosome | position | contribution | magnitude |
|------|--------|------------|----------|--------------|-----------|
| PC7  | 10937  | 22         | 20437004 | 0.2848       | 0.2848    |
| PC7  | 10502  | 22         | 20371297 | 0.2541       | 0.2541    |
| PC7  | 11374  | 22         | 20502771 | 0.2524       | 0.2524    |
| PC8  | 13593  | 22         | 20753041 | 0.2191       | 0.2191    |
| PC6  | 10937  | 22         | 20437004 | -0.2191      | 0.2191    |
| PC8  | 13990  | 22         | 20765538 | 0.2178       | 0.2178    |
| PC8  | 14246  | 22         | 20773711 | 0.2034       | 0.2034    |
| PC5  | 4933   | 22         | 20159049 | -0.2010      | 0.2010    |
| PC3  | 15906  | 22         | 20832523 | 0.1980       | 0.1980    |
| PC8  | 13078  | 22         | 20733667 | 0.1952       | 0.1952    |
| PC7  | 10415  | 22         | 20360856 | 0.1932       | 0.1932    |
| PC8  | 14080  | 22         | 20768038 | -0.1925      | 0.1925    |
| PC9  | 3724   | 22         | 20121080 | -0.1900      | 0.1900    |
| PC6  | 10502  | 22         | 20371297 | -0.1887      | 0.1887    |
| PC10 | 11570  | 22         | 20630754 | 0.1883       | 0.1883    |
| PC9  | 1516   | 22         | 20047587 | -0.1869      | 0.1869    |
| PC3  | 14848  | 22         | 20794119 | 0.1865       | 0.1865    |
| PC9  | 2006   | 22         | 20065314 | -0.1859      | 0.1859    |
| PC8  | 14377  | 22         | 20777533 | -0.1806      | 0.1806    |
| PC10 | 11593  | 22         | 20639324 | 0.1763       | 0.1763    |

## 7 Population differentiation

| population_1 | population_2 | n_1 | n_2 | fst    | nm      | jost_d | jost_d_n_snps | dxy    | dxy_n_snps |
|--------------|--------------|-----|-----|--------|---------|--------|---------------|--------|------------|
| CHB          | GBR          | 20  | 20  | 0.1210 | 1.8153  | 0.0547 | 1969          | 0.3233 | 1969       |
| CHB          | ITU          | 20  | 20  | 0.0631 | 3.7098  | 0.0274 | 1969          | 0.3082 | 1969       |
| CHB          | LWK          | 20  | 20  | 0.1631 | 1.2832  | 0.0754 | 1969          | 0.3335 | 1969       |
| CHB          | PEL          | 20  | 20  | 0.0794 | 2.9000  | 0.0322 | 1969          | 0.2952 | 1969       |
| CHB          | PUR          | 20  | 20  | 0.1101 | 2.0199  | 0.0480 | 1969          | 0.3138 | 1969       |
| CHB          | STU          | 20  | 20  | 0.0865 | 2.6390  | 0.0379 | 1969          | 0.3130 | 1969       |
| CHB          | YRI          | 20  | 20  | 0.1631 | 1.2824  | 0.0751 | 1969          | 0.3322 | 1969       |
| GBR          | ITU          | 20  | 20  | 0.0232 | 10.5031 | 0.0096 | 1969          | 0.2942 | 1969       |
| GBR          | LWK          | 20  | 20  | 0.1198 | 1.8364  | 0.0524 | 1969          | 0.3156 | 1969       |
| GBR          | PEL          | 20  | 20  | 0.0874 | 2.6113  | 0.0355 | 1969          | 0.2962 | 1969       |
| GBR          | PUR          | 20  | 20  | 0.0267 | 9.1026  | 0.0106 | 1969          | 0.2855 | 1969       |
| GBR          | STU          | 20  | 20  | 0.0350 | 6.8881  | 0.0144 | 1969          | 0.2949 | 1969       |
| GBR          | YRI          | 20  | 20  | 0.1344 | 1.6100  | 0.0594 | 1969          | 0.3196 | 1969       |
| ITU          | LWK          | 20  | 20  | 0.1057 | 2.1159  | 0.0465 | 1969          | 0.3157 | 1969       |
| ITU          | PEL          | 20  | 20  | 0.0400 | 5.9960  | 0.0158 | 1969          | 0.2864 | 1969       |
| ITU          | PUR          | 20  | 20  | 0.0269 | 9.0319  | 0.0109 | 1969          | 0.2903 | 1969       |
| ITU          | STU          | 20  | 20  | 0.0052 | 48.2538 | 0.0021 | 1969          | 0.2906 | 1969       |
| ITU          | YRI          | 20  | 20  | 0.1189 | 1.8530  | 0.0528 | 1969          | 0.3192 | 1969       |
| LWK          | PEL          | 20  | 20  | 0.1737 | 1.1891  | 0.0759 | 1969          | 0.3211 | 1969       |
| LWK          | PUR          | 20  | 20  | 0.1176 | 1.8764  | 0.0500 | 1969          | 0.3092 | 1969       |
| LWK          | STU          | 20  | 20  | 0.1112 | 1.9987  | 0.0485 | 1969          | 0.3145 | 1969       |
| LWK          | YRI          | 20  | 20  | 0.0052 | 47.5634 | 0.0020 | 1969          | 0.2731 | 1969       |
| PEL          | PUR          | 20  | 20  | 0.0621 | 3.7736  | 0.0239 | 1969          | 0.2831 | 1969       |
| PEL          | STU          | 20  | 20  | 0.0609 | 3.8555  | 0.0242 | 1969          | 0.2898 | 1969       |
| PEL          | YRI          | 20  | 20  | 0.1799 | 1.1393  | 0.0788 | 1969          | 0.3222 | 1969       |
| PUR          | STU          | 20  | 20  | 0.0339 | 7.1152  | 0.0136 | 1969          | 0.2895 | 1969       |
| PUR          | YRI          | 20  | 20  | 0.1226 | 1.7887  | 0.0522 | 1969          | 0.3098 | 1969       |
| STU          | YRI          | 20  | 20  | 0.1187 | 1.8565  | 0.0520 | 1969          | 0.3159 | 1969       |

8 Nei's standard genetic distance between populations (see 46\_population\_genetic\_distance.tsv; a neighbour-joining tree with bootstrap support at each node, alongside the individual-level IBS tree, is written and rendered as a figure when at least three populations are present – see 52\_IBS\_tree and 53\_population\_tree)

| population | CHB    | GBR    | ITU    | LWK    | PEL    | PUR    | STU    | YRI    |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|
| CHB        | 0.0000 | 0.0661 | 0.0378 | 0.0880 | 0.0418 | 0.0588 | 0.0486 | 0.0876 |
| GBR        | 0.0661 | 0.0000 | 0.0197 | 0.0633 | 0.0452 | 0.0202 | 0.0244 | 0.0707 |
| ITU        | 0.0378 | 0.0197 | 0.0000 | 0.0573 | 0.0251 | 0.0207 | 0.0122 | 0.0639 |
| LWK        | 0.0880 | 0.0633 | 0.0573 | 0.0000 | 0.0879 | 0.0606 | 0.0594 | 0.0112 |
| PEL        | 0.0418 | 0.0452 | 0.0251 | 0.0879 | 0.0000 | 0.0332 | 0.0337 | 0.0910 |
| PUR        | 0.0588 | 0.0202 | 0.0207 | 0.0606 | 0.0332 | 0.0000 | 0.0234 | 0.0629 |
| STU        | 0.0486 | 0.0244 | 0.0122 | 0.0594 | 0.0337 | 0.0234 | 0.0000 | 0.0629 |
| YRI        | 0.0876 | 0.0707 | 0.0639 | 0.0112 | 0.0910 | 0.0629 | 0.0629 | 0.0000 |

## 9 Genome scan FST outliers (top 20 windows by FST; see 39\_genome\_scan\_fst.tsv for the complete list)

| chromosome | window_start | window_end | n_snps | global_fst |
|------------|--------------|------------|--------|------------|
| 22         | 20800428     | 20850427   | 104    | 0.1718     |
| 22         | 20750428     | 20800427   | 213    | 0.1319     |
| 22         | 20850428     | 20900427   | 113    | 0.1118     |
| 22         | 20250428     | 20300427   | 102    | 0.1097     |
| 22         | 20900428     | 20950427   | 138    | 0.1076     |
| 22         | 20150428     | 20200427   | 196    | 0.1007     |
| 22         | 20200428     | 20250427   | 111    | 0.0952     |
| 22         | 20000428     | 20050427   | 160    | 0.0875     |
| 22         | 20700428     | 20750427   | 135    | 0.0845     |
| 22         | 20400428     | 20450427   | 24     | 0.0838     |
| 22         | 20100428     | 20150427   | 115    | 0.0694     |
| 22         | 20950428     | 21000427   | 193    | 0.0693     |
| 22         | 20350428     | 20400427   | 64     | 0.0666     |
| 22         | 20450428     | 20500427   | 48     | 0.0639     |
| 22         | 20050428     | 20100427   | 131    | 0.0512     |
| 22         | 20600428     | 20650427   | 38     | 0.0510     |
| 22         | 20500428     | 20550427   | 9      | 0.0403     |
| 22         | 20300428     | 20350427   | 44     | 0.0351     |
| 22         | 20650428     | 20700427   | 31     | 0.0194     |

**10 pcadapt outlier scan (top 20 significant loci by q-value; see 59\_pcadapt\_outliers.tsv for the complete results and 59b\_pcadapt\_significant\_outliers.tsv for the complete significant list)**

| snp_id | chromosome | position | maf    | mahalanobis_stat | chi2_stat | p_value | q_value | outlier |
|--------|------------|----------|--------|------------------|-----------|---------|---------|---------|
| 10525  | 22         | 20376408 | 0.4625 | 713.9363         | 447.3014  | 0       | 0       | TRUE    |
| 10527  | 22         | 20376436 | 0.4656 | 644.8727         | 404.0311  | 0       | 0       | TRUE    |
| 10807  | 22         | 20401090 | 0.4812 | 567.6480         | 355.6476  | 0       | 0       | TRUE    |
| 10765  | 22         | 20396584 | 0.5000 | 547.7982         | 343.2111  | 0       | 0       | TRUE    |
| 10757  | 22         | 20394994 | 0.4906 | 524.6631         | 328.7164  | 0       | 0       | TRUE    |
| 10806  | 22         | 20401066 | 0.4812 | 500.7546         | 313.7370  | 0       | 0       | TRUE    |
| 10758  | 22         | 20395128 | 0.4844 | 495.2806         | 310.3073  | 0       | 0       | TRUE    |
| 10675  | 22         | 20388698 | 0.4688 | 489.4984         | 306.6847  | 0       | 0       | TRUE    |
| 10767  | 22         | 20396614 | 0.4656 | 452.1966         | 283.3140  | 0       | 0       | TRUE    |
| 10793  | 22         | 20400371 | 0.4469 | 441.6063         | 276.6789  | 0       | 0       | TRUE    |
| 10812  | 22         | 20401719 | 0.4688 | 437.6783         | 274.2179  | 0       | 0       | TRUE    |
| 10783  | 22         | 20399959 | 0.4500 | 436.5882         | 273.5349  | 0       | 0       | TRUE    |
| 10680  | 22         | 20390126 | 0.4375 | 432.5119         | 270.9810  | 0       | 0       | TRUE    |
| 10561  | 22         | 20378905 | 0.4906 | 429.5010         | 269.0946  | 0       | 0       | TRUE    |
| 10562  | 22         | 20378965 | 0.4969 | 420.9905         | 263.7625  | 0       | 0       | TRUE    |
| 10937  | 22         | 20437004 | 0.4719 | 420.8293         | 263.6615  | 0       | 0       | TRUE    |
| 10823  | 22         | 20402404 | 0.4781 | 409.4727         | 256.5463  | 0       | 0       | TRUE    |
| 10669  | 22         | 20388262 | 0.4250 | 390.2509         | 244.5033  | 0       | 0       | TRUE    |
| 10974  | 22         | 20449384 | 0.4719 | 388.0593         | 243.1302  | 0       | 0       | TRUE    |
| 10829  | 22         | 20403013 | 0.4812 | 384.4381         | 240.8614  | 0       | 0       | TRUE    |

**11 Linkage disequilibrium decay (mean r-squared by physical distance bin; see 43\_LD\_decay.tsv for the complete list)**

| distance_bin_start | distance_bin_end | n_pairs | mean_r2 |
|--------------------|------------------|---------|---------|
| 0                  | 4999             | 29649   | 0.2378  |
| 5000               | 9999             | 26955   | 0.1428  |
| 10000              | 14999            | 26344   | 0.1186  |
| 15000              | 19999            | 24701   | 0.1005  |
| 20000              | 24999            | 19601   | 0.0945  |
| 25000              | 29999            | 14745   | 0.0945  |
| 30000              | 34999            | 12472   | 0.0908  |
| 35000              | 39999            | 8631    | 0.1008  |
| 40000              | 44999            | 5739    | 0.1013  |
| 45000              | 49999            | 3807    | 0.0782  |
| 50000              | 54999            | 2603    | 0.0340  |
| 55000              | 59999            | 1467    | 0.0194  |
| 60000              | 64999            | 1189    | 0.0399  |
| 65000              | 69999            | 1594    | 0.0293  |
| 70000              | 74999            | 1744    | 0.0346  |
| 75000              | 79999            | 983     | 0.0459  |
| 80000              | 84999            | 1010    | 0.0379  |

| distance_bin_start | distance_bin_end | n_pairs | mean_r2 |
|--------------------|------------------|---------|---------|
| 85000              | 89999            | 915     | 0.0645  |
| 90000              | 94999            | 779     | 0.0601  |
| 95000              | 99999            | 748     | 0.0492  |
| 100000             | 104999           | 402     | 0.0710  |
| 105000             | 109999           | 349     | 0.0425  |
| 110000             | 114999           | 266     | 0.0550  |
| 115000             | 119999           | 107     | 0.0121  |
| 120000             | 124999           | 111     | 0.0139  |
| 125000             | 129999           | 72      | 0.0122  |
| 130000             | 134999           | 158     | 0.0092  |
| 135000             | 139999           | 192     | 0.0176  |
| 140000             | 144999           | 139     | 0.0142  |
| 145000             | 149999           | 263     | 0.0208  |
| 150000             | 154999           | 372     | 0.0142  |
| 155000             | 159999           | 241     | 0.0104  |
| 160000             | 164999           | 317     | 0.0113  |
| 165000             | 169999           | 363     | 0.0157  |
| 170000             | 174999           | 455     | 0.0130  |
| 175000             | 179999           | 245     | 0.0136  |
| 180000             | 184999           | 308     | 0.0087  |
| 185000             | 189999           | 190     | 0.0078  |
| 190000             | 194999           | 318     | 0.0091  |
| 195000             | 199999           | 65      | 0.0104  |
| 200000             | 204999           | 214     | 0.0084  |
| 205000             | 209999           | 216     | 0.0067  |
| 210000             | 214999           | 116     | 0.0069  |
| 215000             | 219999           | 243     | 0.0055  |
| 220000             | 224999           | 150     | 0.0072  |
| 225000             | 229999           | 173     | 0.0058  |
| 230000             | 234999           | 63      | 0.0073  |
| 235000             | 239999           | 65      | 0.0133  |
| 240000             | 244999           | 1       | 0.0077  |

## 12 LD-based effective population size (per population; see 45\_Ne\_LD.tsv for full diagnostics)

| population | n_samples | n_snps | n_chromosomes | n_pairs | harmonic_mean_n | mean_r2 | mean_r2_drift | ne | ne_status                  |
|------------|-----------|--------|---------------|---------|-----------------|---------|---------------|----|----------------------------|
| CHB        | 20        | 0      | 0             | 0       | NA              | NA      | NA            | NA | fewer_than_two_chromosomes |
| GBR        | 20        | 0      | 0             | 0       | NA              | NA      | NA            | NA | fewer_than_two_chromosomes |
| ITU        | 20        | 0      | 0             | 0       | NA              | NA      | NA            | NA | fewer_than_two_chromosomes |
| LWK        | 20        | 0      | 0             | 0       | NA              | NA      | NA            | NA | fewer_than_two_chromosomes |
| PEL        | 20        | 0      | 0             | 0       | NA              | NA      | NA            | NA | fewer_than_two_chromosomes |
| PUR        | 20        | 0      | 0             | 0       | NA              | NA      | NA            | NA | fewer_than_two_chromosomes |
| STU        | 20        | 0      | 0             | 0       | NA              | NA      | NA            | NA | fewer_than_two_chromosomes |
| YRI        | 20        | 0      | 0             | 0       | NA              | NA      | NA            | NA | fewer_than_two_chromosomes |



### 13 Close relatives (up to 20 shown; see 36\_close\_relatives.tsv for the complete list)

| sample_1 | sample_2 | population_1 | population_2 | IBS0   | kinship | relationship_degree |
|----------|----------|--------------|--------------|--------|---------|---------------------|
| HG03873  | HG03998  | ITU          | STU          | 0.0000 | 0.4525  | duplicate/MZ twin   |
| NA19331  | NA19334  | LWK          | LWK          | 0.0028 | 0.4459  | duplicate/MZ twin   |
| HG01938  | HG02278  | PEL          | PEL          | 0.0056 | 0.2781  | 1st-degree          |
| HG00742  | HG01054  | PUR          | PUR          | 0.0140 | 0.2468  | 1st-degree          |
| NA19172  | NA19247  | YRI          | YRI          | 0.0168 | 0.2448  | 1st-degree          |
| NA19360  | NA18516  | LWK          | YRI          | 0.0112 | 0.2313  | 1st-degree          |
| HG01067  | HG01182  | PUR          | PUR          | 0.0140 | 0.2225  | 1st-degree          |
| HG01067  | HG01086  | PUR          | PUR          | 0.0196 | 0.2125  | 1st-degree          |
| HG00099  | HG00141  | GBR          | GBR          | 0.0112 | 0.2110  | 1st-degree          |
| HG01067  | HG01305  | PUR          | PUR          | 0.0168 | 0.2100  | 1st-degree          |
| HG00122  | HG01308  | GBR          | PUR          | 0.0140 | 0.1971  | 1st-degree          |
| NA18516  | NA19222  | YRI          | YRI          | 0.0140 | 0.1959  | 1st-degree          |
| HG01396  | HG03740  | PUR          | STU          | 0.0084 | 0.1944  | 1st-degree          |
| NA18516  | NA19210  | YRI          | YRI          | 0.0196 | 0.1910  | 1st-degree          |
| HG00253  | HG01067  | GBR          | PUR          | 0.0168 | 0.1900  | 1st-degree          |
| HG01938  | HG02345  | PEL          | PEL          | 0.0084 | 0.1875  | 1st-degree          |
| NA18516  | NA19247  | YRI          | YRI          | 0.0196 | 0.1856  | 1st-degree          |
| HG00250  | HG00253  | GBR          | GBR          | 0.0168 | 0.1852  | 1st-degree          |
| HG00141  | HG00240  | GBR          | GBR          | 0.0168 | 0.1825  | 1st-degree          |
| HG00253  | HG01396  | GBR          | PUR          | 0.0168 | 0.1818  | 1st-degree          |

### 14 Sex check (reported vs inferred sex from X-chromosome heterozygosity and Y-chromosome call rate; up to 20 shown, discordant and ambiguous calls first; see 42\_sex\_check.tsv for the complete list)

| sample  | vcf_sample | n_x_snps | x_heterozygosity_F | x_inferred_sex | n_y_snps | y_call_rate | y_inferred_sex | inferred_sex | reported_sex | status     |
|---------|------------|----------|--------------------|----------------|----------|-------------|----------------|--------------|--------------|------------|
| NA18907 | NA18907    | 1148     | 0.9205             | male           | 60468    | 0           | female         | discordant   | female       | discordant |
| HG04059 | HG04059    | 1148     | 0.9277             | male           | 60468    | 0           | female         | discordant   | female       | discordant |
| HG00250 | HG00250    | 1148     | 1.0637             | male           | 60468    | 0           | female         | discordant   | female       | discordant |
| HG01061 | HG01061    | 1148     | 1.0872             | male           | 60468    | 0           | female         | discordant   | female       | discordant |
| HG01303 | HG01303    | 1148     | 1.1831             | male           | 60468    | 0           | female         | discordant   | female       | discordant |
| HG02425 | HG02425    | 1148     | -0.2899            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG03873 | HG03873    | 1148     | -0.2706            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG03760 | HG03760    | 1148     | -0.2472            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG01942 | HG01942    | 1148     | -0.2301            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG01566 | HG01566    | 1148     | -0.2275            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG03888 | HG03888    | 1148     | -0.2272            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG00125 | HG00125    | 1148     | -0.1966            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG01098 | HG01098    | 1148     | -0.1649            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG00099 | HG00099    | 1148     | -0.1573            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG01396 | HG01396    | 1148     | -0.1504            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG03858 | HG03858    | 1148     | -0.1370            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG00258 | HG00258    | 1148     | -0.1324            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG03857 | HG03857    | 1148     | -0.1303            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG04070 | HG04070    | 1148     | -0.1297            | female         | 60468    | 0           | female         | female       | female       | match      |
| HG02215 | HG02215    | 1148     | -0.1258            | female         | 60468    | 0           | female         | female       | female       | match      |

## 15 Runs of homozygosity (highest FROH samples, up to 20 shown; see 38\_ROH\_sample\_summary.tsv for the complete list)

| sample  | population | n_runs | total_length_bp | mean_length_bp | longest_run_bp | froh   | total_length_bp_short | total_length_bp_intermediate | total_length_bp_long | froh_short | froh_intermediate | froh_long |
|---------|------------|--------|-----------------|----------------|----------------|--------|-----------------------|------------------------------|----------------------|------------|-------------------|-----------|
| HG02146 | PEL        | 4      | 788356          | 197089.00      | 574060         | 0.7884 | 214296                | 574060                       | 0                    | 0.2143     | 0.5741            | 0         |
| HG01942 | PEL        | 4      | 595694          | 148923.50      | 235561         | 0.5957 | 595694                | 0                            | 0                    | 0.5957     | 0.0000            | 0         |
| HG01933 | PEL        | 4      | 537370          | 134342.50      | 311897         | 0.5374 | 537370                | 0                            | 0                    | 0.5374     | 0.0000            | 0         |
| HG01578 | PEL        | 6      | 528877          | 88146.17       | 194872         | 0.5289 | 528877                | 0                            | 0                    | 0.5289     | 0.0000            | 0         |
| HG03989 | STU        | 2      | 469894          | 234947.00      | 265232         | 0.4699 | 469894                | 0                            | 0                    | 0.4699     | 0.0000            | 0         |
| HG00151 | GBR        | 4      | 438830          | 109707.50      | 207650         | 0.4388 | 438830                | 0                            | 0                    | 0.4388     | 0.0000            | 0         |
| HG01950 | PEL        | 4      | 438096          | 109524.00      | 217282         | 0.4381 | 438096                | 0                            | 0                    | 0.4381     | 0.0000            | 0         |
| NA18611 | CHB        | 3      | 437054          | 145684.67      | 249970         | 0.4371 | 437054                | 0                            | 0                    | 0.4371     | 0.0000            | 0         |
| HG00637 | PUR        | 4      | 407361          | 101840.25      | 155344         | 0.4074 | 407361                | 0                            | 0                    | 0.4074     | 0.0000            | 0         |
| HG00123 | GBR        | 3      | 406869          | 135623.00      | 221114         | 0.4069 | 406869                | 0                            | 0                    | 0.4069     | 0.0000            | 0         |
| NA18612 | CHB        | 3      | 400265          | 133421.67      | 226775         | 0.4003 | 400265                | 0                            | 0                    | 0.4003     | 0.0000            | 0         |
| NA18616 | CHB        | 5      | 391340          | 78268.00       | 268046         | 0.3914 | 391340                | 0                            | 0                    | 0.3914     | 0.0000            | 0         |
| HG01054 | PUR        | 3      | 379034          | 126344.67      | 228087         | 0.3790 | 379034                | 0                            | 0                    | 0.3790     | 0.0000            | 0         |
| HG01086 | PUR        | 3      | 364023          | 121341.00      | 162866         | 0.3640 | 364023                | 0                            | 0                    | 0.3640     | 0.0000            | 0         |
| HG03836 | STU        | 2      | 359185          | 179592.50      | 284841         | 0.3592 | 359185                | 0                            | 0                    | 0.3592     | 0.0000            | 0         |
| HG03990 | STU        | 3      | 349620          | 116540.00      | 175138         | 0.3496 | 349620                | 0                            | 0                    | 0.3496     | 0.0000            | 0         |
| NA18647 | CHB        | 3      | 339679          | 113226.33      | 182839         | 0.3397 | 339679                | 0                            | 0                    | 0.3397     | 0.0000            | 0         |
| HG01917 | PEL        | 3      | 337023          | 112341.00      | 261589         | 0.3370 | 337023                | 0                            | 0                    | 0.3370     | 0.0000            | 0         |
| HG01927 | PEL        | 2      | 330751          | 165375.50      | 191744         | 0.3308 | 330751                | 0                            | 0                    | 0.3308     | 0.0000            | 0         |
| NA18564 | CHB        | 3      | 323282          | 107760.67      | 181407         | 0.3233 | 323282                | 0                            | 0                    | 0.3233     | 0.0000            | 0         |

## 16 Discriminant analysis of principal components

| K  | n_pca | n_da | BIC | assignment_accuracy | mean_success | calinski_harabasz | davies_bouldin | replicate_max_rmse | replicate_min_cluster_correlation |
|----|-------|------|-----|---------------------|--------------|-------------------|----------------|--------------------|-----------------------------------|
| 2  | 30    | 1    | NA  | 0.2500              | 1.0000       | 14.1061           | 2.9137         | 0.0000             | 1.0000                            |
| 3  | 50    | 2    | NA  | 0.3562              | 0.9722       | 11.4855           | 3.2275         | 0.0000             | 1.0000                            |
| 4  | 10    | 3    | NA  | 0.4000              | 0.9833       | 9.3288            | 3.5914         | 0.1514             | 0.8760                            |
| 5  | 30    | 4    | NA  | 0.4562              | 0.9623       | 8.0883            | 3.3545         | 0.2692             | 0.5281                            |
| 6  | 20    | 5    | NA  | 0.5000              | 0.9444       | 7.2266            | 3.2839         | 0.2732             | 0.2926                            |
| 7  | 20    | 6    | NA  | 0.5000              | 0.9683       | 6.6813            | 3.0430         | 0.1490             | 0.7587                            |
| 8  | 10    | 7    | NA  | 0.5250              | 0.9278       | 6.1282            | 3.0267         | 0.2186             | -0.1573                           |
| 9  | 20    | 8    | NA  | 0.4812              | 0.9309       | 5.6760            | 3.3201         | 0.2513             | 0.0984                            |
| 10 | 30    | 9    | NA  | 0.4812              | 0.9350       | 5.3617            | 3.1843         | 0.2714             | -0.0563                           |

## 17 DAPC SNP loadings (top 20 by contribution across all K; see 22f\_DAPC\_loadings\_K.tsv for the complete per-K top-N lists)

| K  | axis | snp_id | chromosome | position | contribution |
|----|------|--------|------------|----------|--------------|
| 8  | LD7  | 10937  | 22         | 20437004 | 0.0844       |
| 8  | LD7  | 11374  | 22         | 20502771 | 0.0788       |
| 8  | LD7  | 10502  | 22         | 20371297 | 0.0662       |
| 10 | LD7  | 7866   | 22         | 20246081 | 0.0634       |
| 9  | LD3  | 5277   | 22         | 20168641 | 0.0555       |
| 7  | LD3  | 7866   | 22         | 20246081 | 0.0547       |
| 5  | LD4  | 14793  | 22         | 20791821 | 0.0536       |
| 9  | LD6  | 10937  | 22         | 20437004 | 0.0511       |
| 9  | LD6  | 13990  | 22         | 20765538 | 0.0499       |
| 10 | LD8  | 102    | 22         | 20003266 | 0.0497       |
| 9  | LD8  | 20613  | 22         | 20977077 | 0.0497       |
| 7  | LD3  | 7946   | 22         | 20248391 | 0.0476       |
| 4  | LD3  | 10937  | 22         | 20437004 | 0.0471       |
| 9  | LD6  | 14246  | 22         | 20773711 | 0.0470       |
| 6  | LD4  | 14793  | 22         | 20791821 | 0.0469       |
| 5  | LD3  | 5706   | 22         | 20180455 | 0.0467       |
| 8  | LD4  | 21207  | 22         | 20993766 | 0.0456       |
| 5  | LD3  | 4933   | 22         | 20159049 | 0.0454       |
| 9  | LD5  | 17500  | 22         | 20889926 | 0.0453       |
| 6  | LD4  | 19620  | 22         | 20952226 | 0.0441       |

## 18 Analysis of molecular variance

| component                                    | Sigma    | %        |
|----------------------------------------------|----------|----------|
| Variations Between population                | 27.1959  | 9.0406   |
| Variations Between samples Within population | 12.8319  | 4.2657   |
| Variations Within samples                    | 260.7906 | 86.6937  |
| Total variations                             | 300.8184 | 100.0000 |

| statistic              | Phi    |
|------------------------|--------|
| Phi-samples-total      | 0.1331 |
| Phi-samples-population | 0.0469 |
| Phi-population-total   | 0.0904 |

## 19 Multilocus genotype (MLG) and clonal diversity

| population | n   | mlg | emlg | emlg_se | shannon_h | stoddart_taylor_g | simpson_lambda | evenness_e5 | expected_heterozygosity | ia     | rbard  |
|------------|-----|-----|------|---------|-----------|-------------------|----------------|-------------|-------------------------|--------|--------|
| CHB        | 20  | 20  | 20   | 0       | 2.9957    | 20                | 0.9500         | 1           | 0.2451                  | 3.5008 | 0.0130 |
| GBR        | 20  | 20  | 20   | 0       | 2.9957    | 20                | 0.9500         | 1           | 0.2569                  | 1.7093 | 0.0062 |
| ITU        | 20  | 20  | 20   | 0       | 2.9957    | 20                | 0.9500         | 1           | 0.2635                  | 1.0261 | 0.0035 |
| LWK        | 20  | 20  | 20   | 0       | 2.9957    | 20                | 0.9500         | 1           | 0.2492                  | 1.7865 | 0.0064 |
| PEL        | 20  | 20  | 20   | 0       | 2.9957    | 20                | 0.9500         | 1           | 0.2396                  | 5.8010 | 0.0204 |
| PUR        | 20  | 20  | 20   | 0       | 2.9957    | 20                | 0.9500         | 1           | 0.2548                  | 2.2954 | 0.0076 |
| STU        | 20  | 20  | 20   | 0       | 2.9957    | 20                | 0.9500         | 1           | 0.2578                  | 1.8962 | 0.0065 |
| YRI        | 20  | 20  | 20   | 0       | 2.9957    | 20                | 0.9500         | 1           | 0.2522                  | 1.1104 | 0.0040 |
| Total      | 160 | 160 | 20   | 0       | 5.0752    | 160               | 0.9938         | 1           | 0.2714                  | 2.2770 | 0.0066 |

## 20 Sex-biased dispersal test

mAIC test: statistic = -0.648, p = 0.5177 (n = 83 female, 77 male). See 60\_sexbias\_AIC\_by\_sample.tsv for per-sample assignment indices.

## 21 Mantel test and isolation by distance

| mantel_r | mantel_p | slope  | r_squared | partial_mantel_r | partial_mantel_p |
|----------|----------|--------|-----------|------------------|------------------|
| 0.3129   | 0.001    | 0.0028 | 0.1366    | 0.1627           | 0.001            |

## 22 Admixture-model cross-validation

| K | cv_error |
|---|----------|
| 2 | 0.47688  |
| 3 | 0.46091  |
| 4 | 0.45539  |
| 5 | 0.45427  |
| 6 | 0.45225  |
| 7 | 0.45211  |
| 8 | 0.45249  |
| 9 | 0.45470  |

## 23 Chromosome-specific results

| chromosome | qc_snps | ld_snps | global_fst | pc1_percent |
|------------|---------|---------|------------|-------------|
| 22         | 1969    | 357     | 0.0915     | 9.1314      |

## 24 Figure gallery

The gallery contains one version of every analysis figure. The HTML report prefers browser-friendly SVG or PNG files. The PDF report prefers the compact vector PDF source for each figure. Original files remain available in the sibling `figures/` directory.

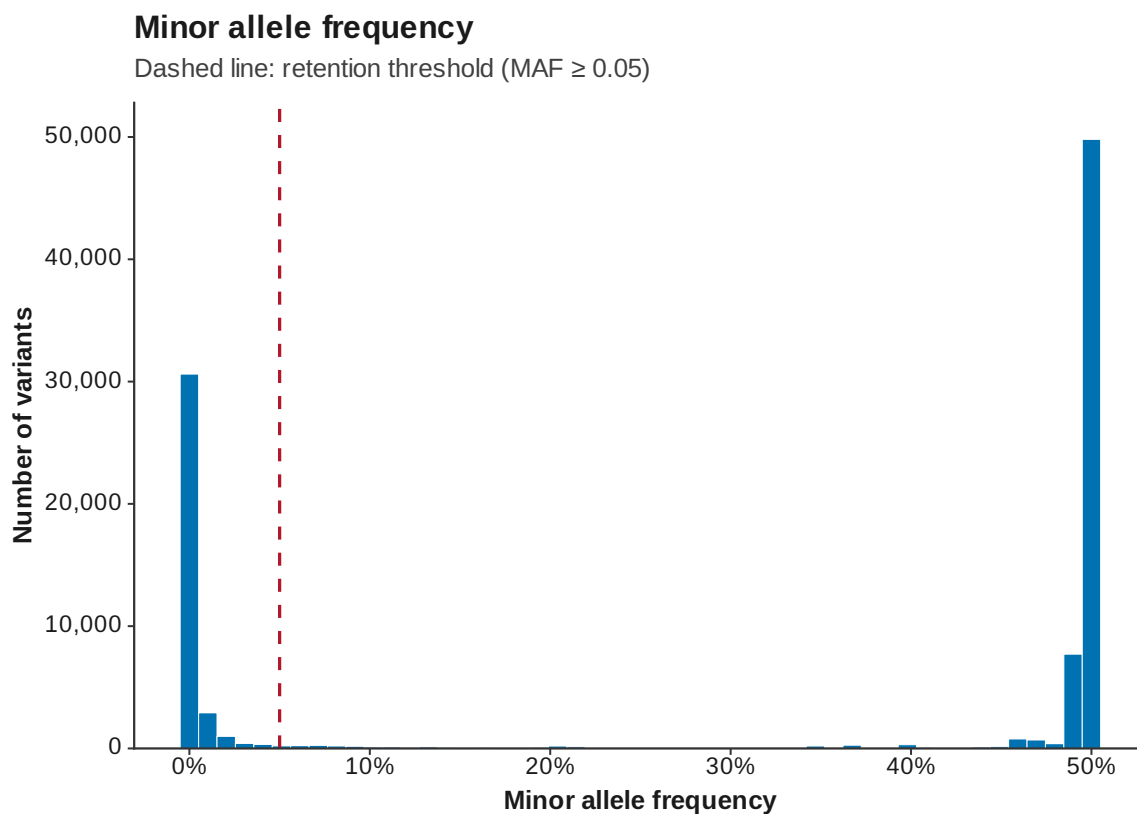


Figure 1: MAF

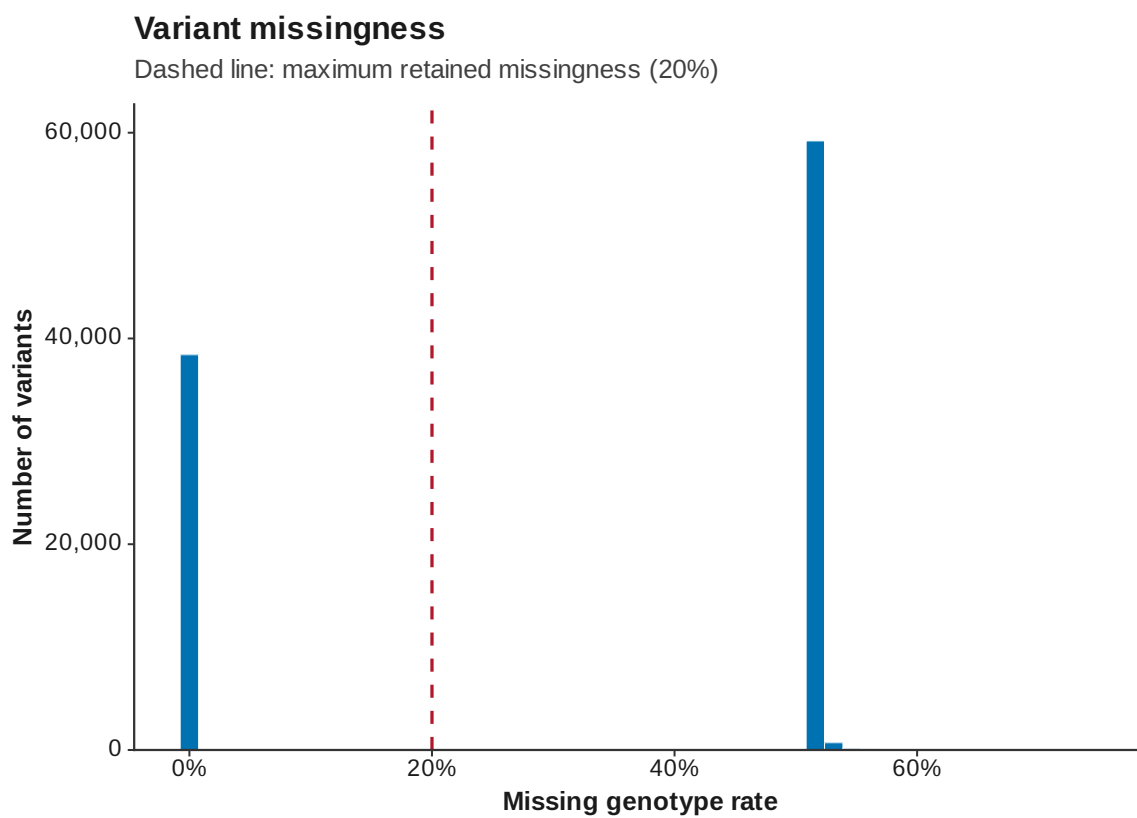
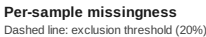


Figure 2: Variant missingness



14

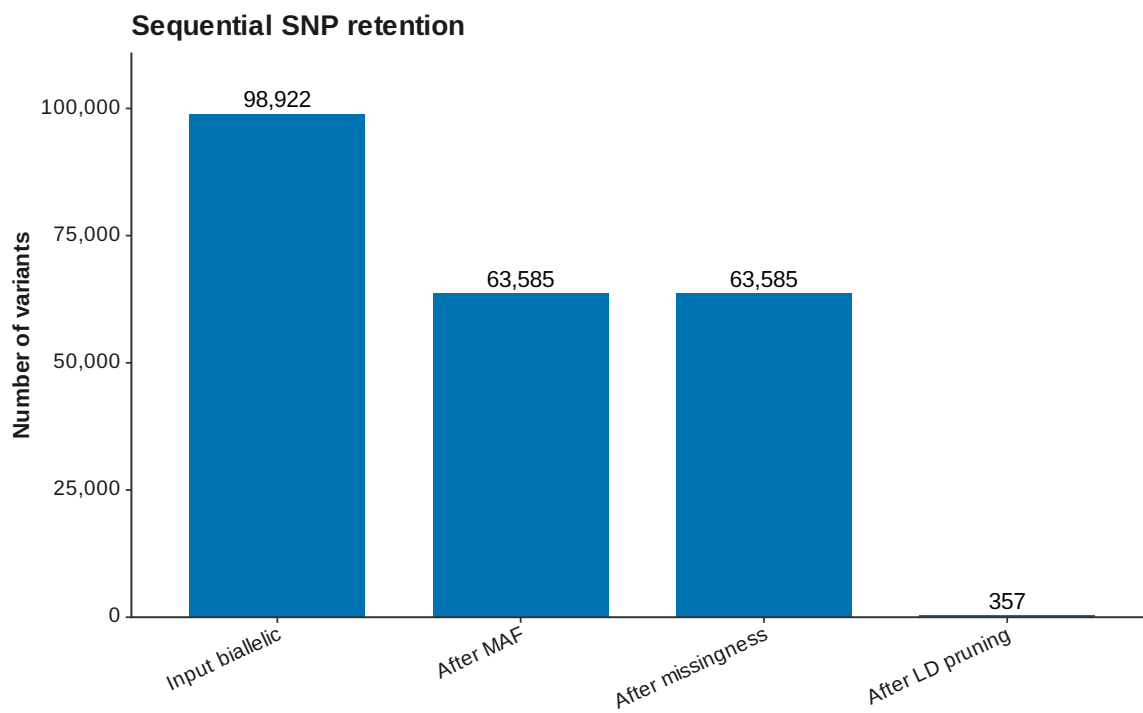


Figure 4: SNP retention

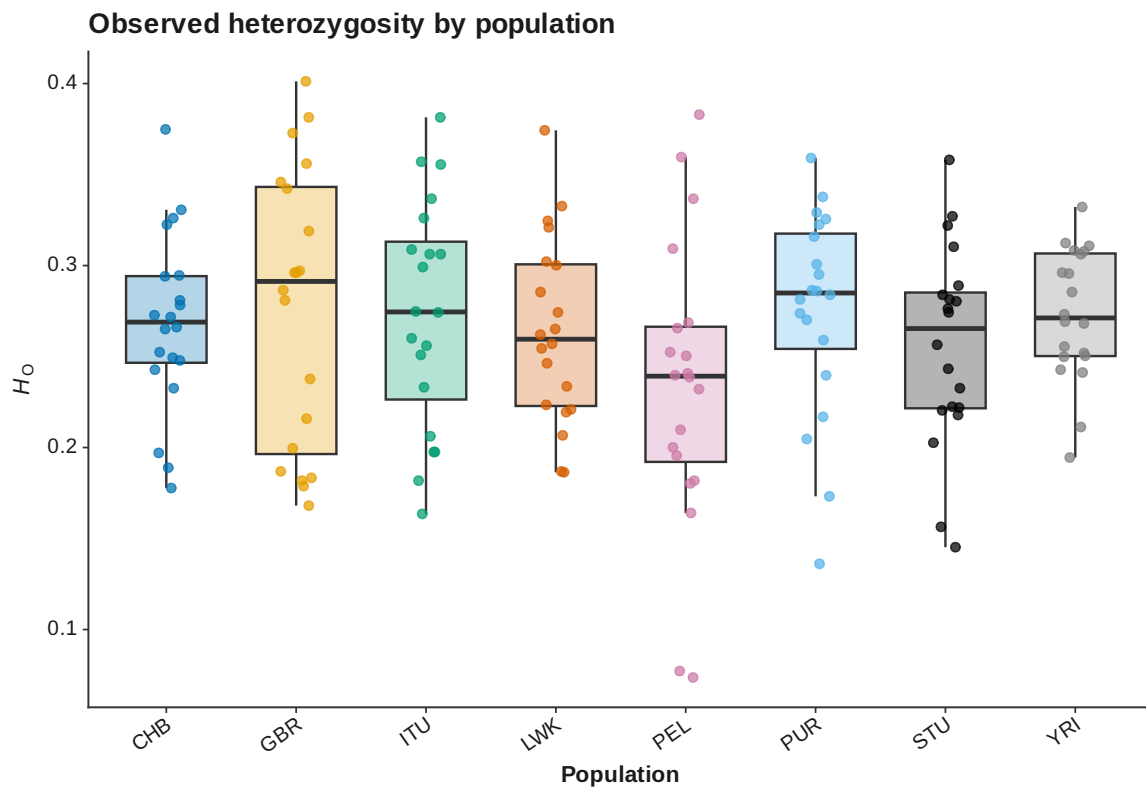


Figure 5: Sample heterozygosity



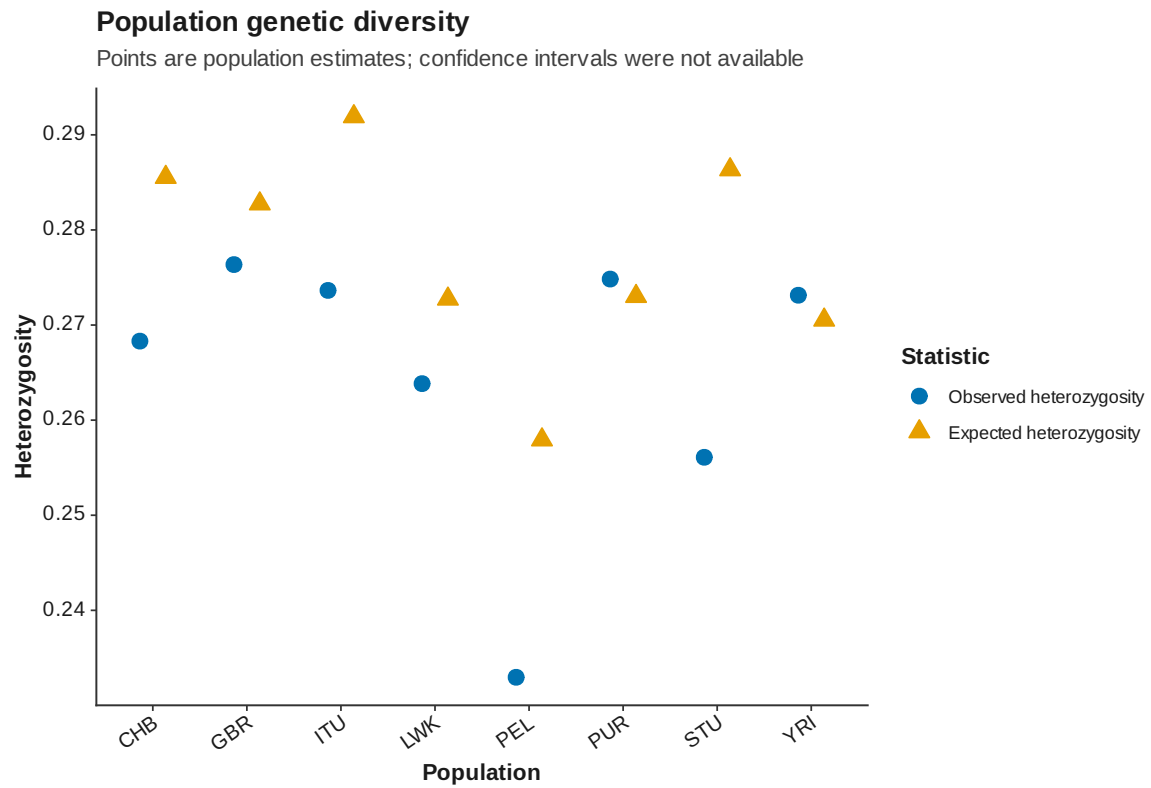


Figure 6: Population diversity

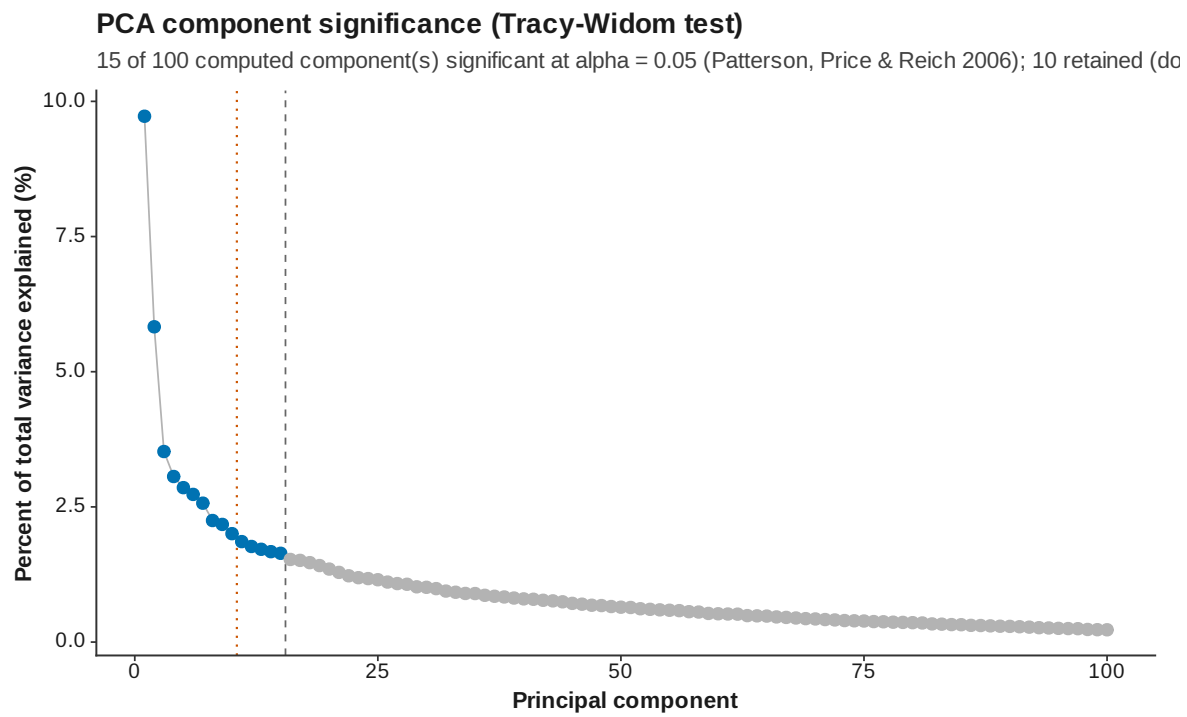


Figure 7: PCA Tracy Widom test

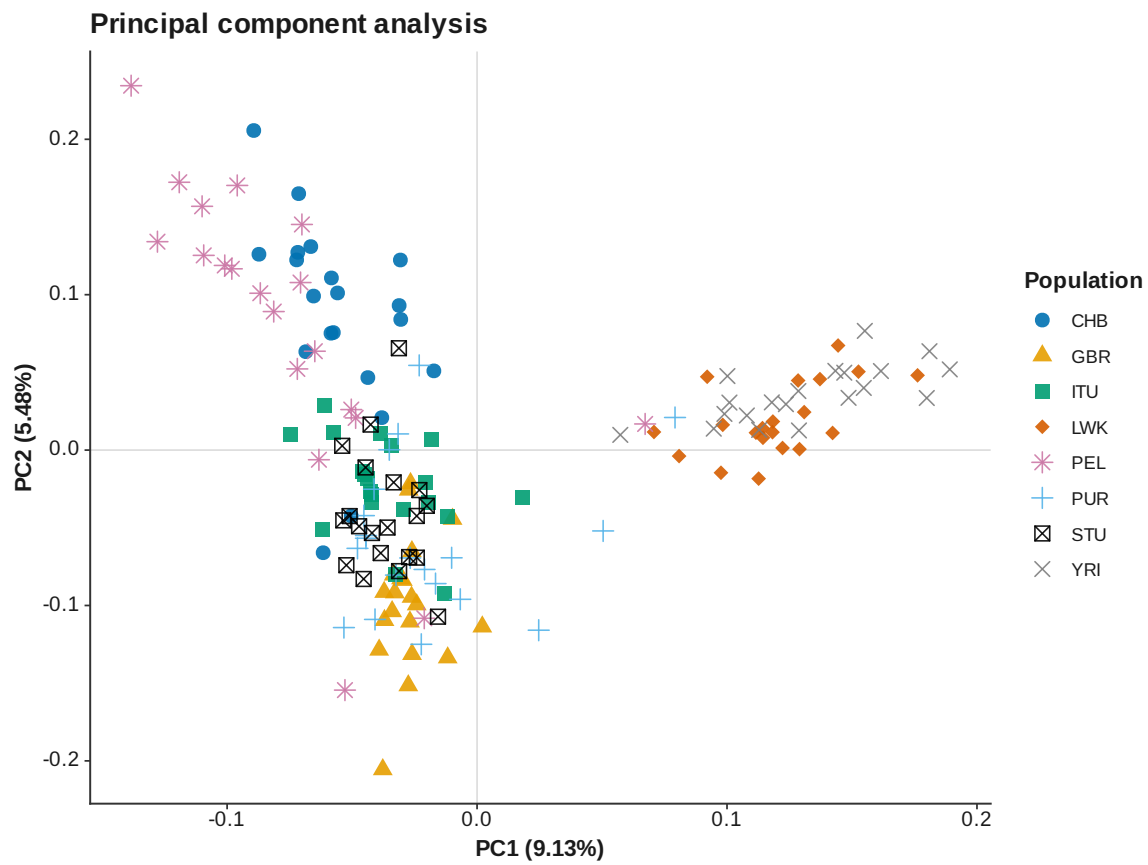


Figure 8: PCA PC 1 PC 2

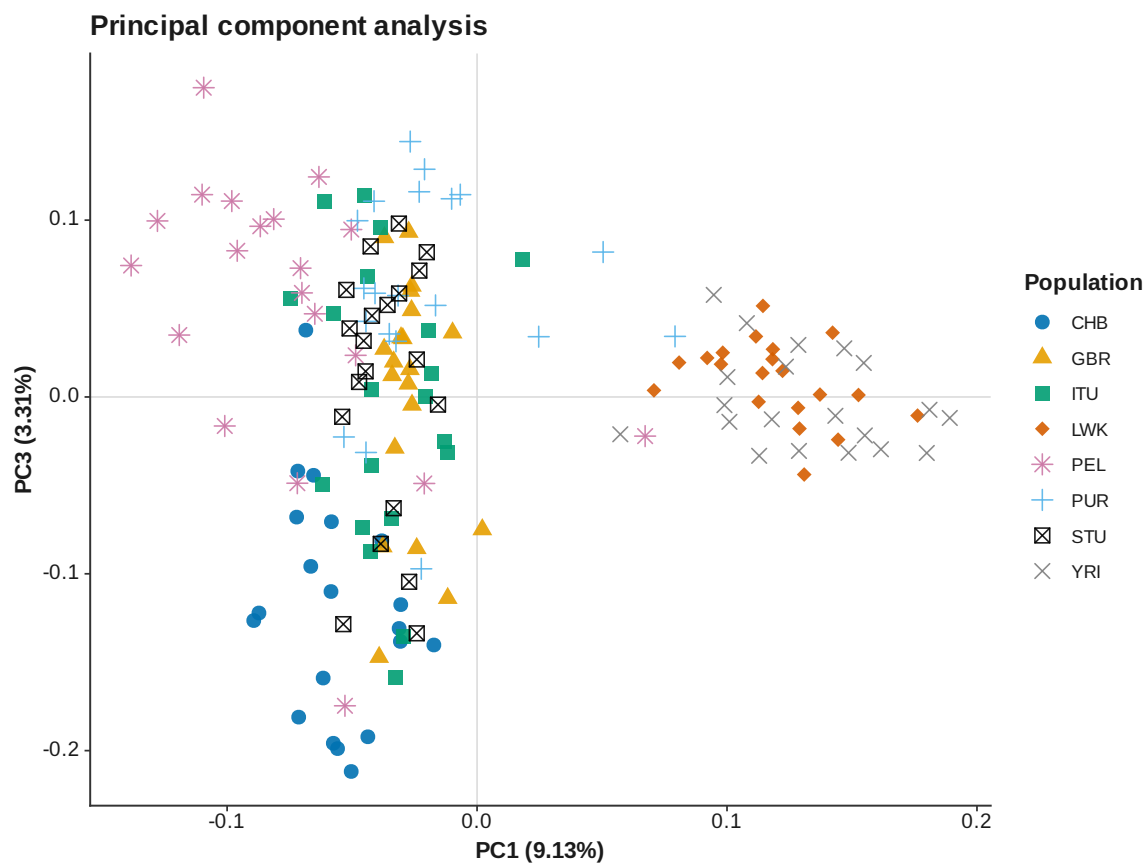


Figure 9: PCA PC 1 PC 3

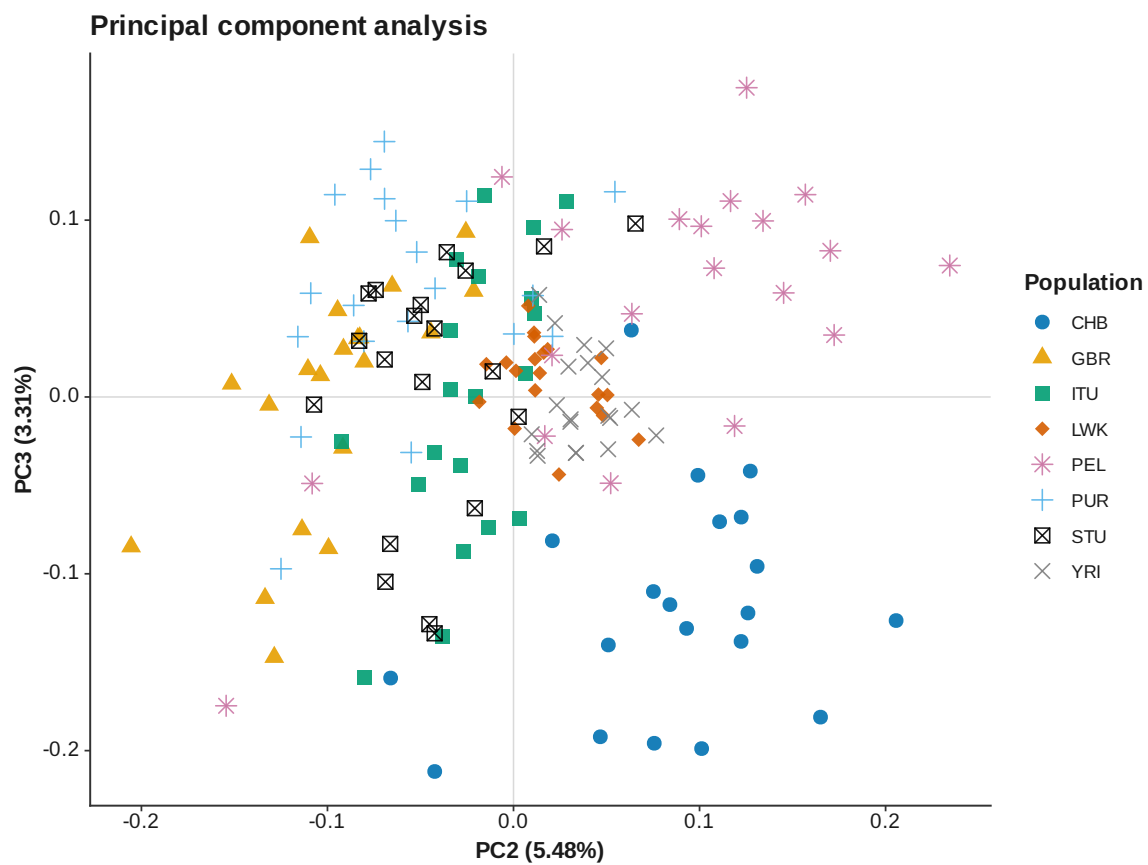


Figure 10: PCA PC 2 PC 3

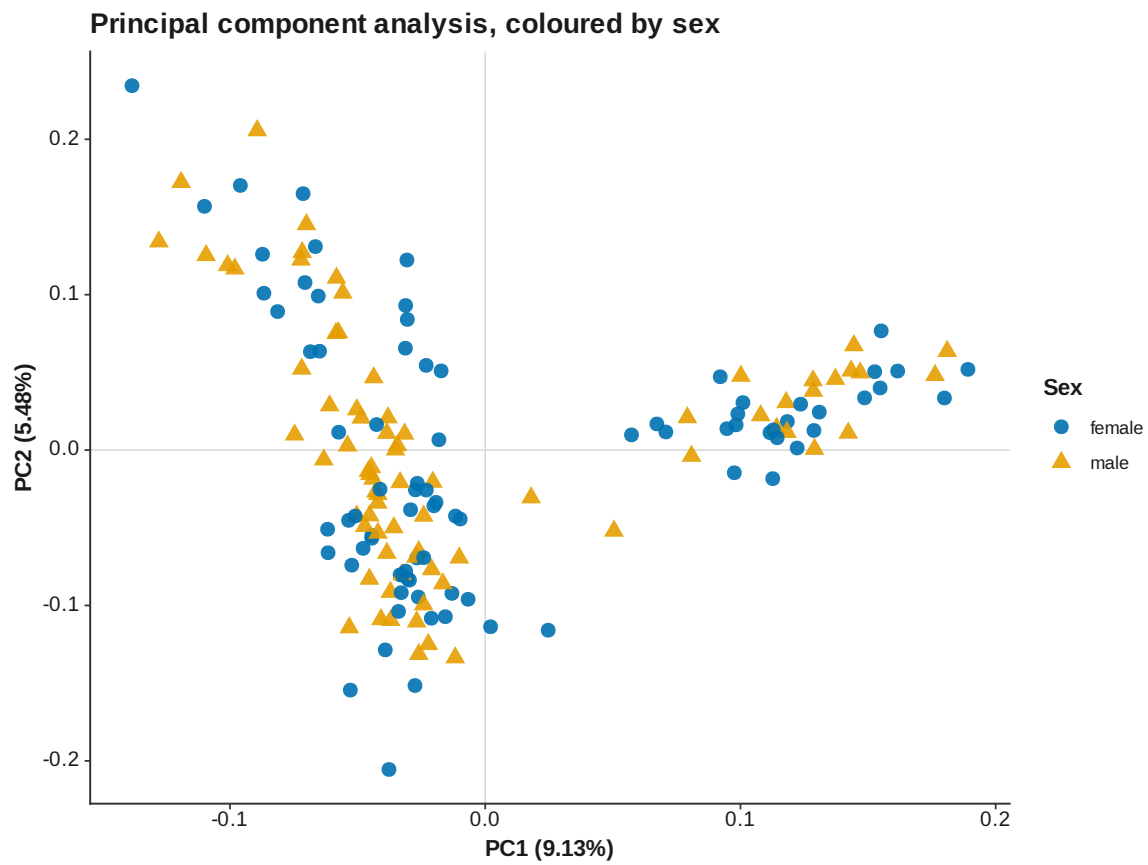


Figure 11: PCA PC 1 PC 2 by sex

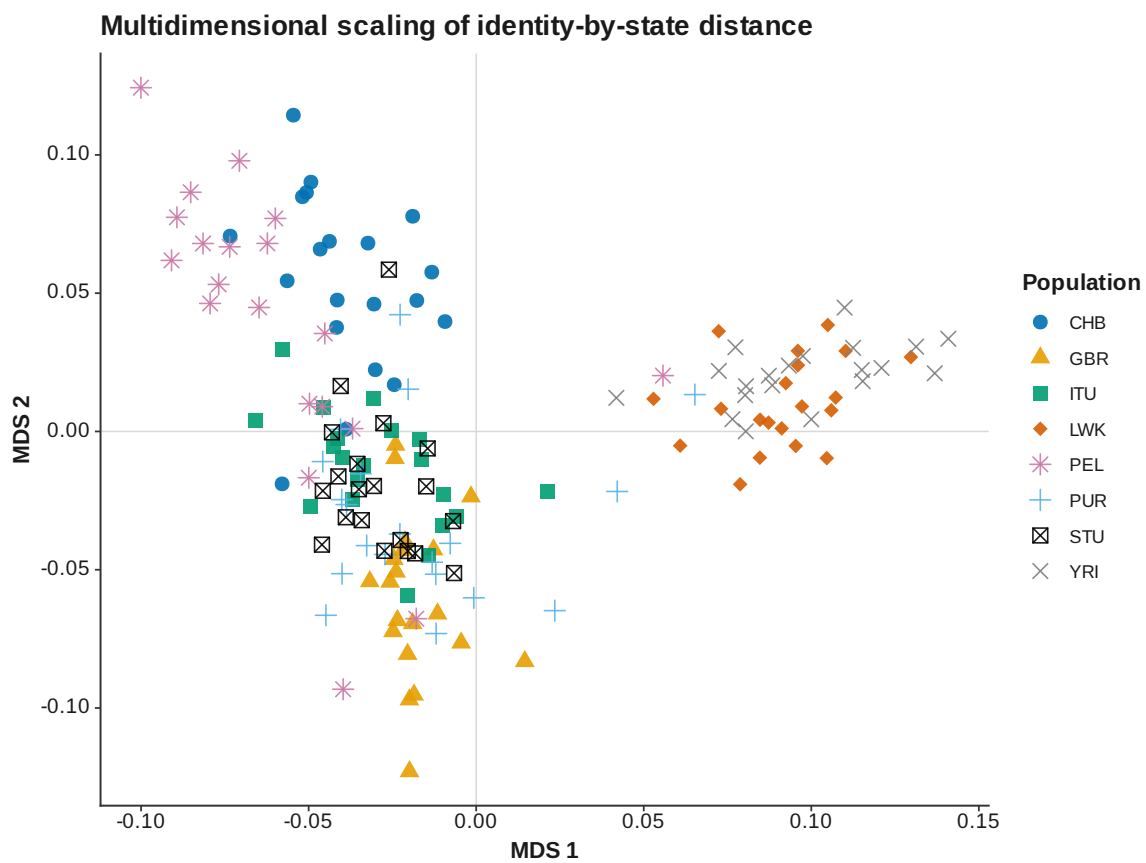


Figure 12: IBS MDS

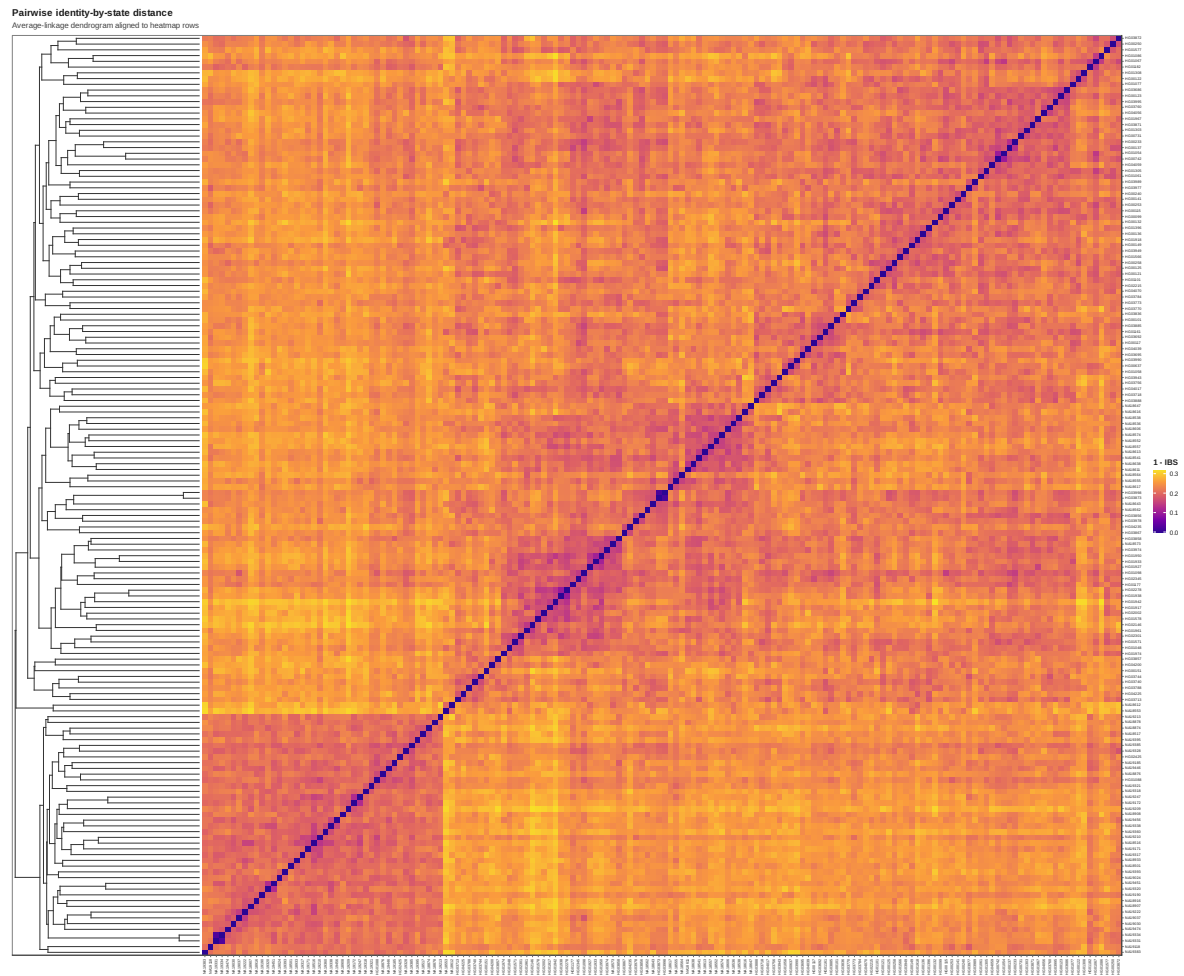


Figure 13: IBS heatmap



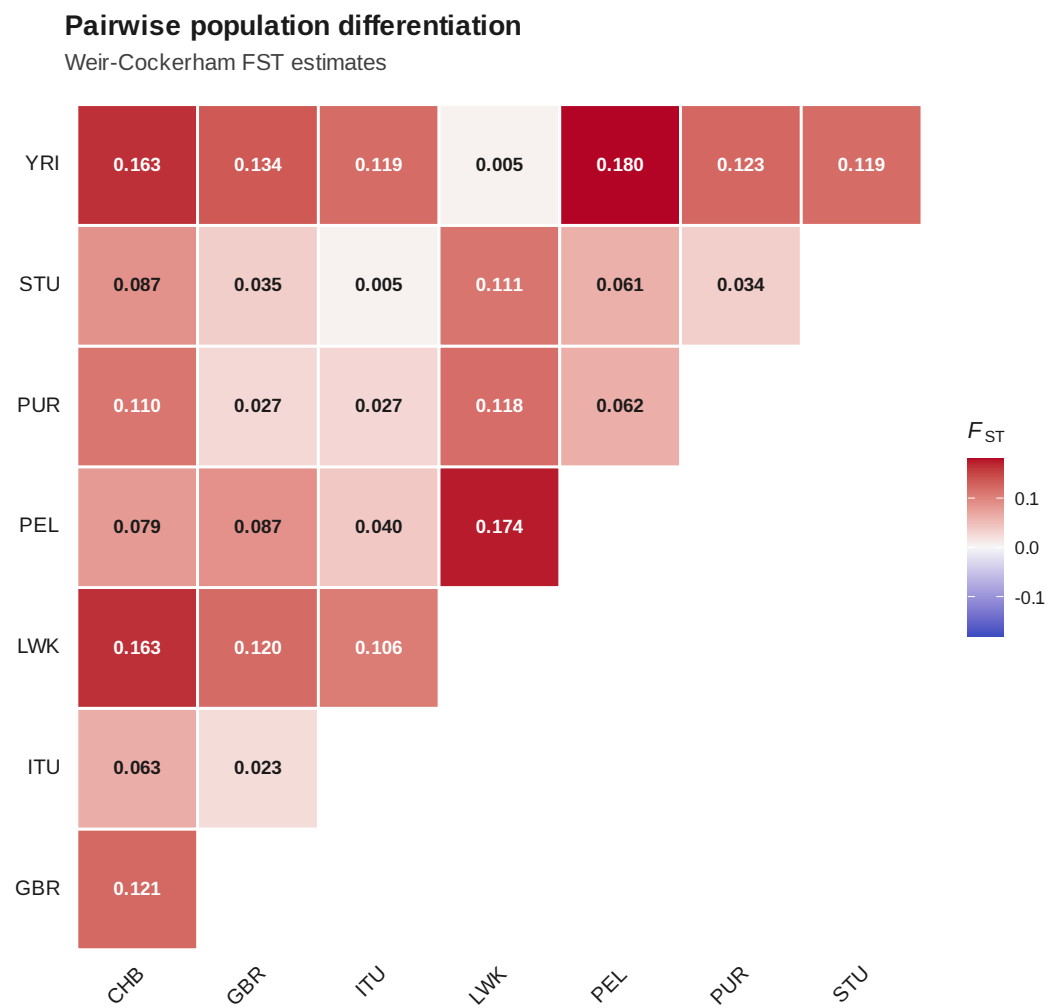


Figure 14: Pairwise  $F_{ST}$

**Discriminant analysis of principal components (K = 3)**

Replicate membership RMSE = 1.491e-17 (stability threshold = 0.05); minimum cluster correlation = 1 (threshold = 0.9).  
50 PCA axis/axes retained for this model.

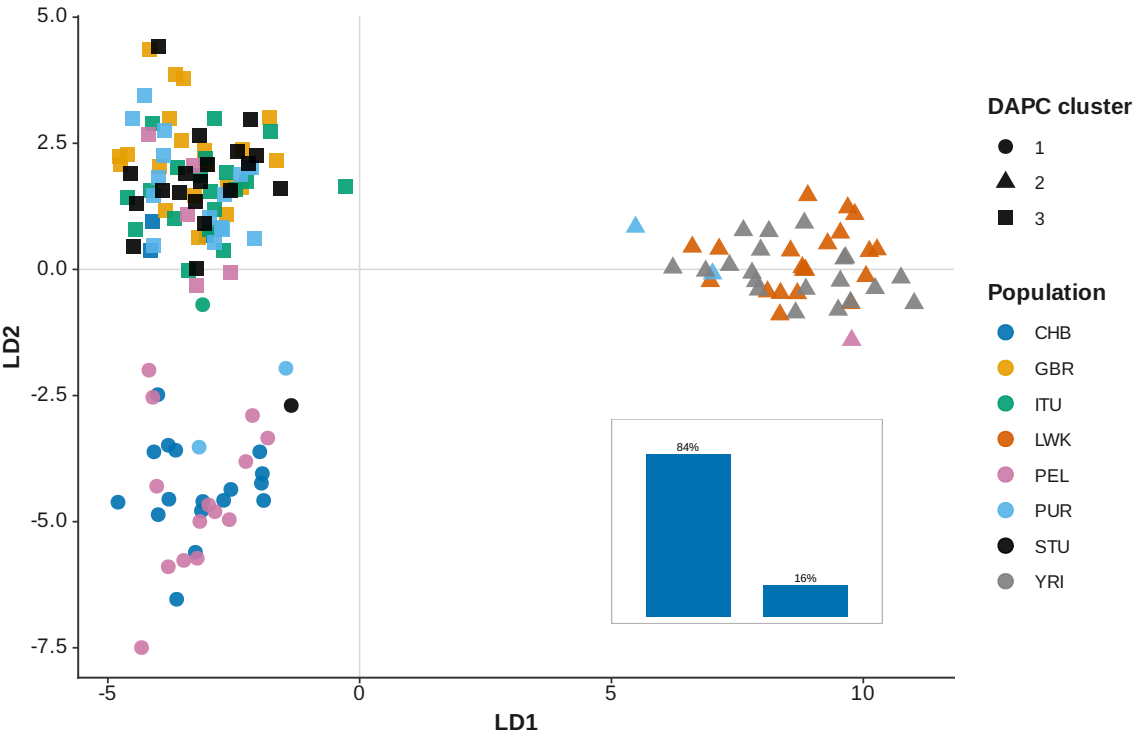


Figure 15: DAPC K 3

### Discriminant analysis of principal components (K = 4)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.1514 > 0.05, minimum cluster correlation = 0.876 < 0.9).**

**Avoid interpreting these assignments.  
10 PCA axis/axes retained for this model.**

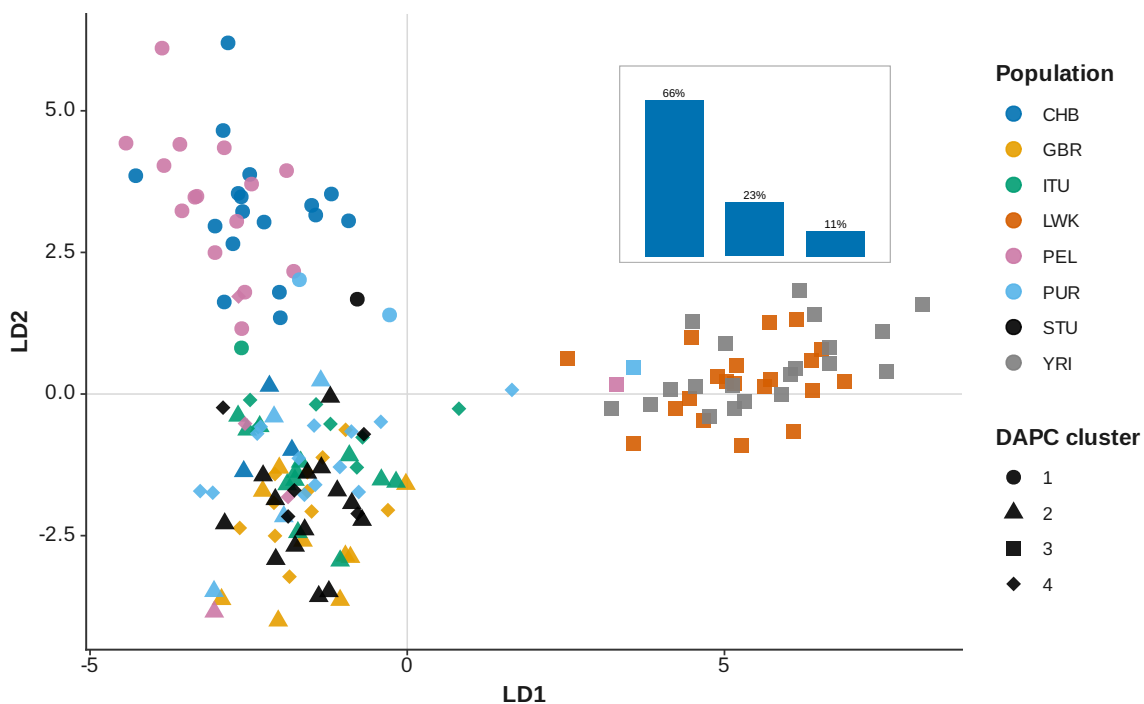


Figure 16: DAPC K 4

### Discriminant analysis of principal components (K = 5)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.2692 > 0.05, minimum cluster correlation = 0.5281 < 0.9).**

**Avoid interpreting these assignments.**  
**30 PCA axis/axes retained for this model.**

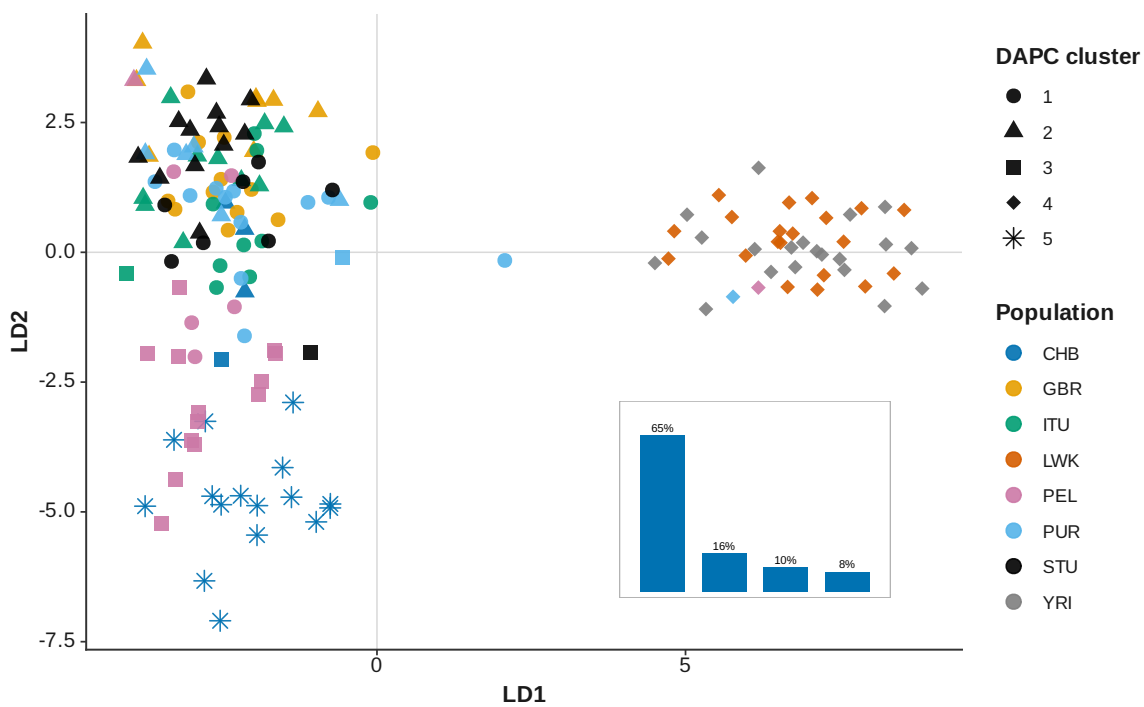


Figure 17: DAPC K 5

### Discriminant analysis of principal components (K = 6)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.2732 > 0.05, minimum cluster correlation = 0.2926 < 0.9).**

**Avoid interpreting these assignments.**  
**20 PCA axis/axes retained for this model.**

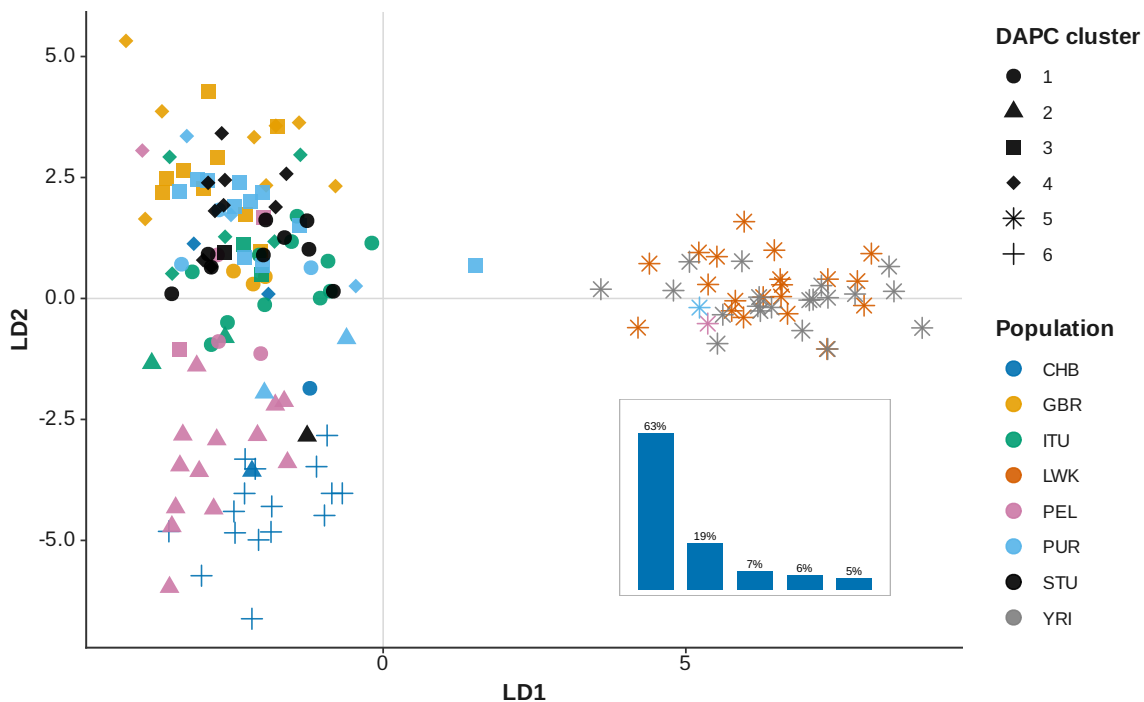


Figure 18: DAPC K 6

### Discriminant analysis of principal components (K = 7)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.149 > 0.05, minimum cluster correlation = 0.7587 < 0.9).**

**Avoid interpreting these assignments.**  
**20 PCA axis/axes retained for this model.**

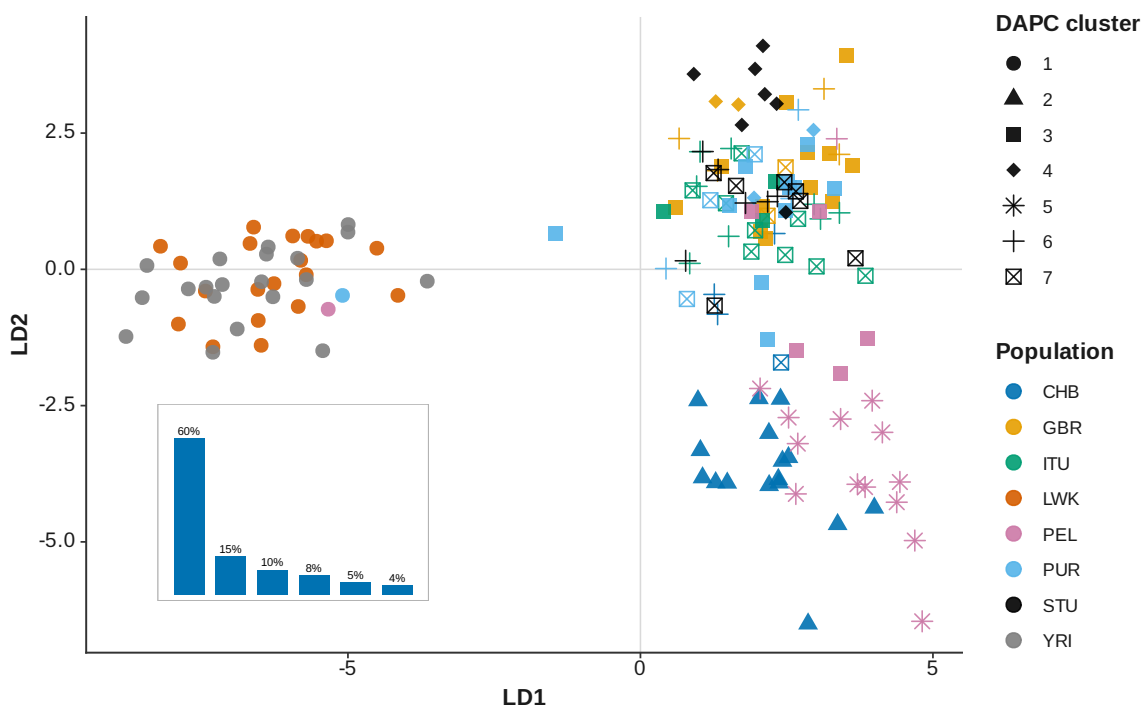


Figure 19: DAPC K 7

### Discriminant analysis of principal components (K = 8)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.2186 > 0.05, minimum cluster correlation = -0.1573 < 0.9).**

**Avoid interpreting these assignments.**

**10 PCA axis/axes retained for this model.**

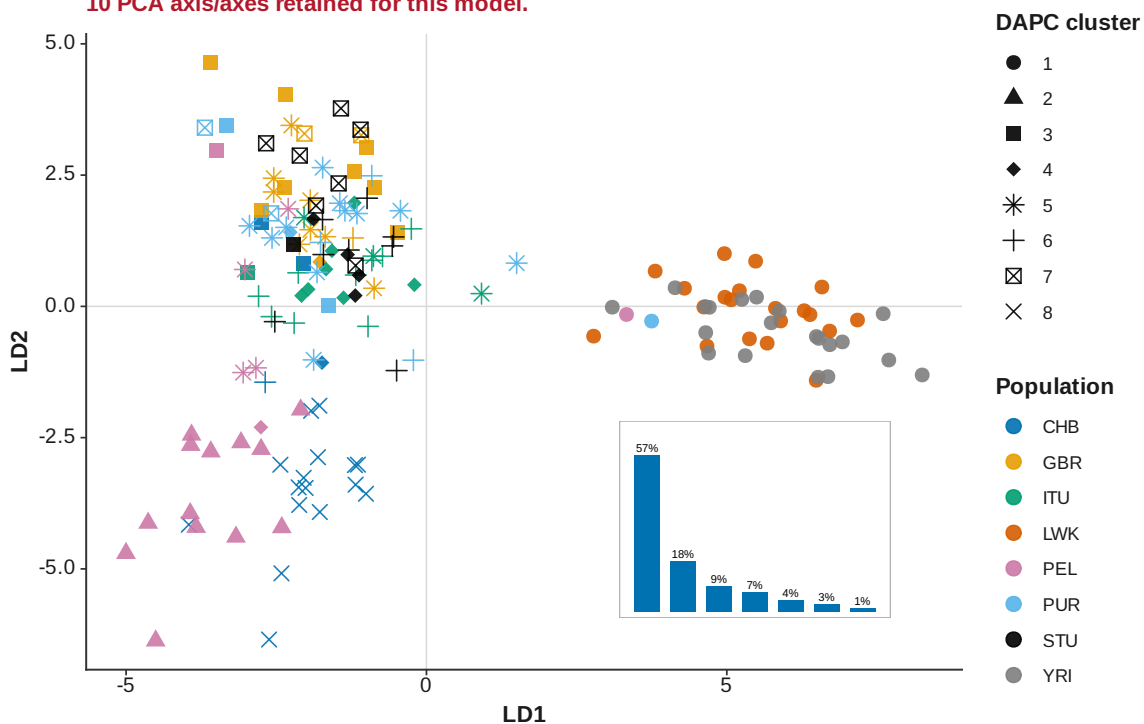


Figure 20: DAPC K 8

### Discriminant analysis of principal components (K = 9)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.2513 > 0.05, minimum cluster correlation = 0.09844 < 0.9).**

**Avoid interpreting these assignments.**

**20 PCA axis/axes retained for this model.**

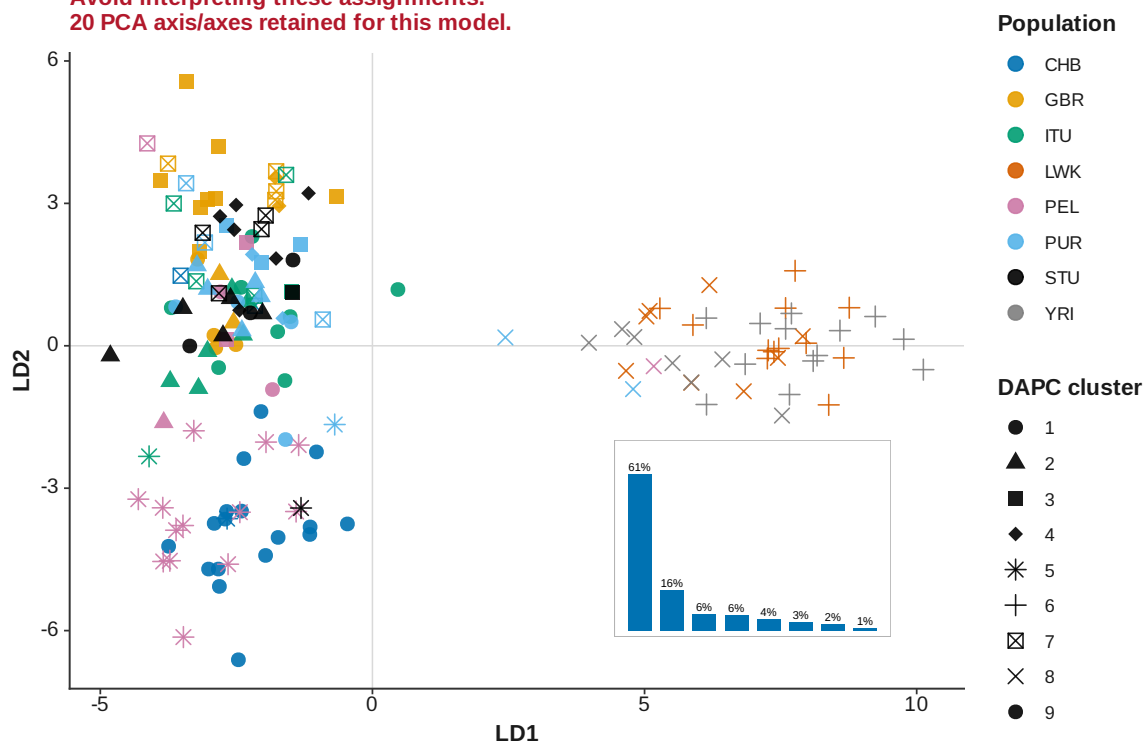


Figure 21: DAPC K 9



# Discriminant analysis of principal components (K = 10)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.2714 > 0.05, minimum cluster correlation = -0.05631 < 0.9).**

**Avoid interpreting these assignments.**

**30 PCA axis/axes retained for this model.**

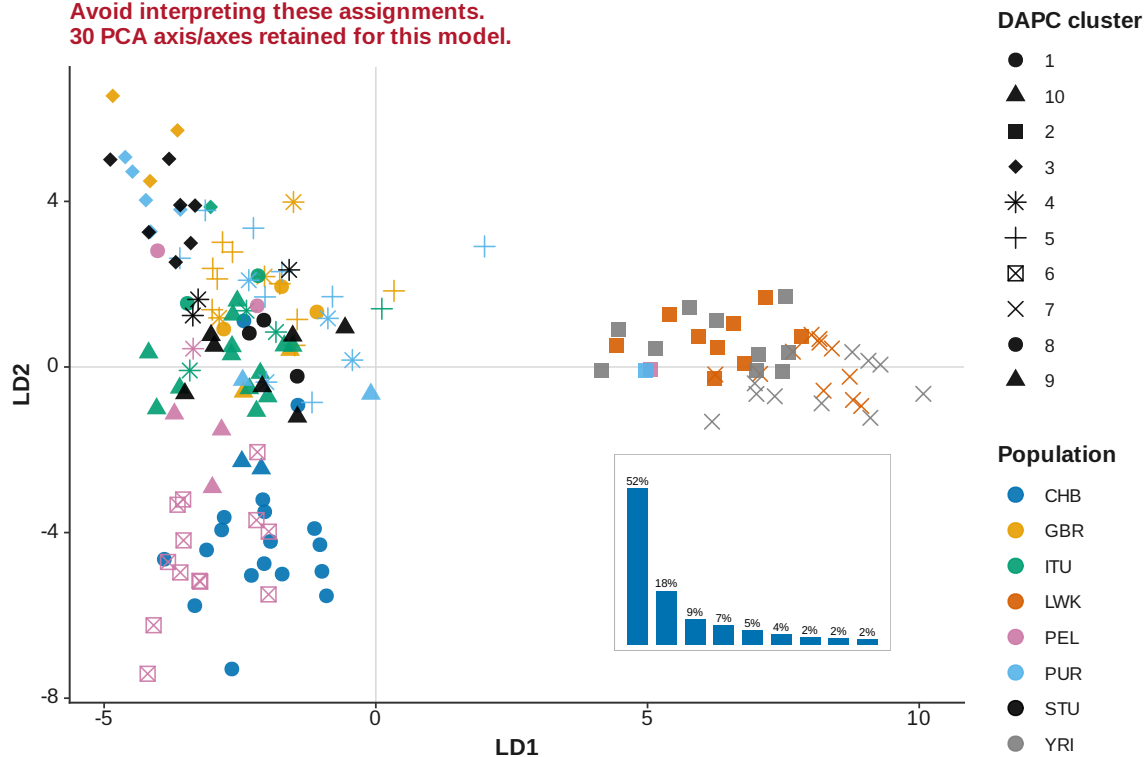


Figure 22: DAPC K 10

### Discriminant analysis of principal components cluster-number selection

Consensus number of clusters = 2; 12 of 12 method votes agree

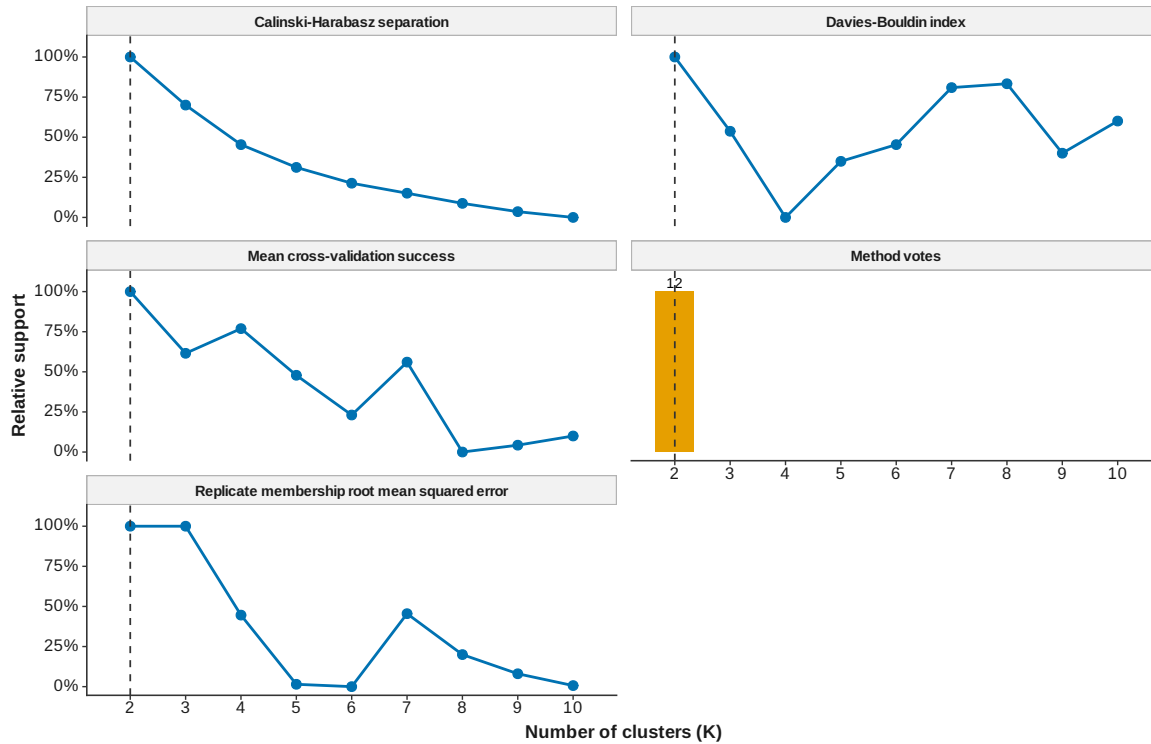
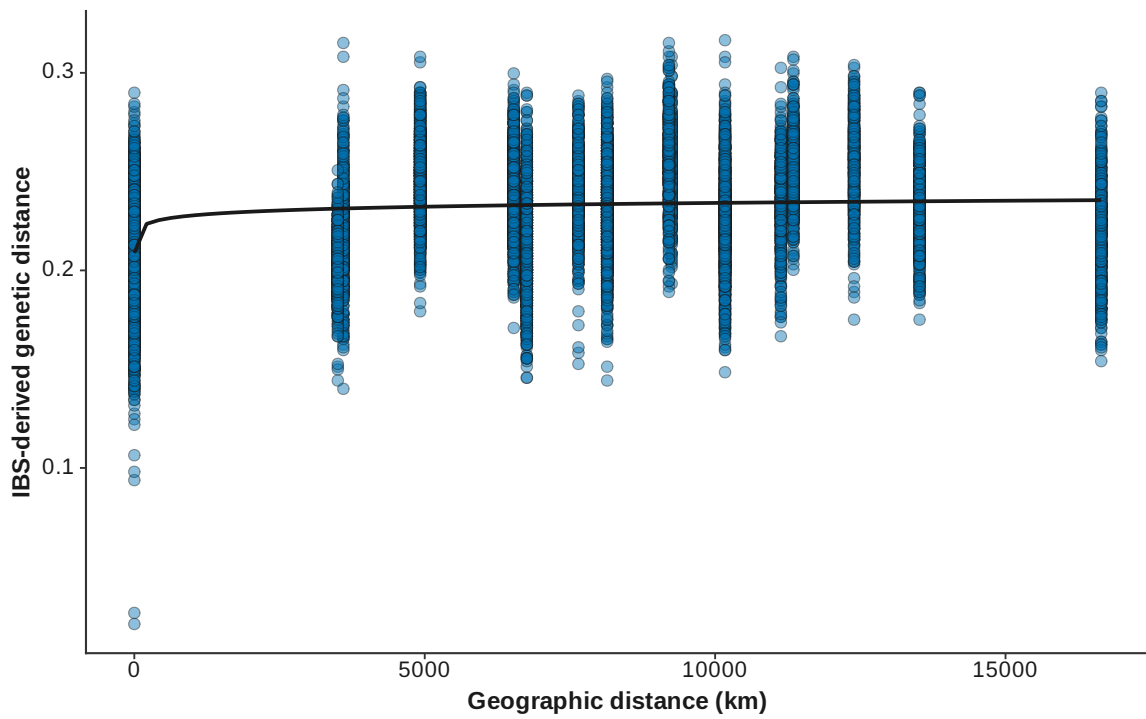


Figure 23: DAPC cluster number selection

### Isolation by distance

Mantel  $r = 0.313$ ,  $p = 0.0010$  (partial, controlling for population:  $r = 0.163$ ,  $p = 0.0010$ )



Curve: linear model of genetic distance on  $\log(1 + \text{geographic distance})$ ; 12,720 pairwise comparisons.

Figure 24: Isolation by distance

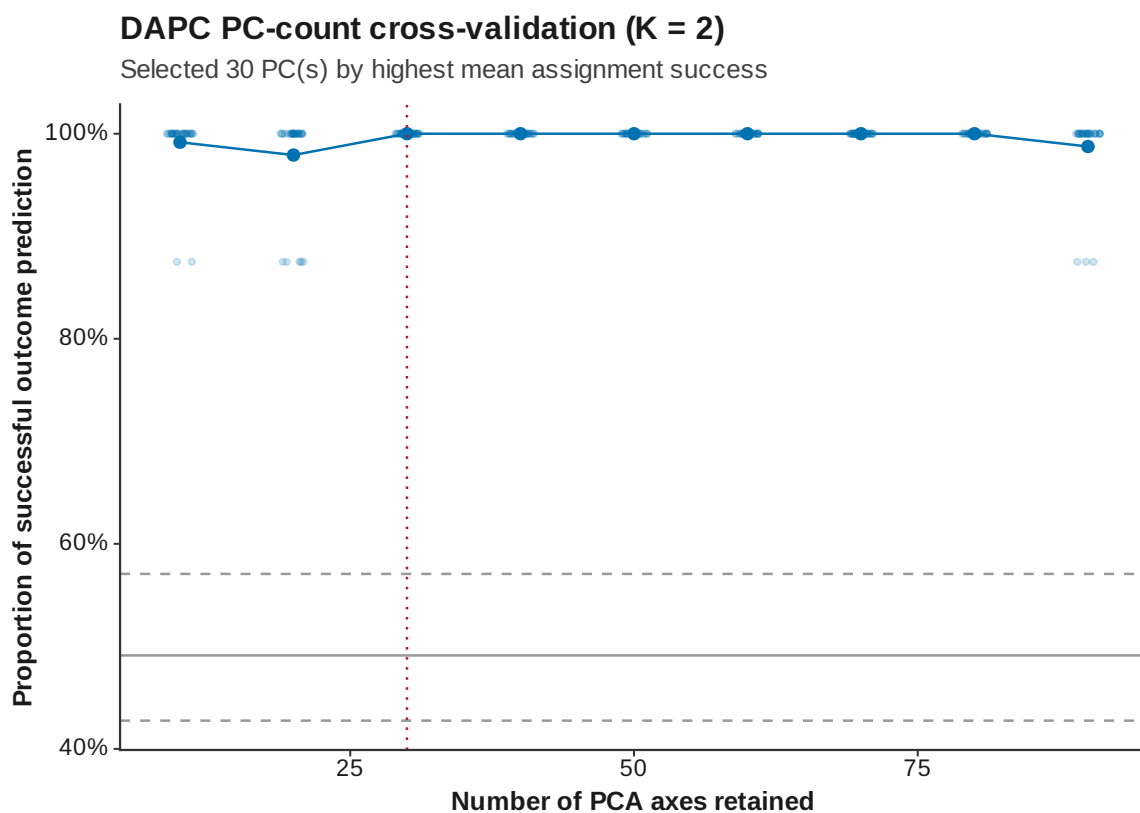


Figure 25: DAPC xval K 2

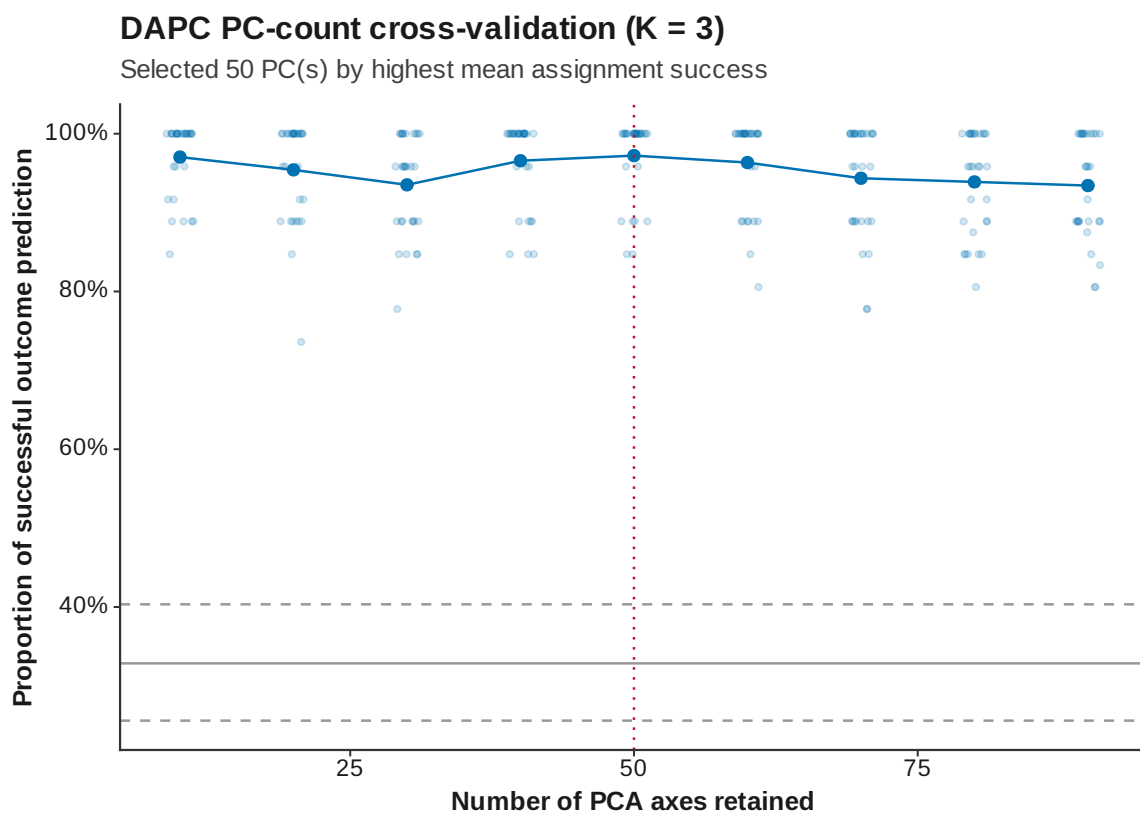


Figure 26: DAPC xval K 3

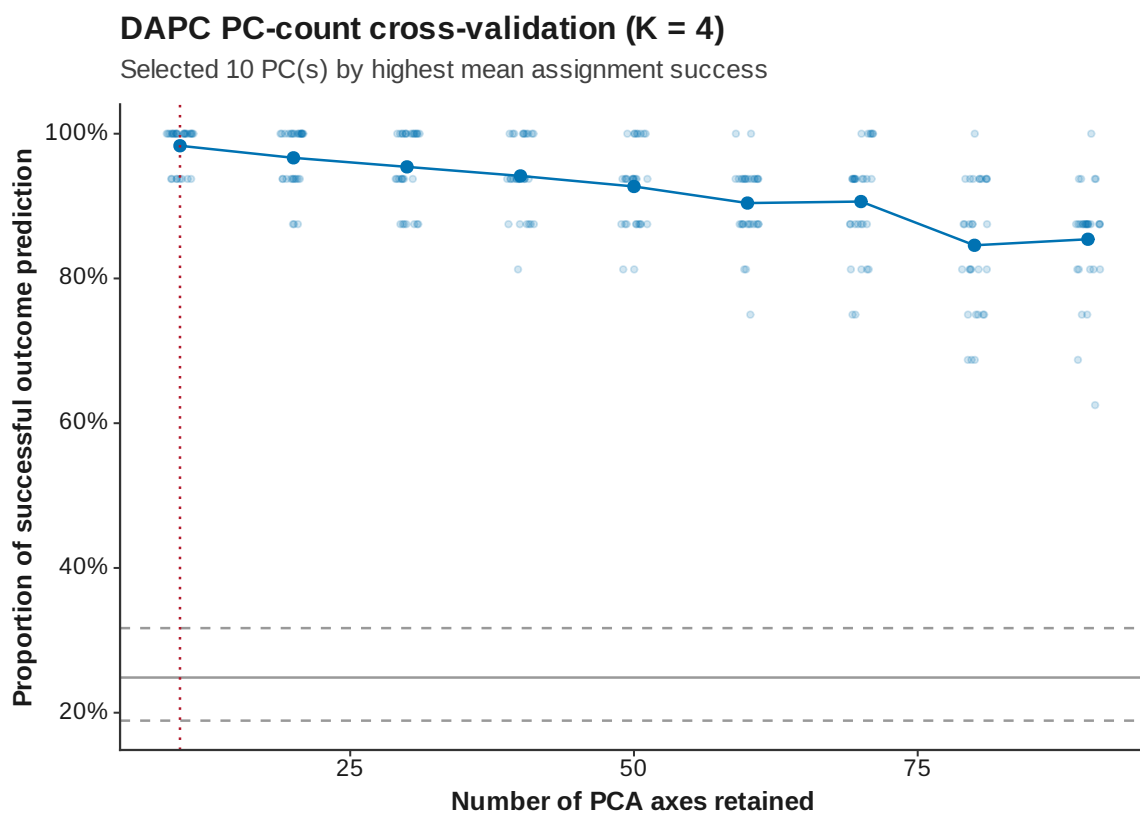


Figure 27: DAPC xval K 4

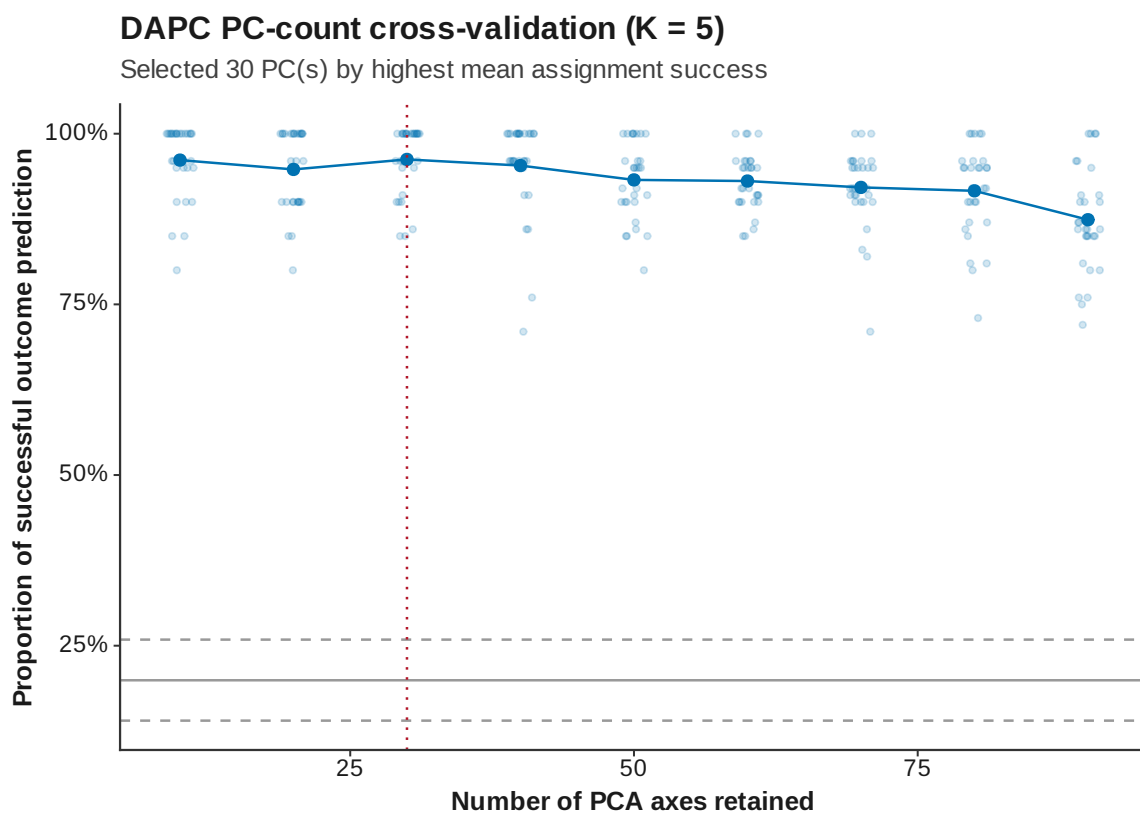


Figure 28: DAPC xval K 5

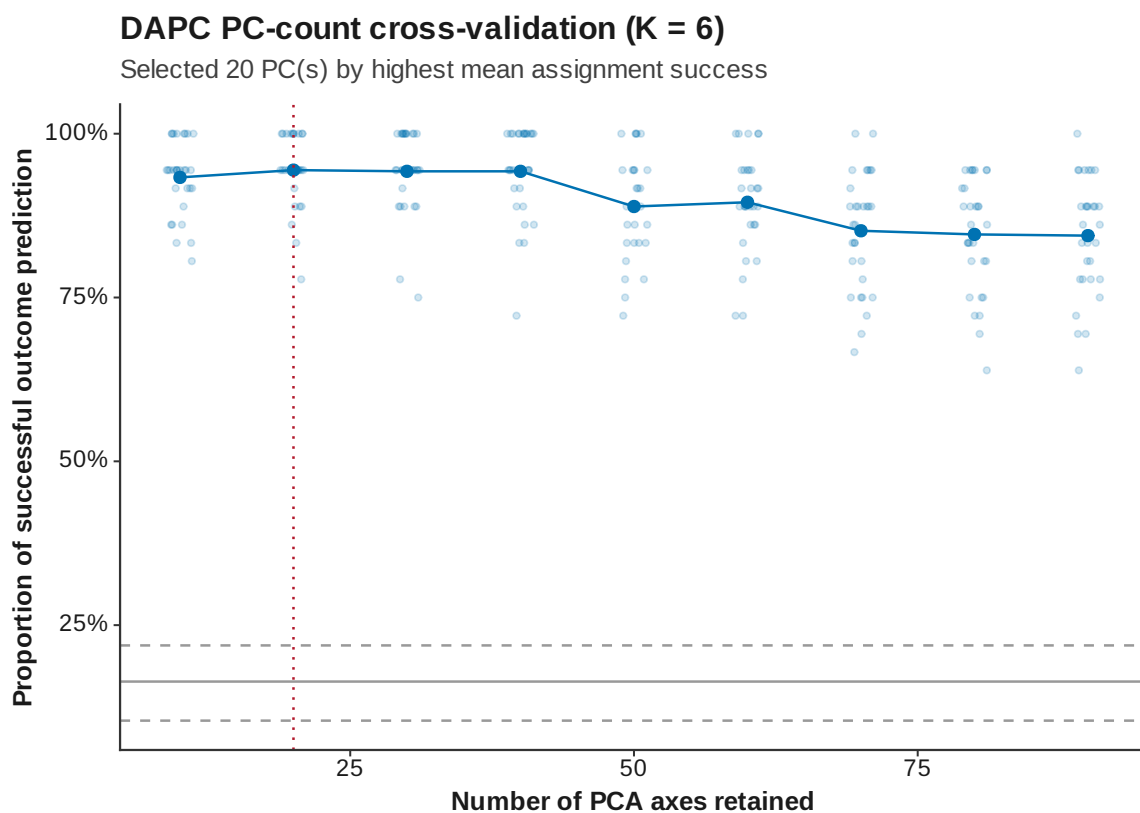


Figure 29: DAPC xval K 6



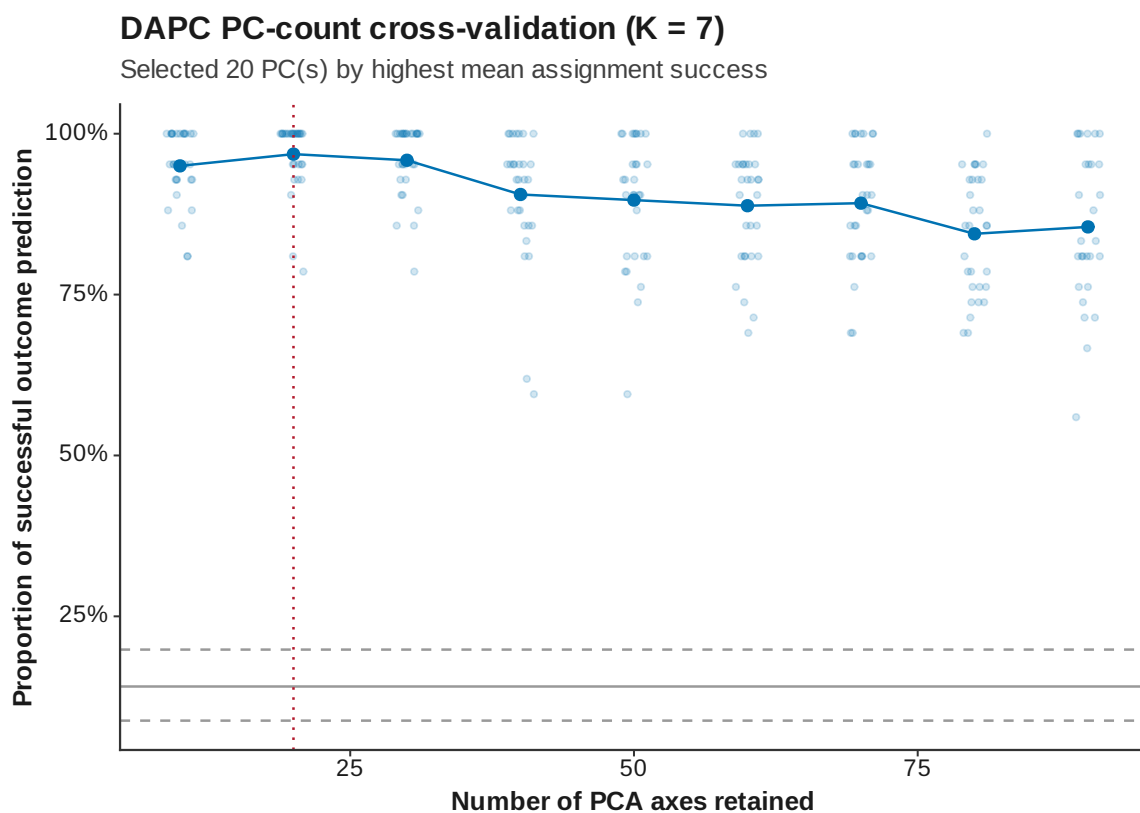


Figure 30: DAPC xval K 7

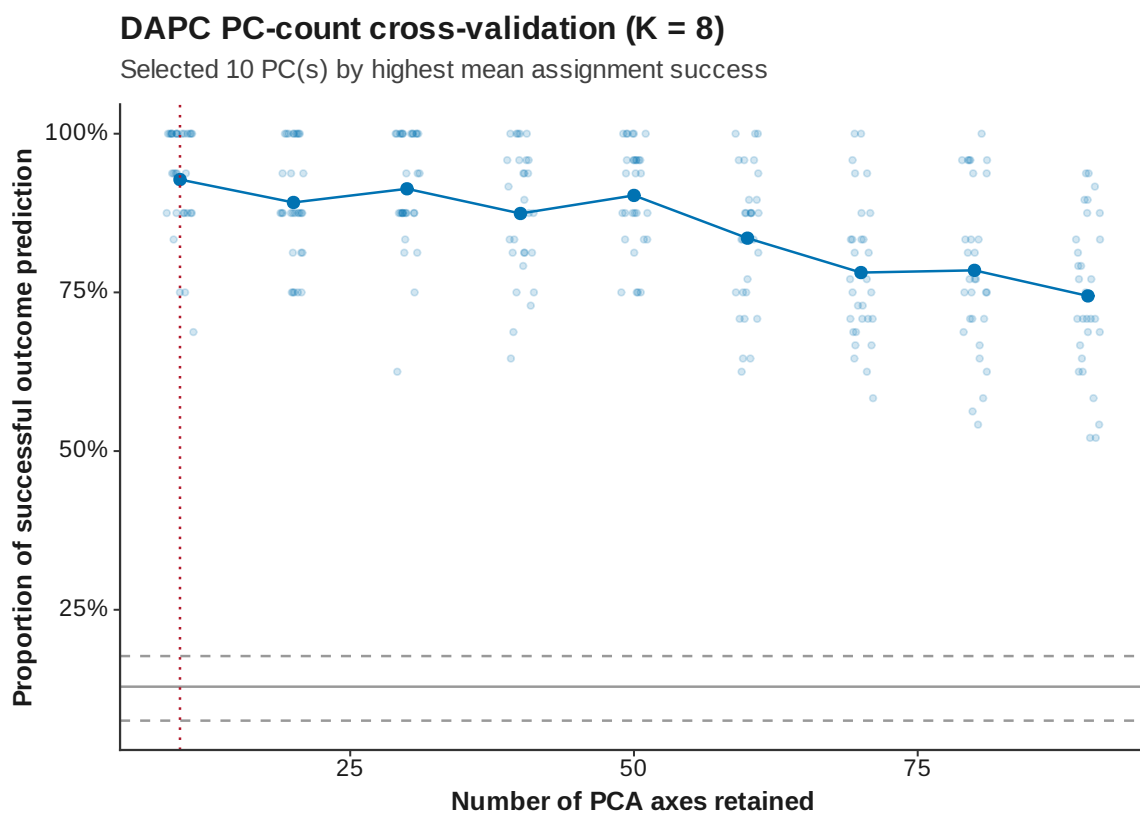


Figure 31: DAPC xval K 8

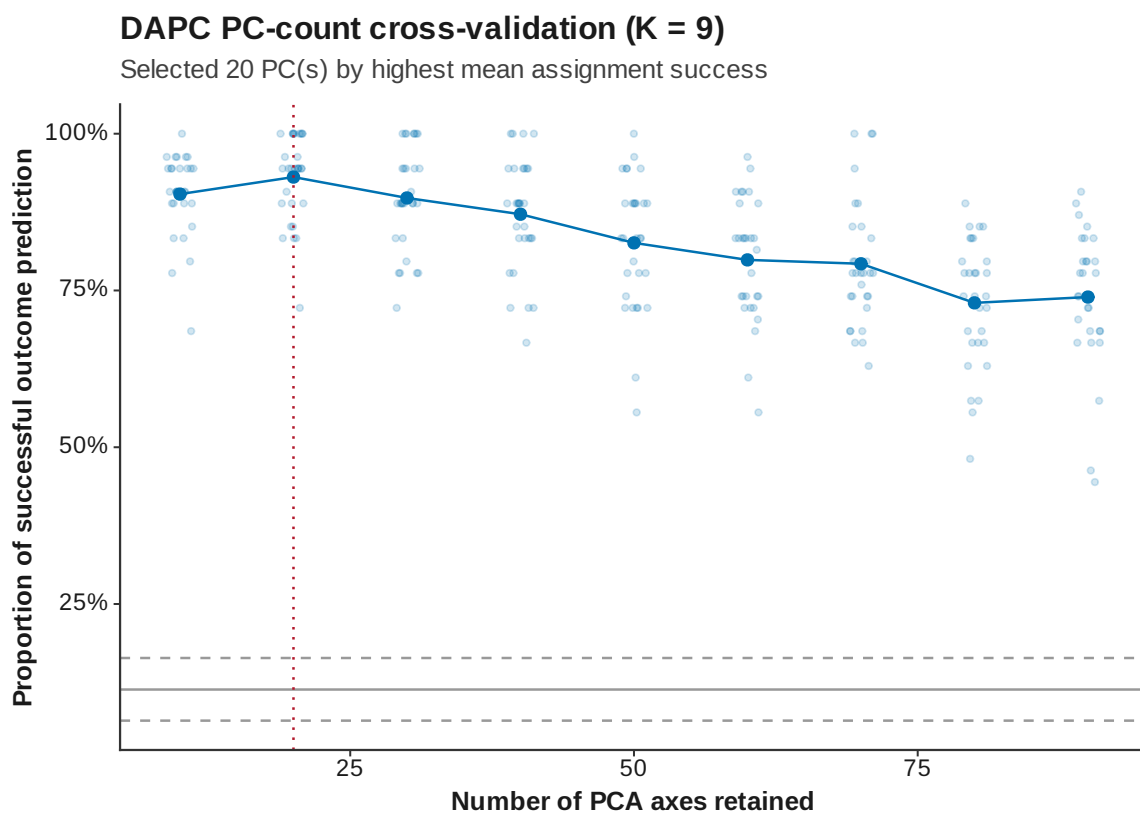


Figure 32: DAPC xval K 9

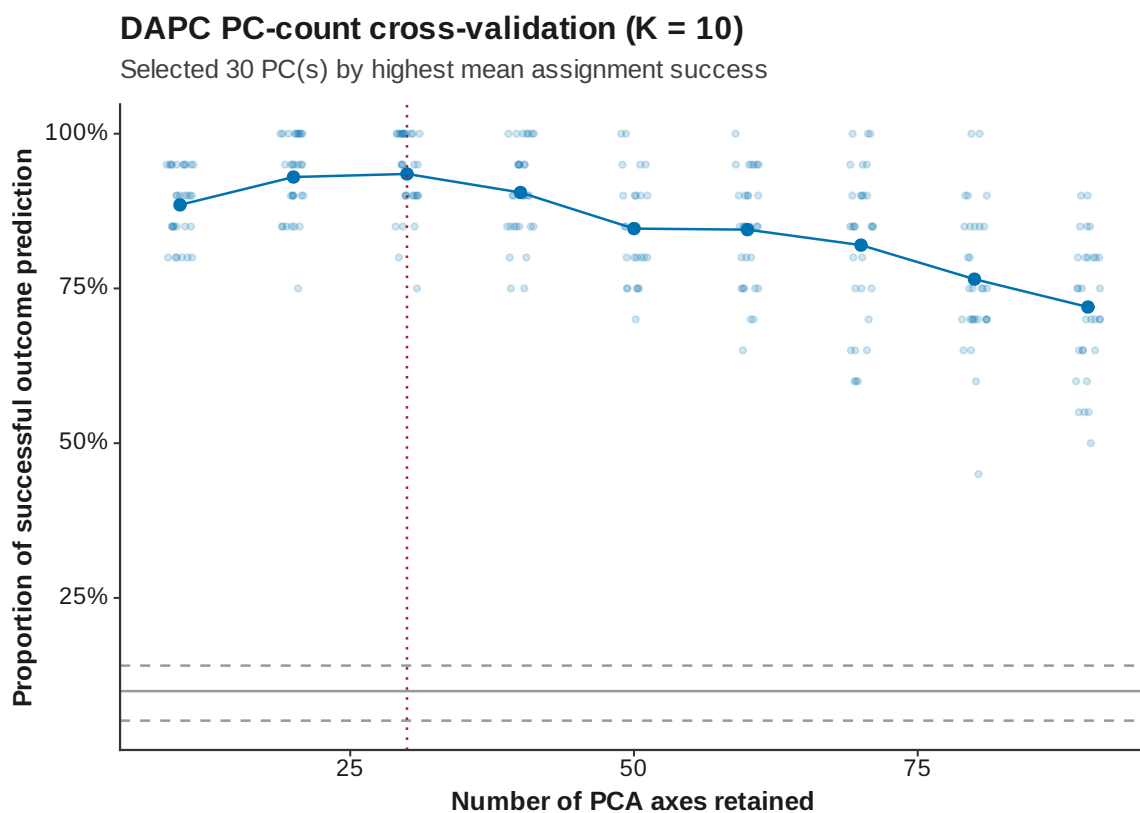


Figure 33: DAPC xval K 10

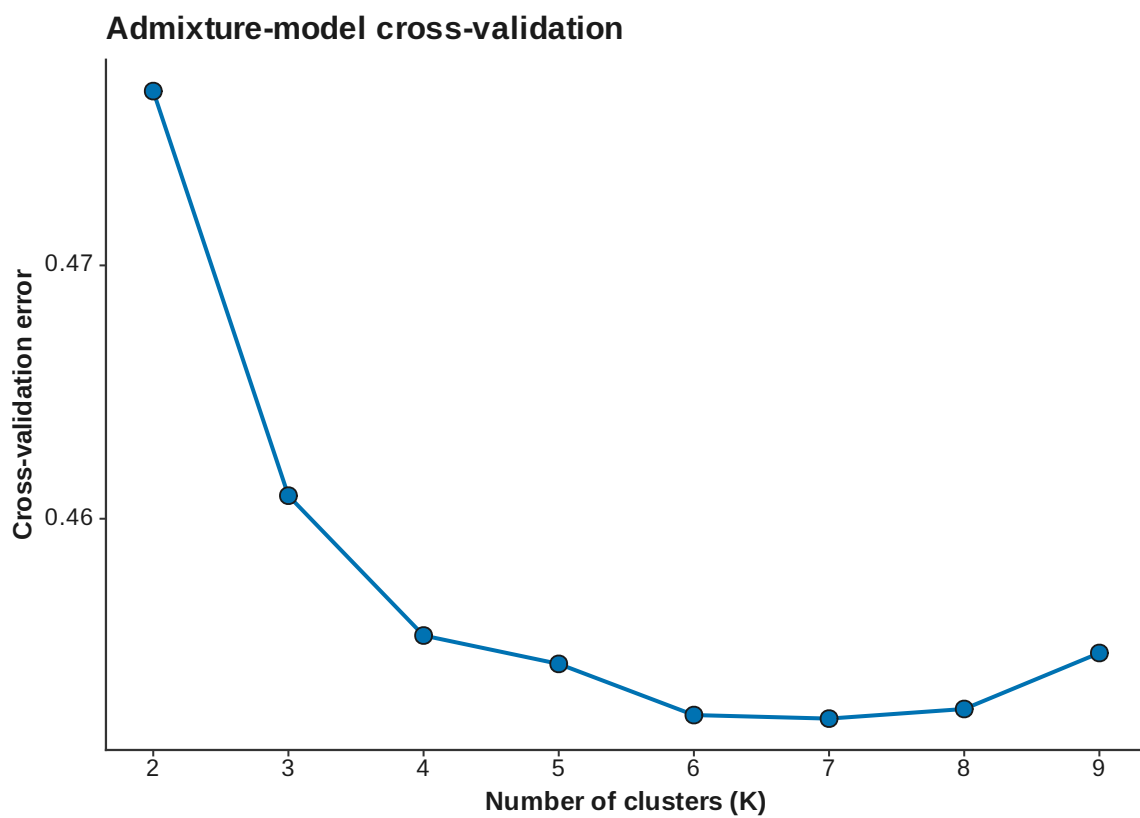


Figure 34: ADMIXTURE CV

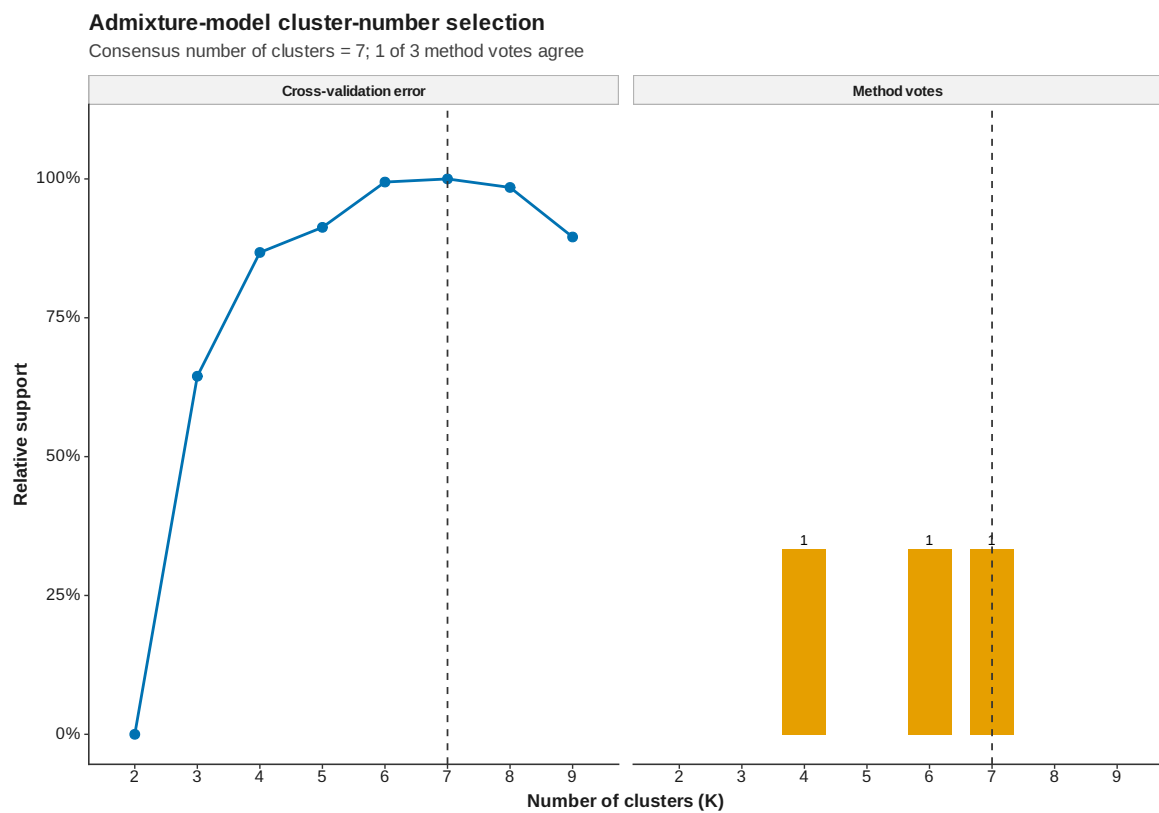


Figure 35: ADMIXTURE cluster number selection

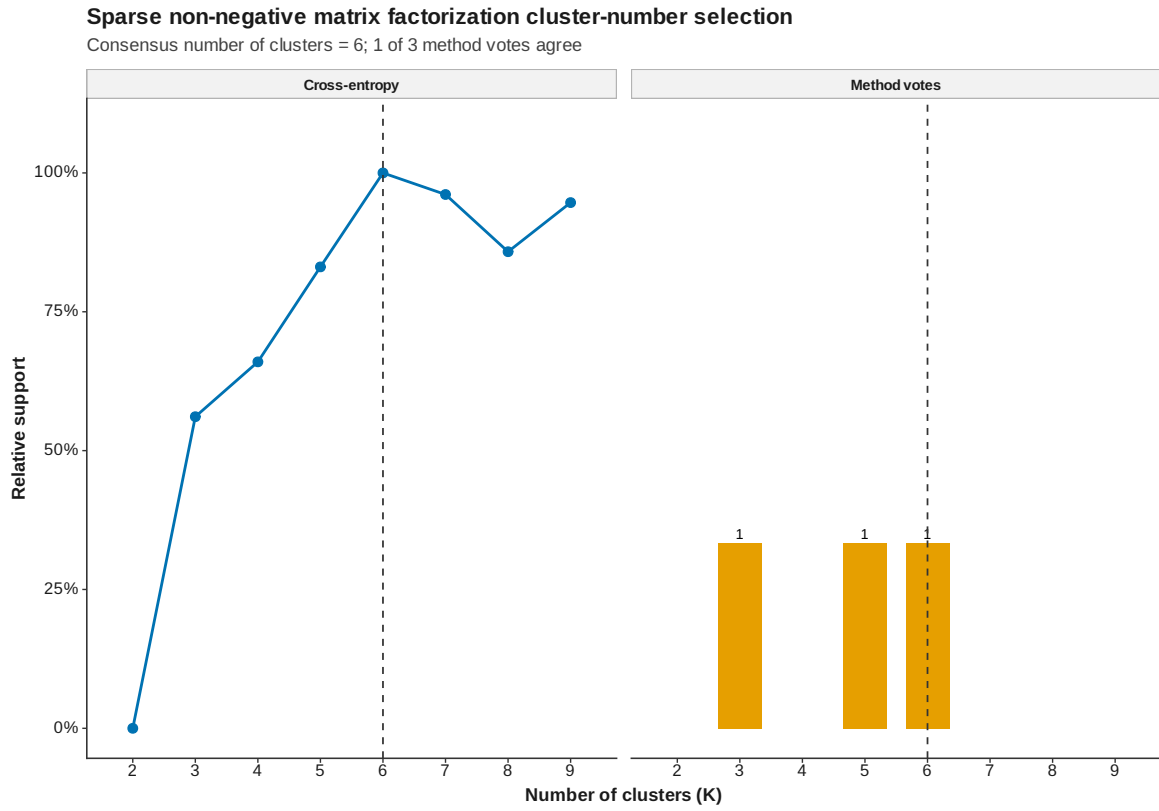


Figure 36: SNMF cluster number selection

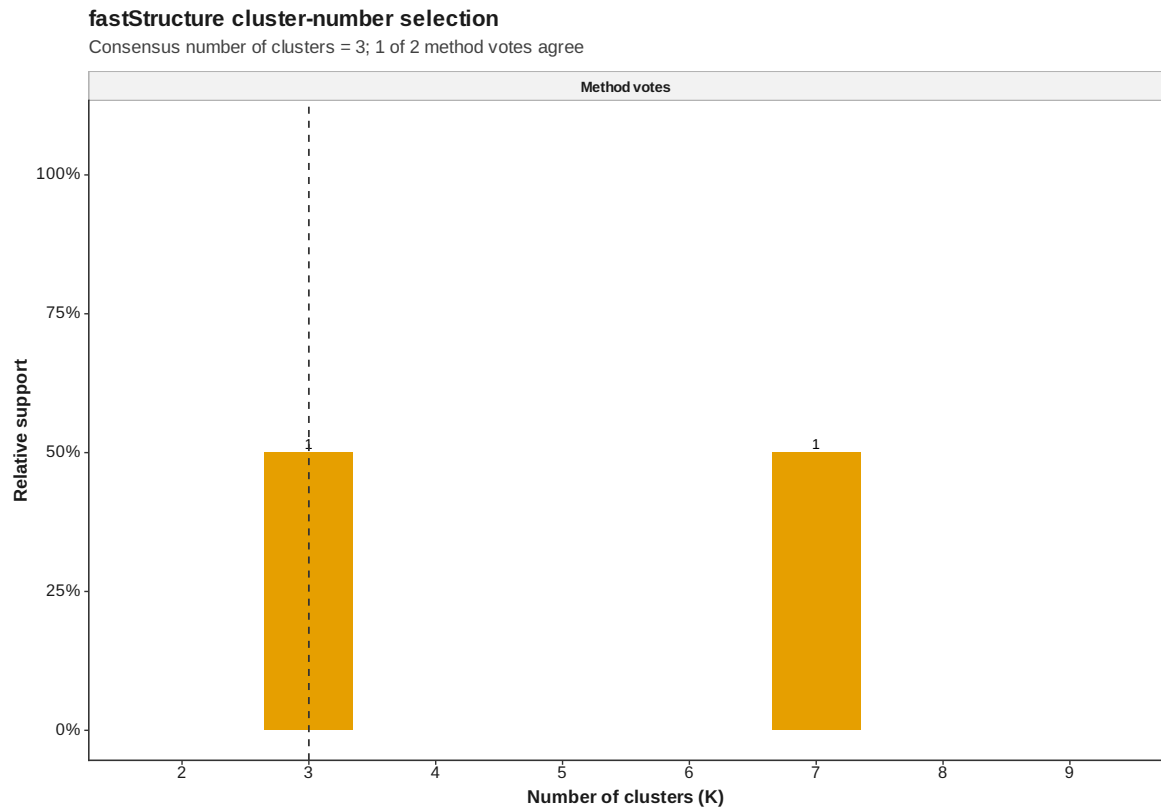


Figure 37: FastStructure cluster number selection





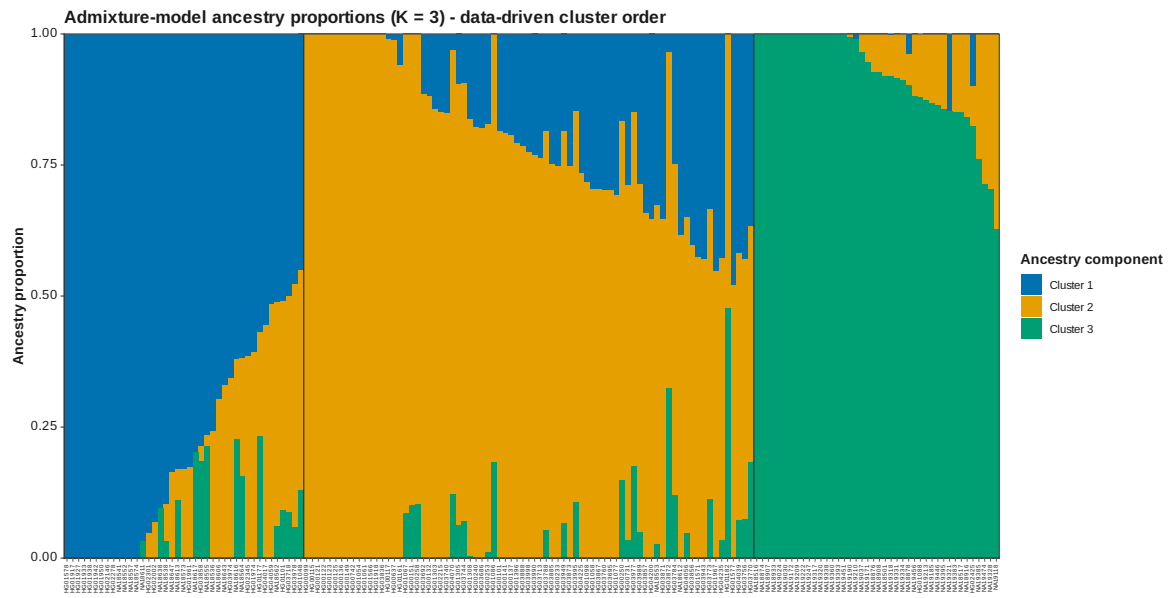


Figure 39: ADMIXTURE Q data driven K 3

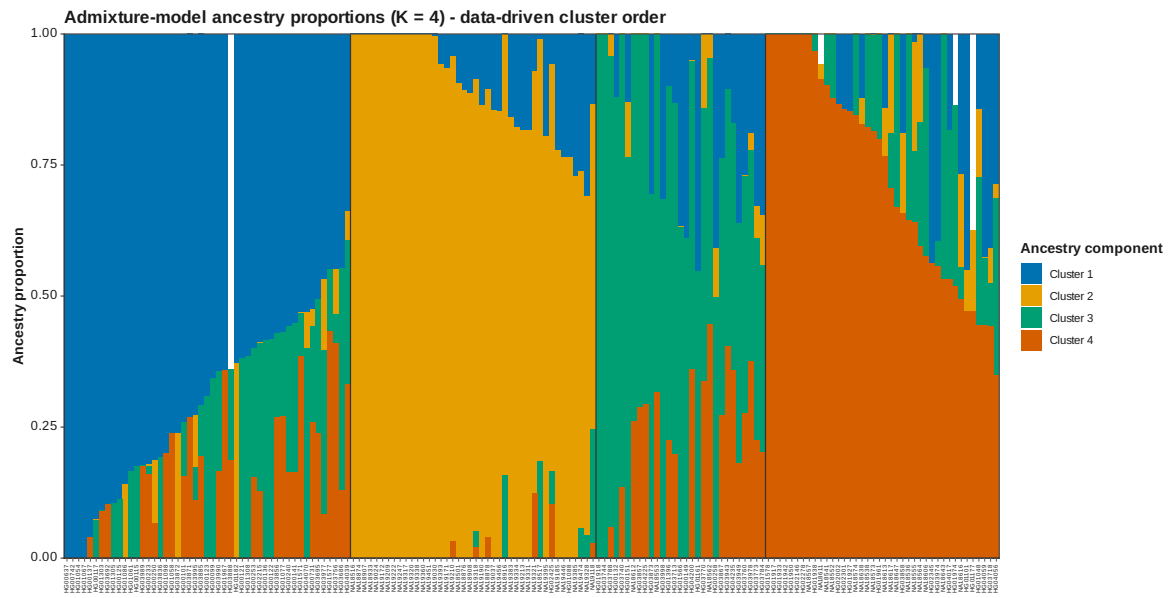


Figure 40: ADMIXTURE Q data driven K 4

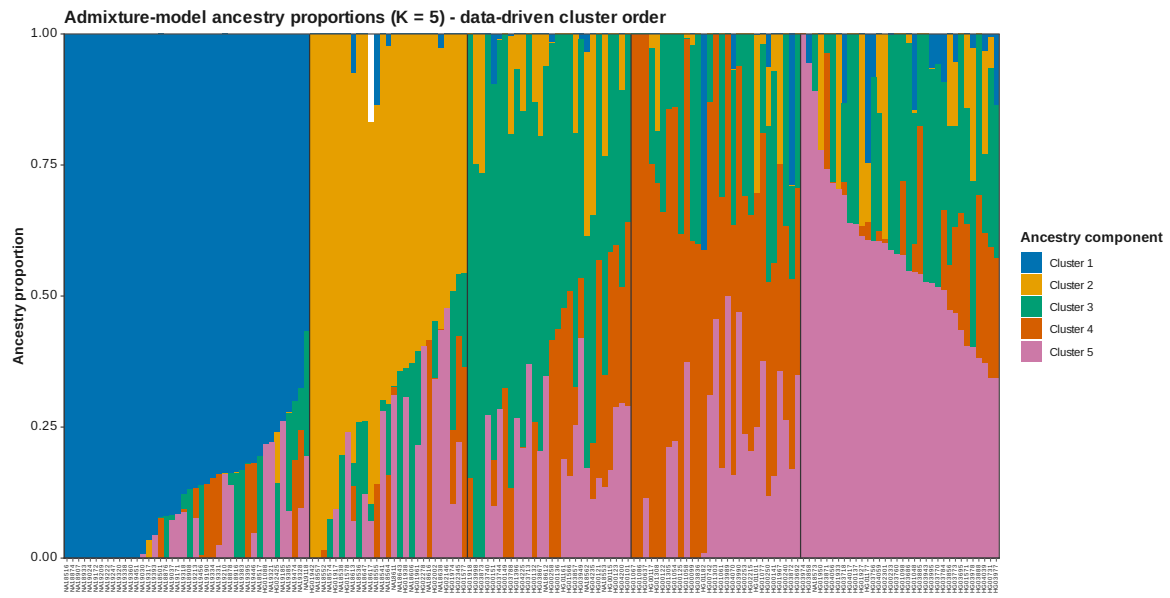


Figure 41: ADMIXTURE Q data driven K 5

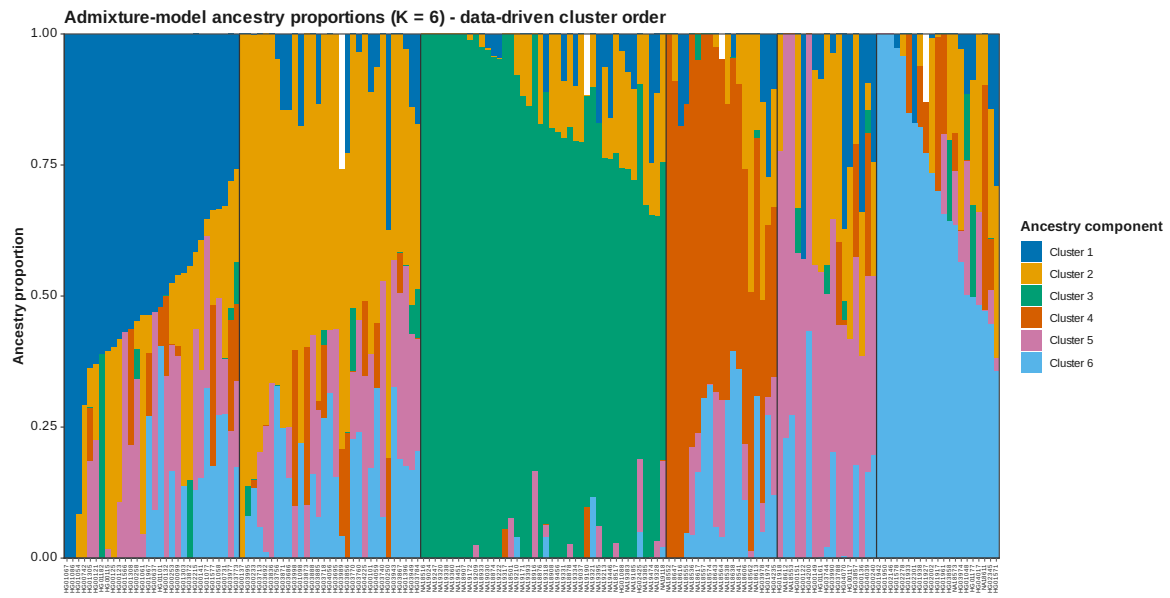


Figure 42: ADMIXTURE Q data driven K 6

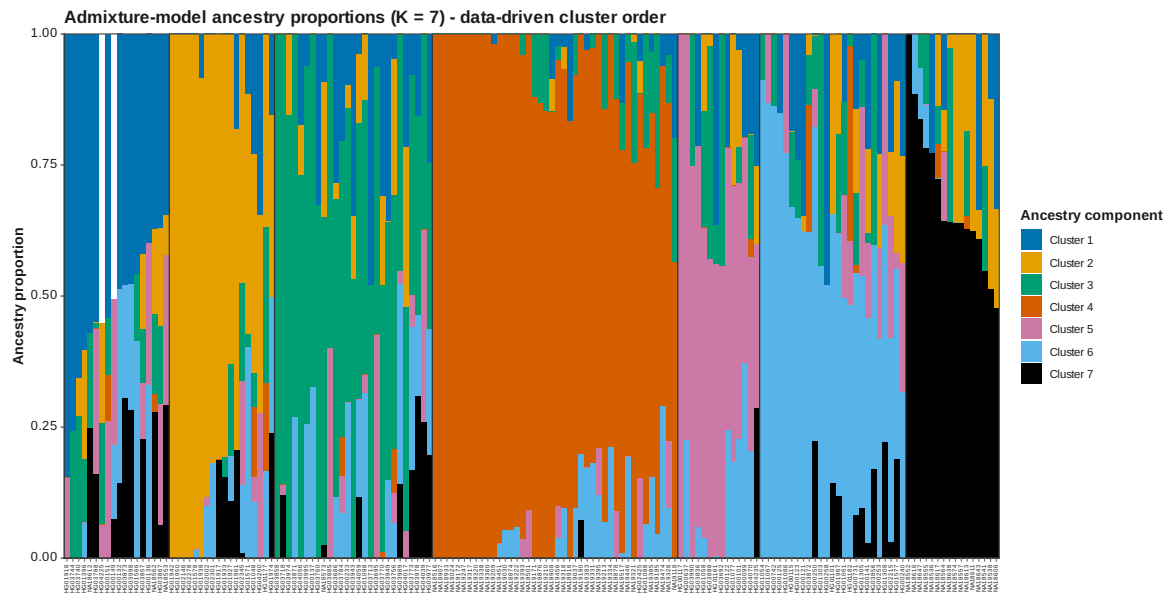


Figure 43: ADMIXTURE Q data driven K 7

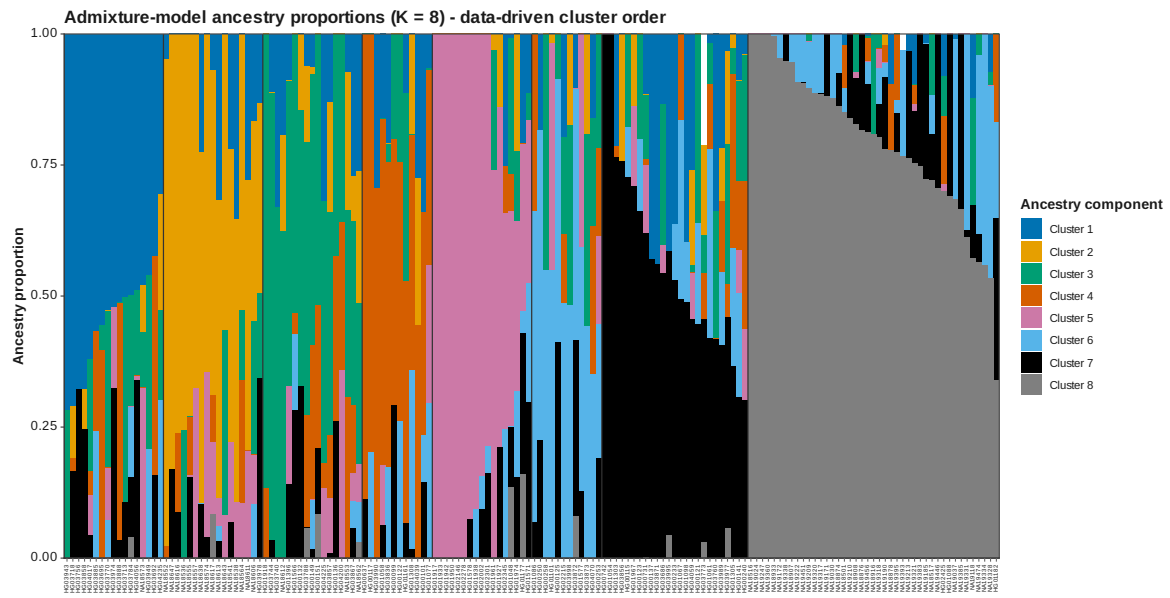


Figure 44: ADMIXTURE Q data driven K 8

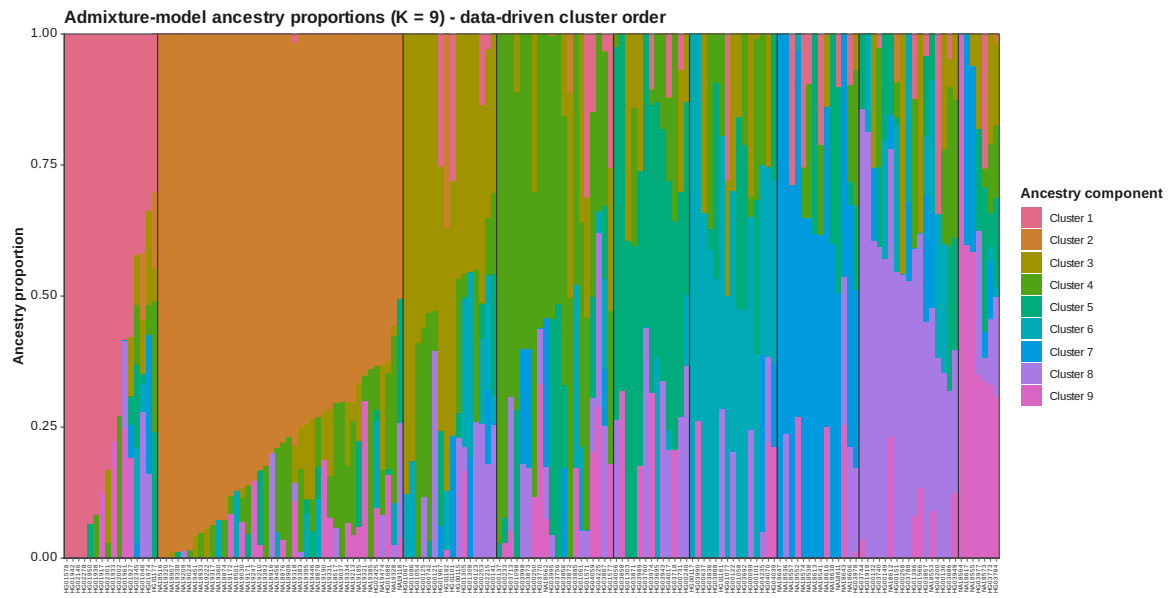


Figure 45: ADMIXTURE Q data driven K 9



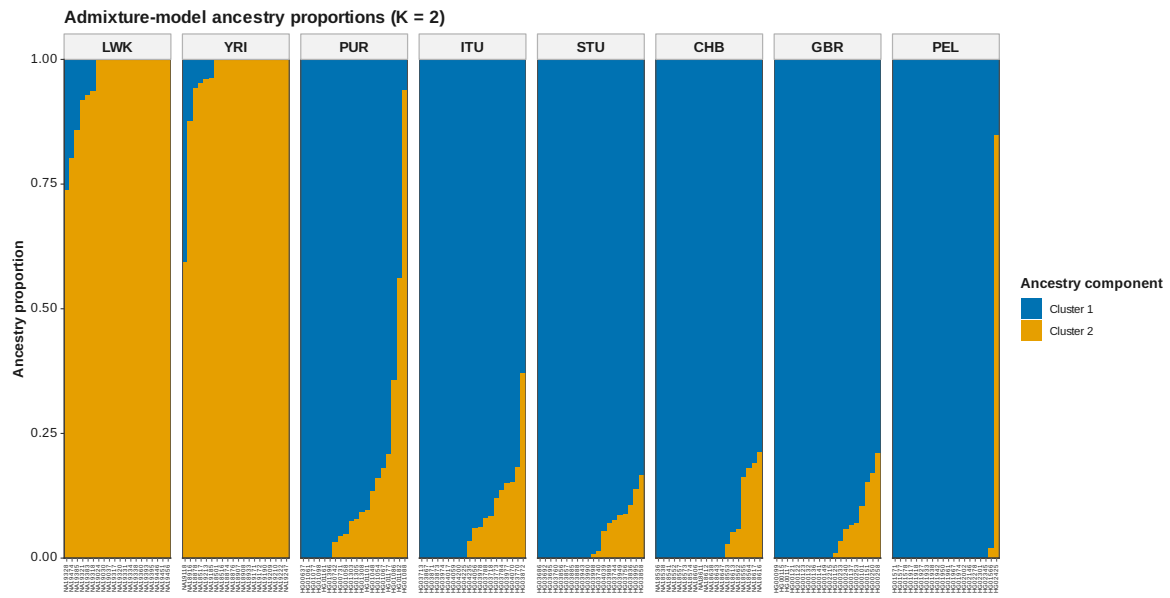


Figure 46: ADMIXTURE Q K 2

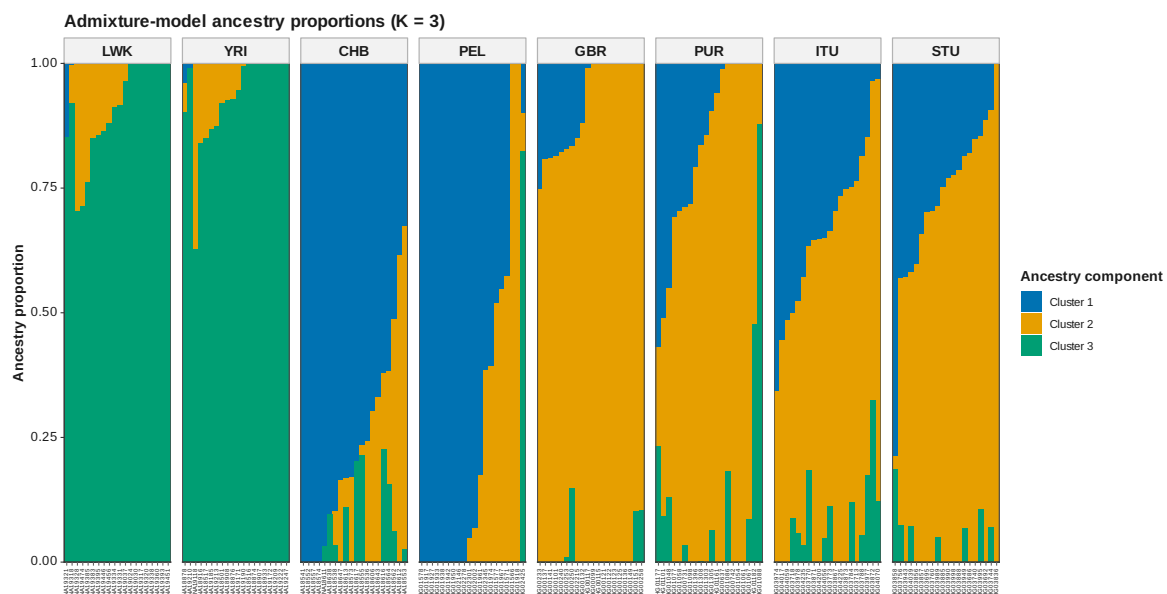


Figure 47: ADMIXTURE Q K 3

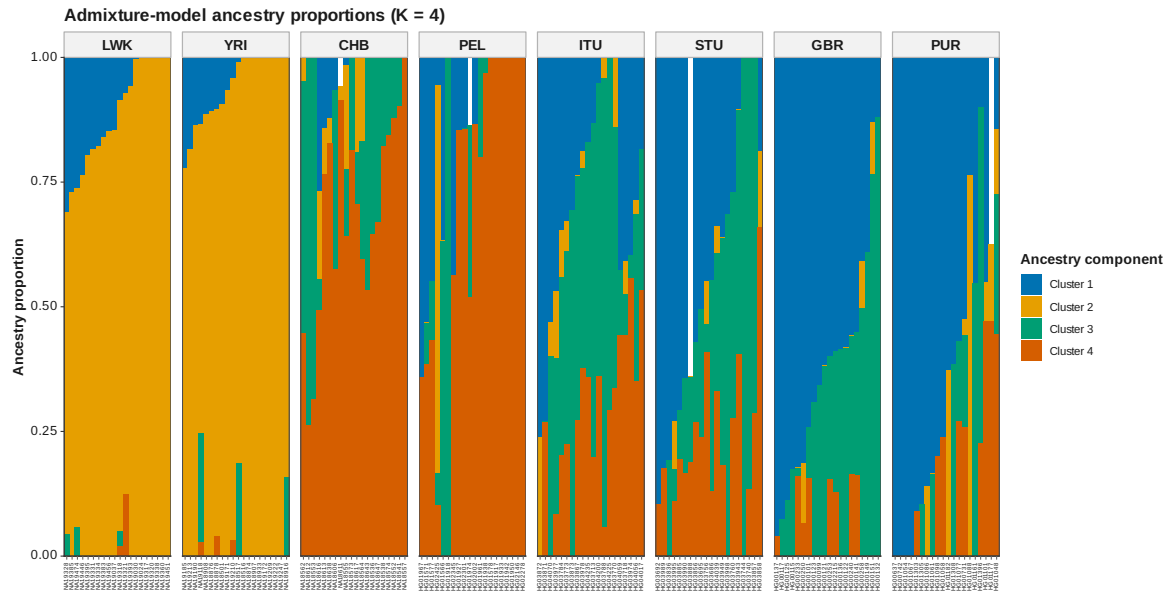


Figure 48: ADMIXTURE Q K 4

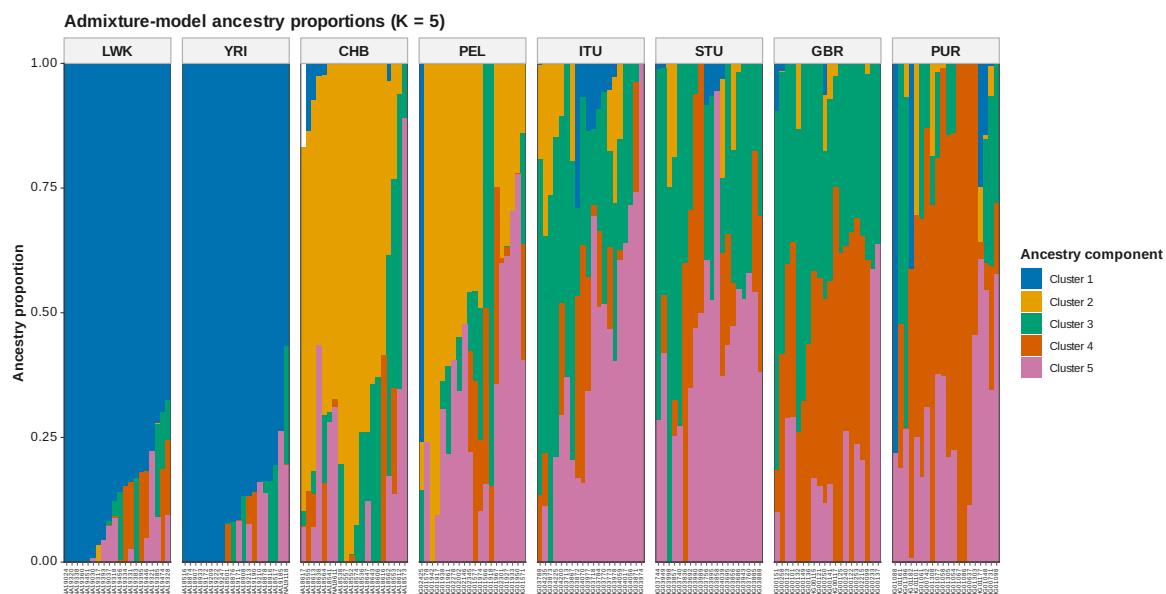


Figure 49: ADMIXTURE Q K 5

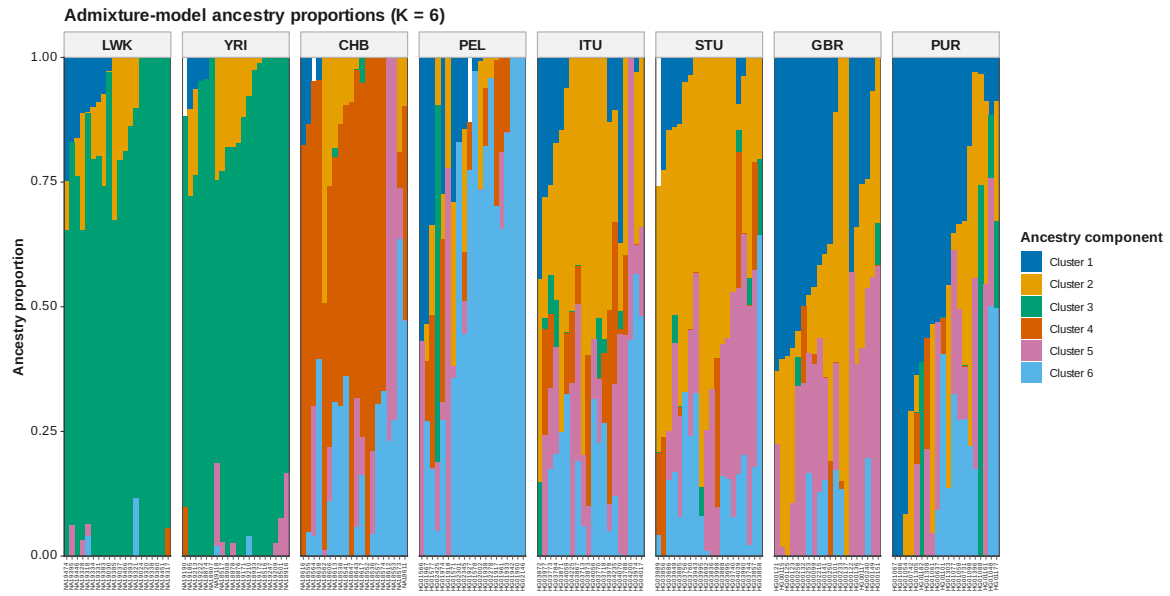


Figure 50: ADMIXTURE Q K 6

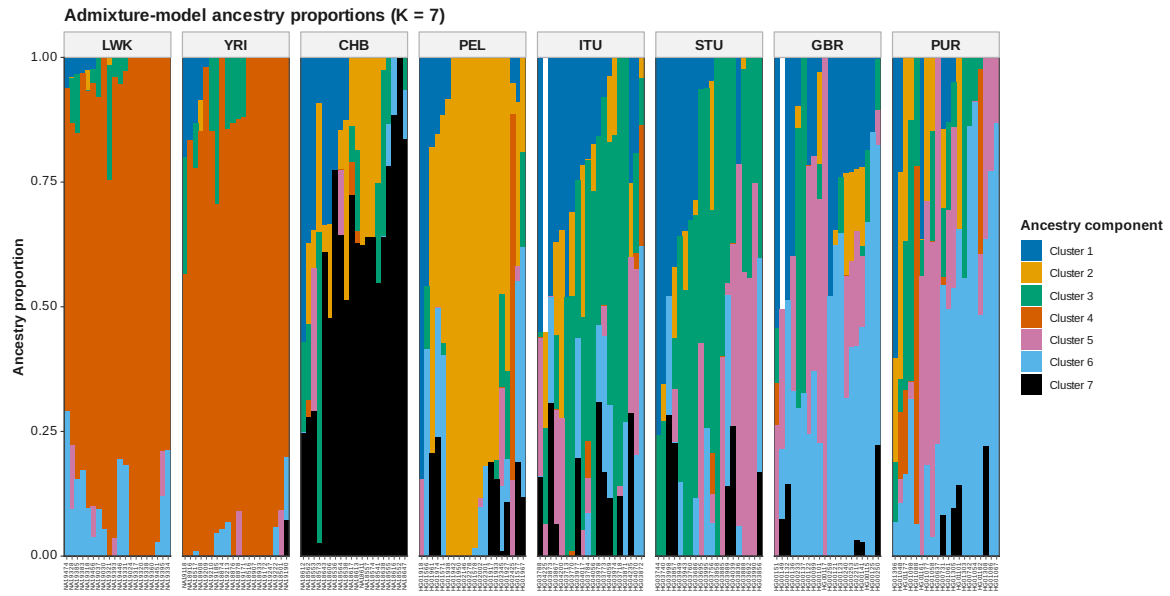


Figure 51: ADMIXTURE Q K 7

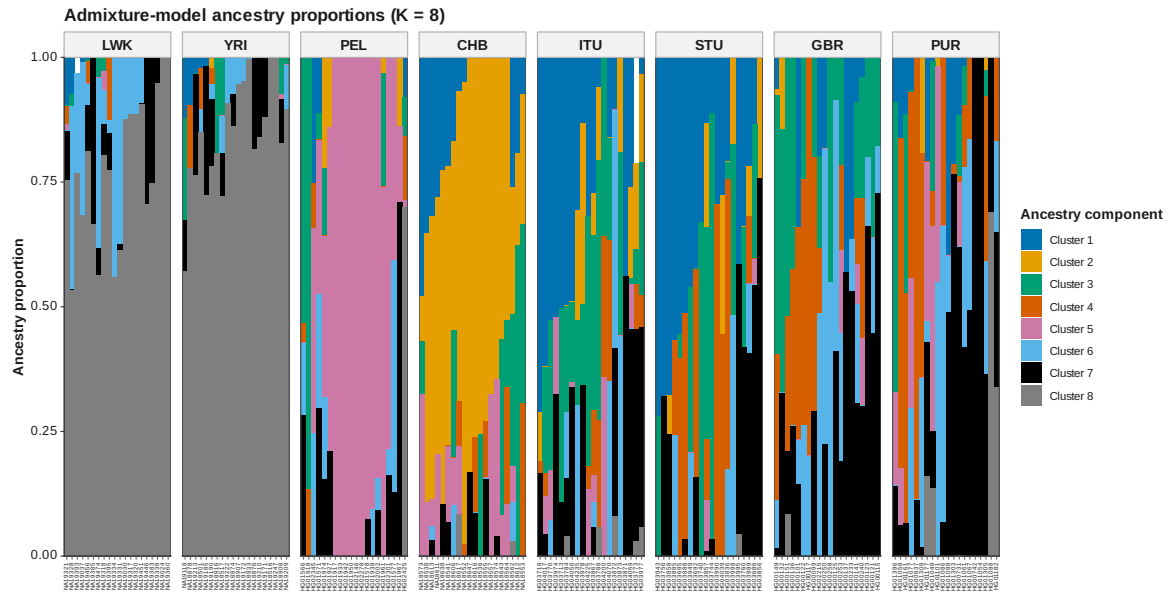


Figure 52: ADMIXTURE Q K 8

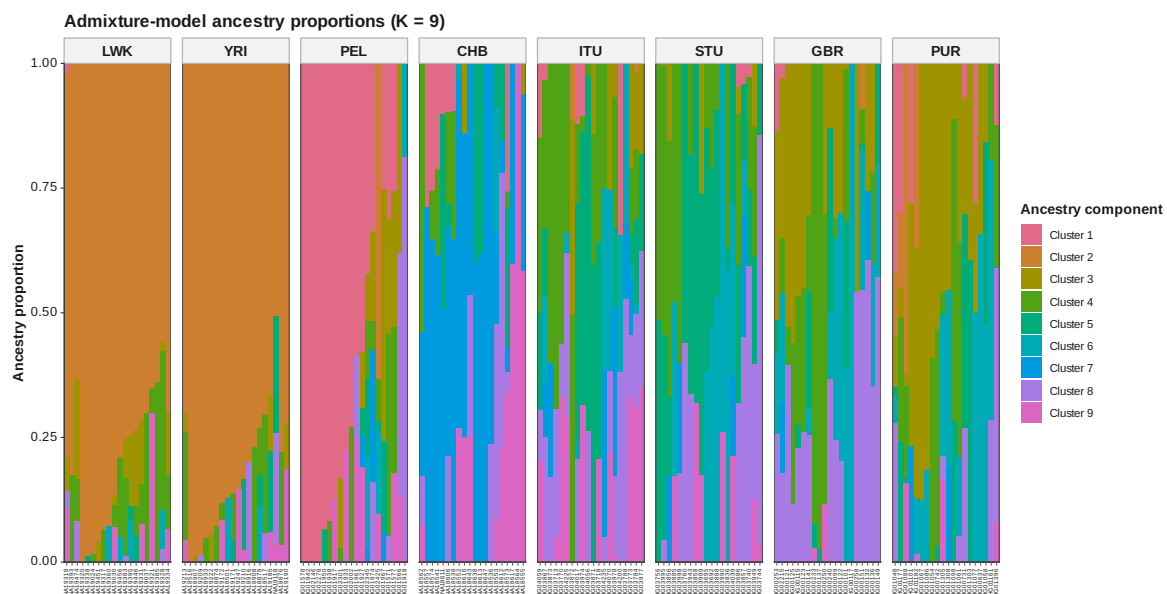


Figure 53: ADMIXTURE Q K 9



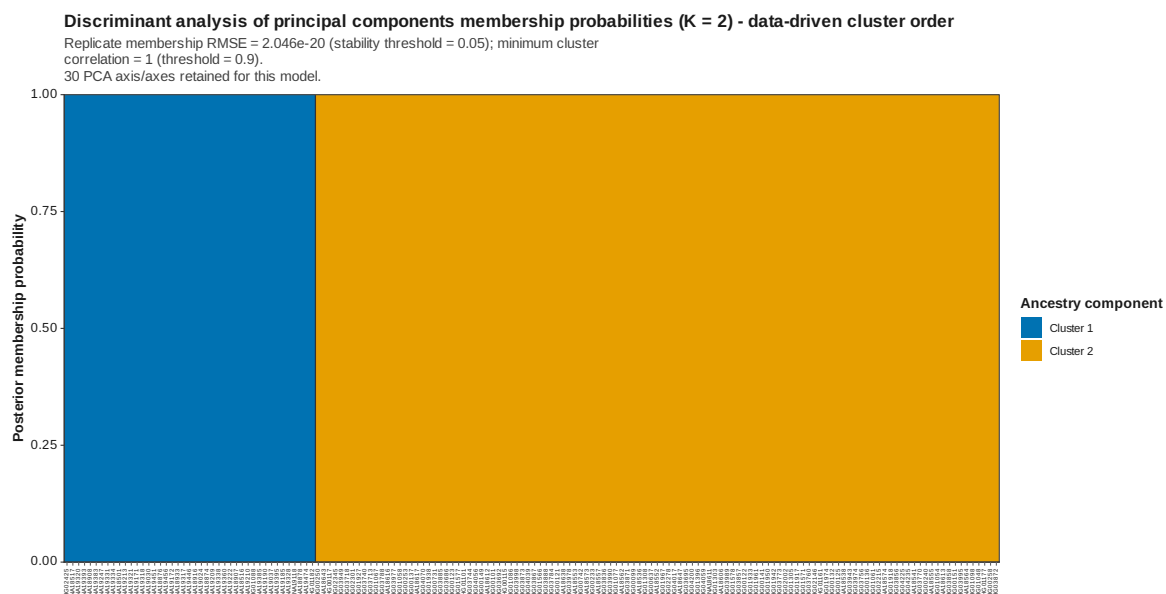


Figure 54: DAPC membership data driven K 2

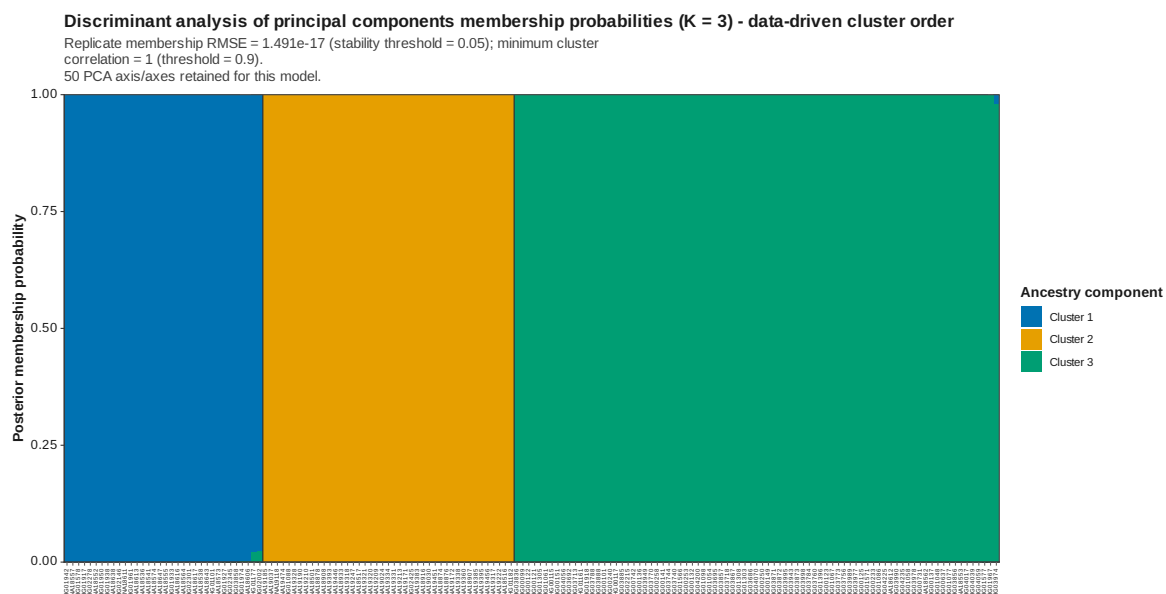


Figure 55: DAPC membership data driven K 3

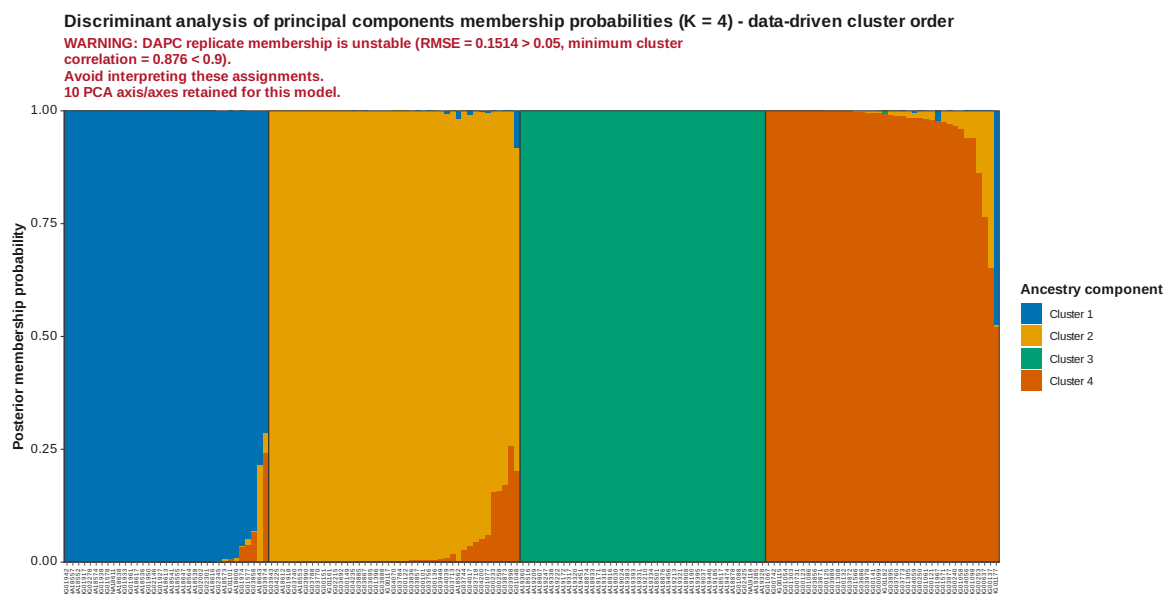


Figure 56: DAPC membership data driven K 4

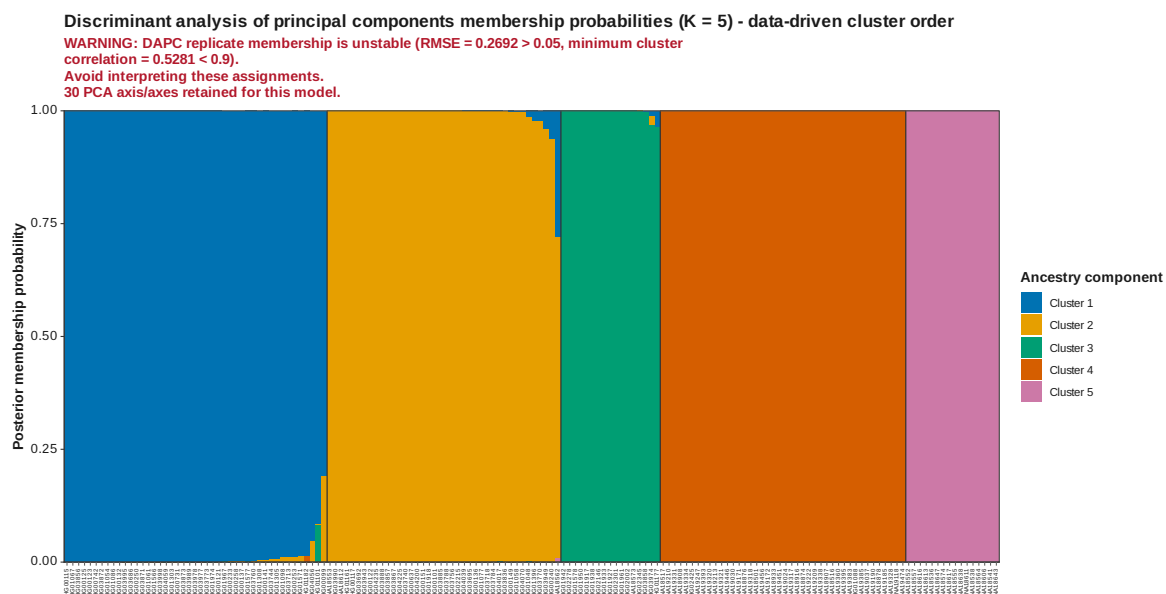


Figure 57: DAPC membership data driven K 5

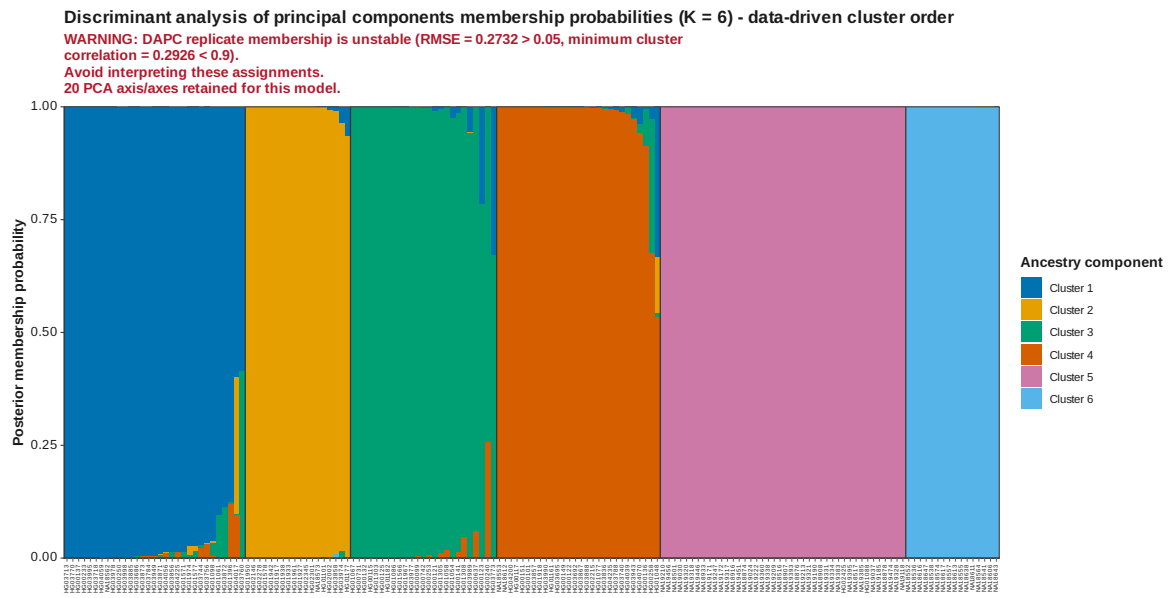


Figure 58: DAPC membership data driven K 6

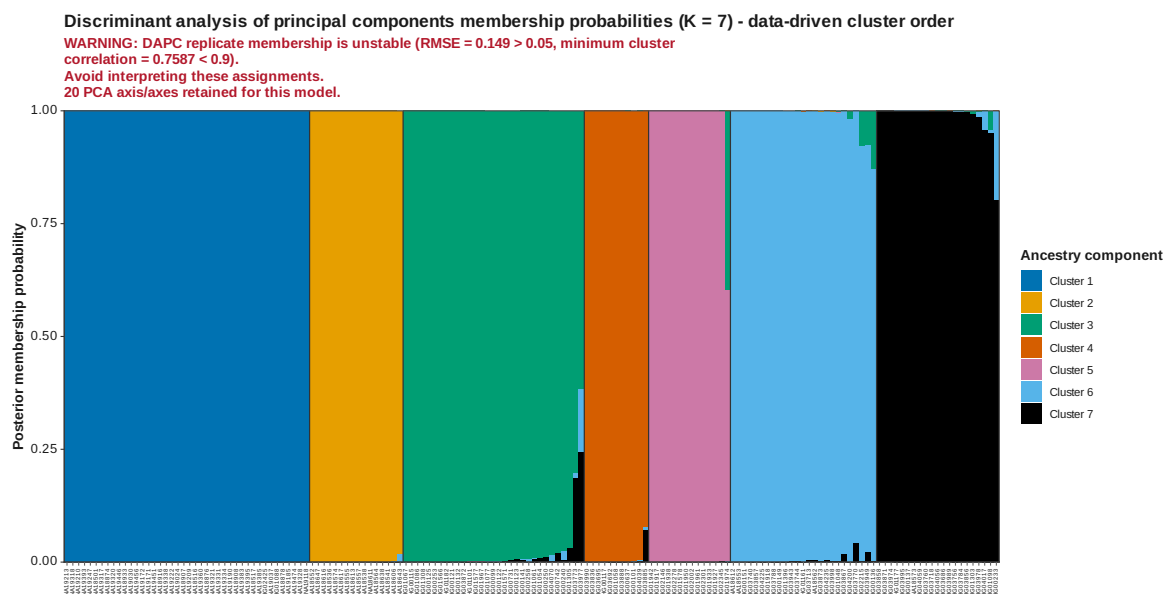


Figure 59: DAPC membership data driven K 7

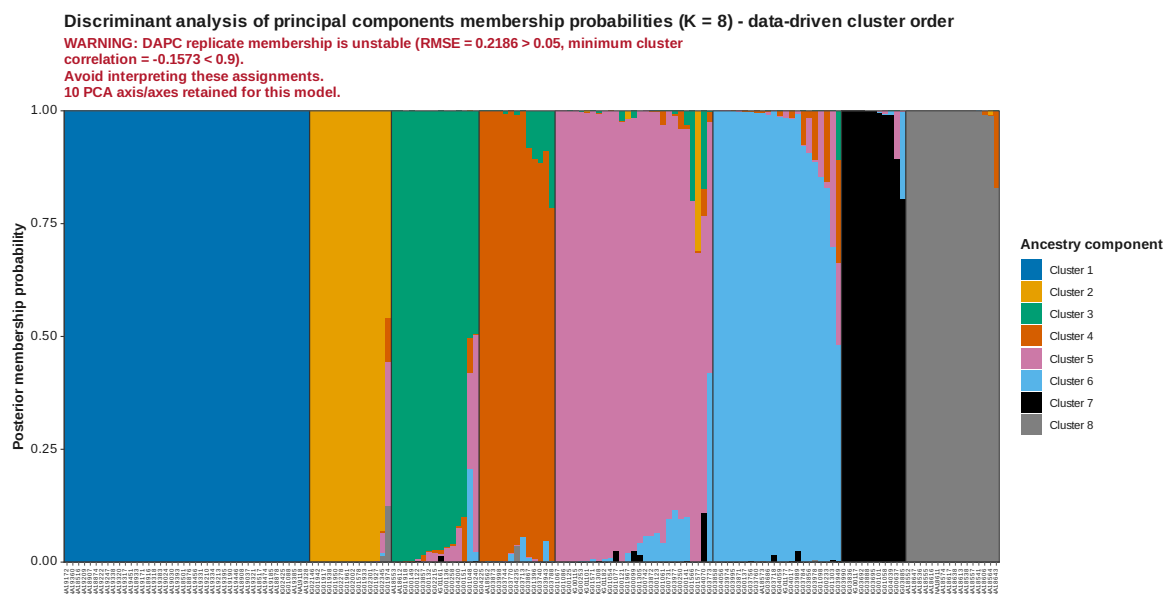


Figure 60: DAPC membership data driven K 8

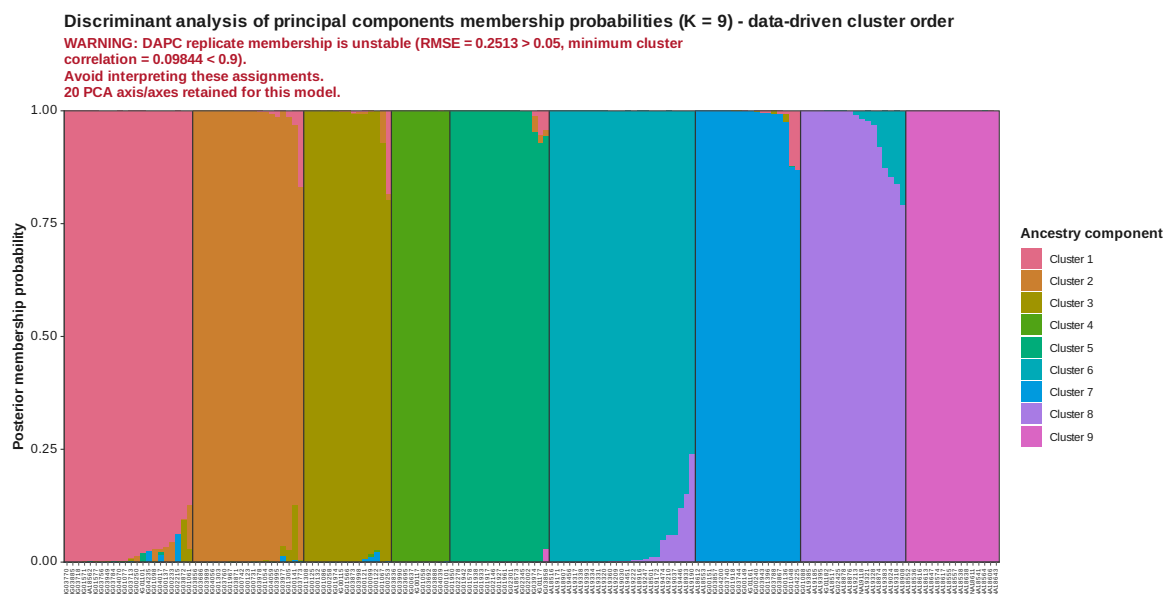


Figure 61: DAPC membership data driven K 9



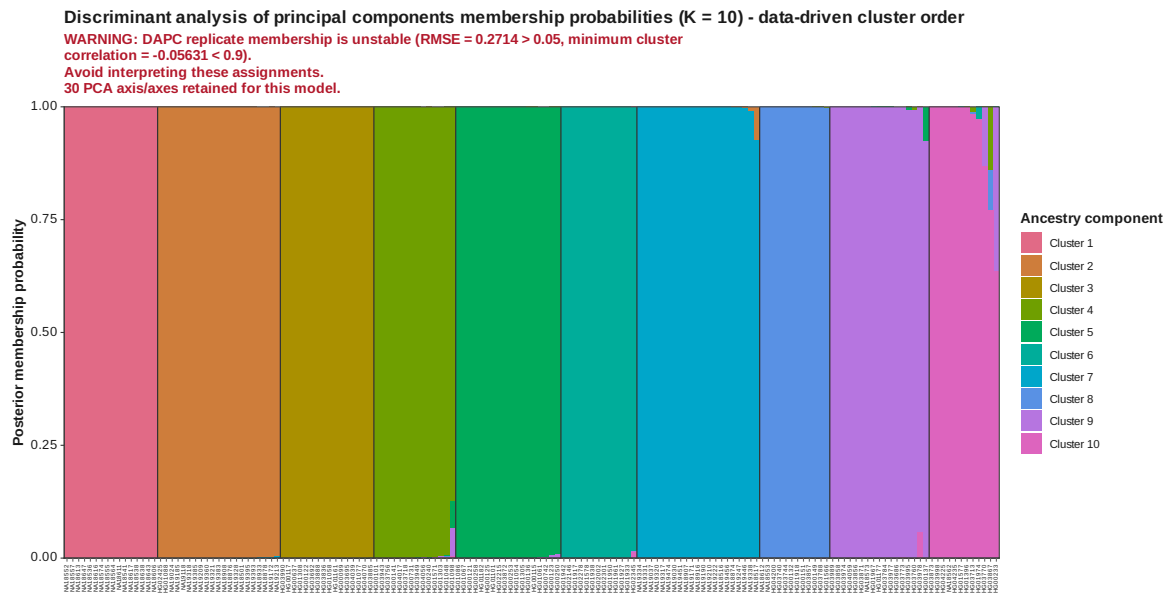


Figure 62: DAPC membership data driven K 10

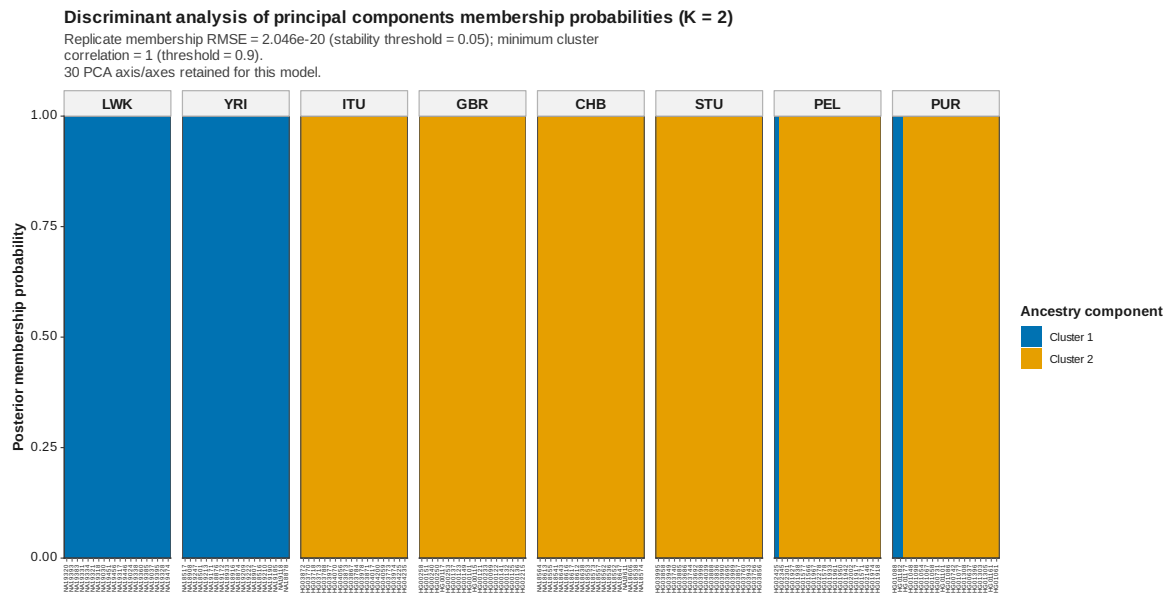


Figure 63: DAPC membership K 2

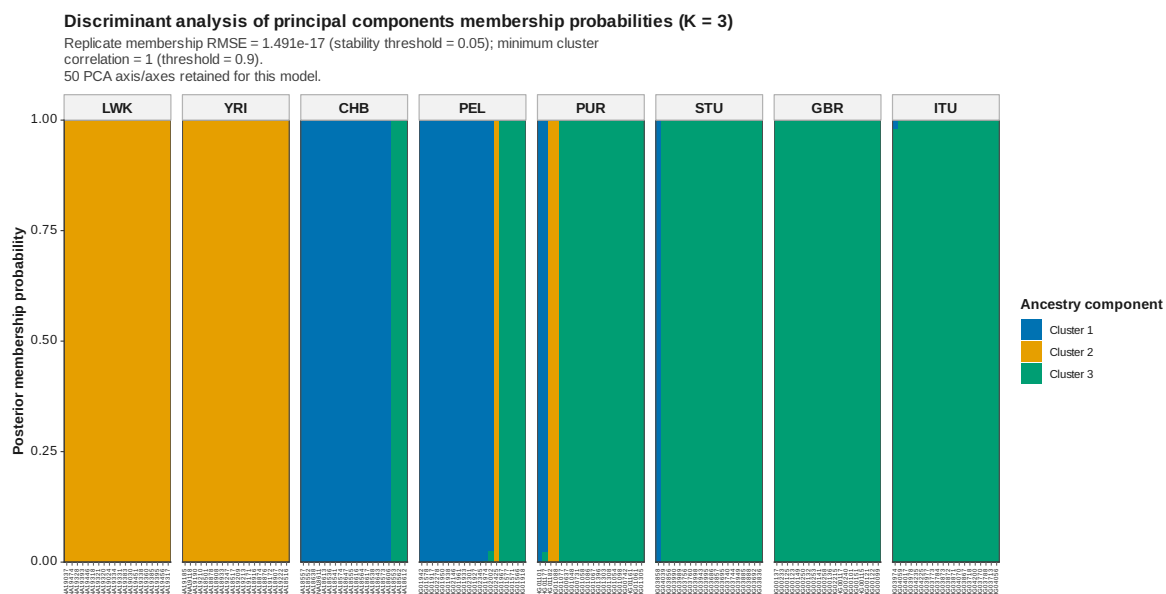


Figure 64: DAPC membership K 3

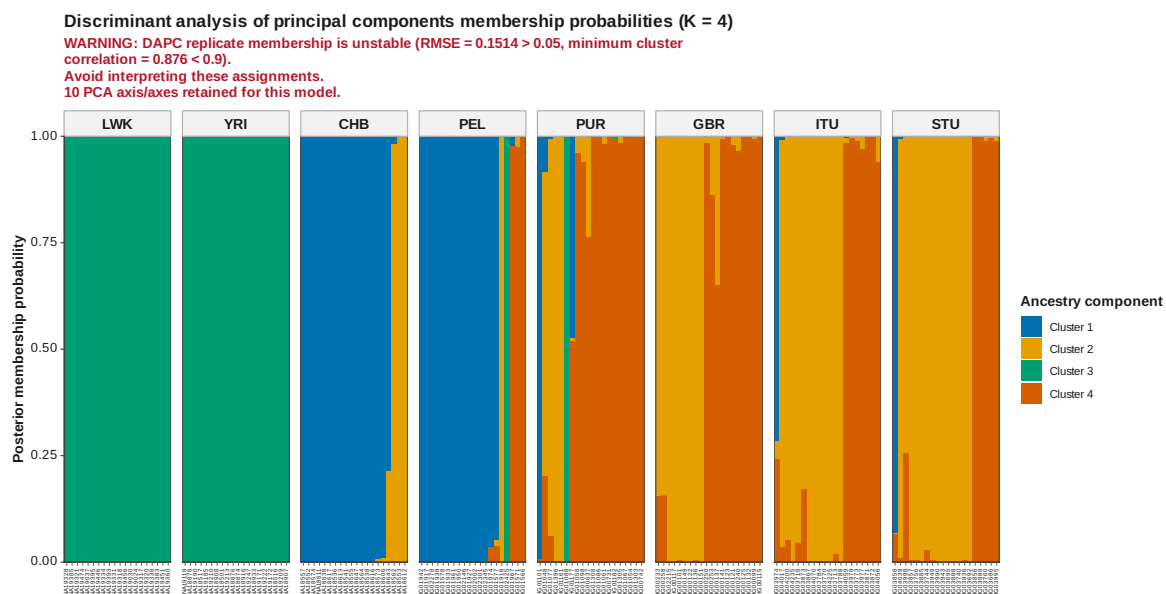


Figure 65: DAPC membership K 4

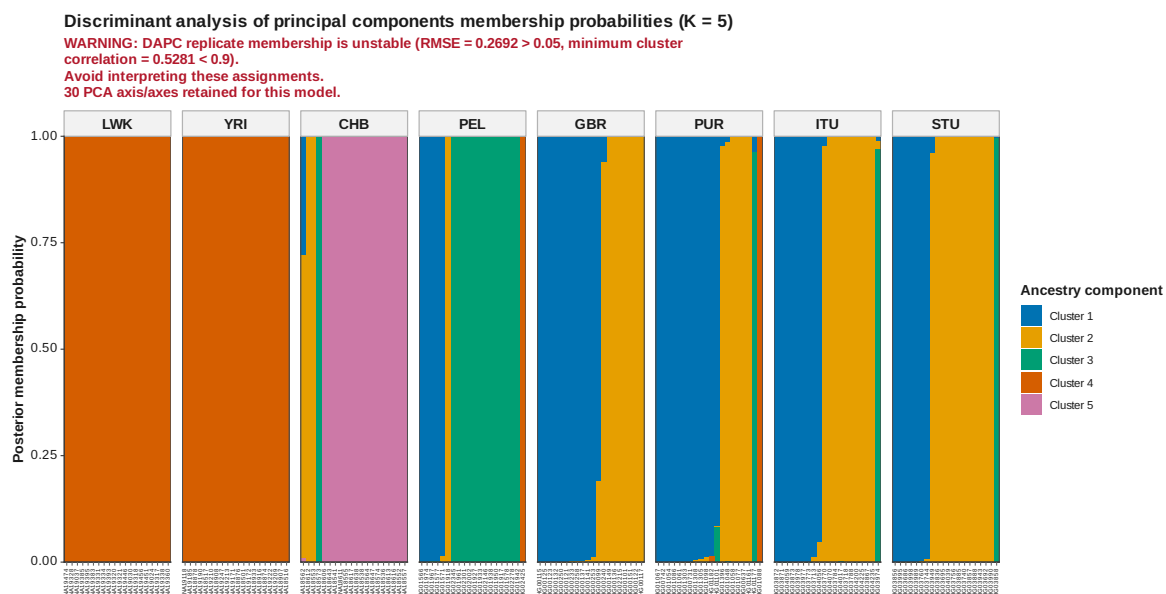


Figure 66: DAPC membership K 5

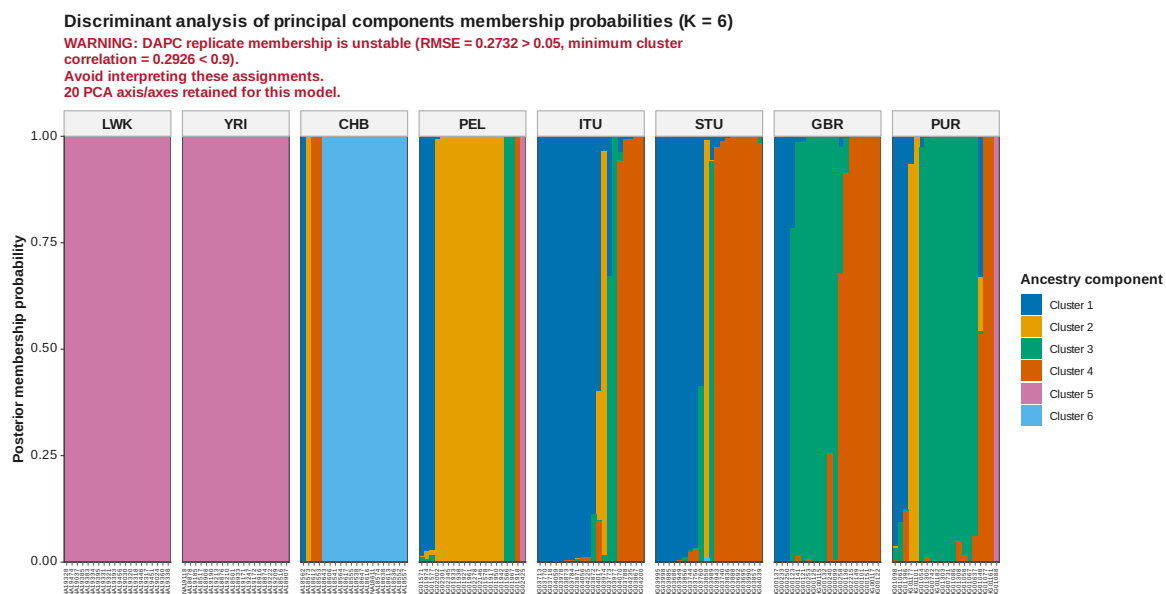


Figure 67: DAPC membership K 6

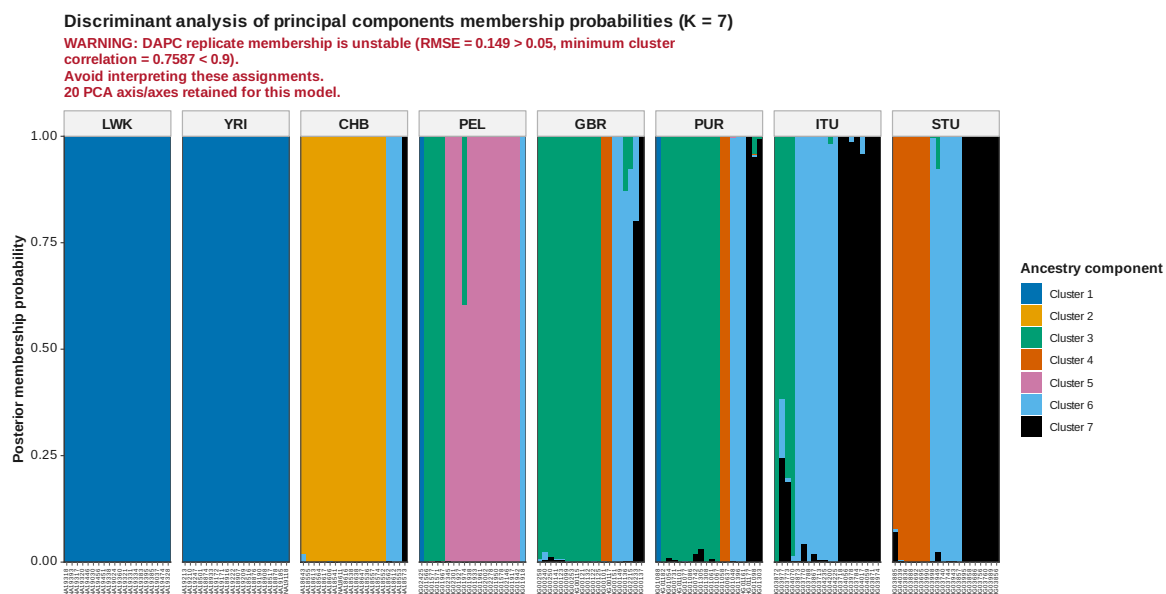


Figure 68: DAPC membership K 7

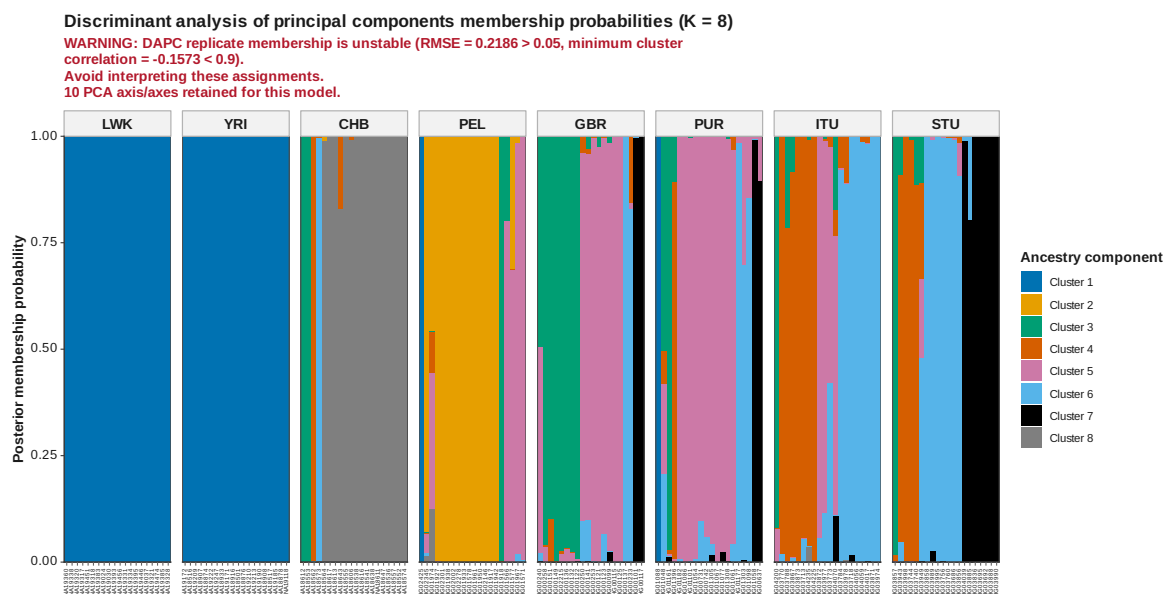


Figure 69: DAPC membership K 8



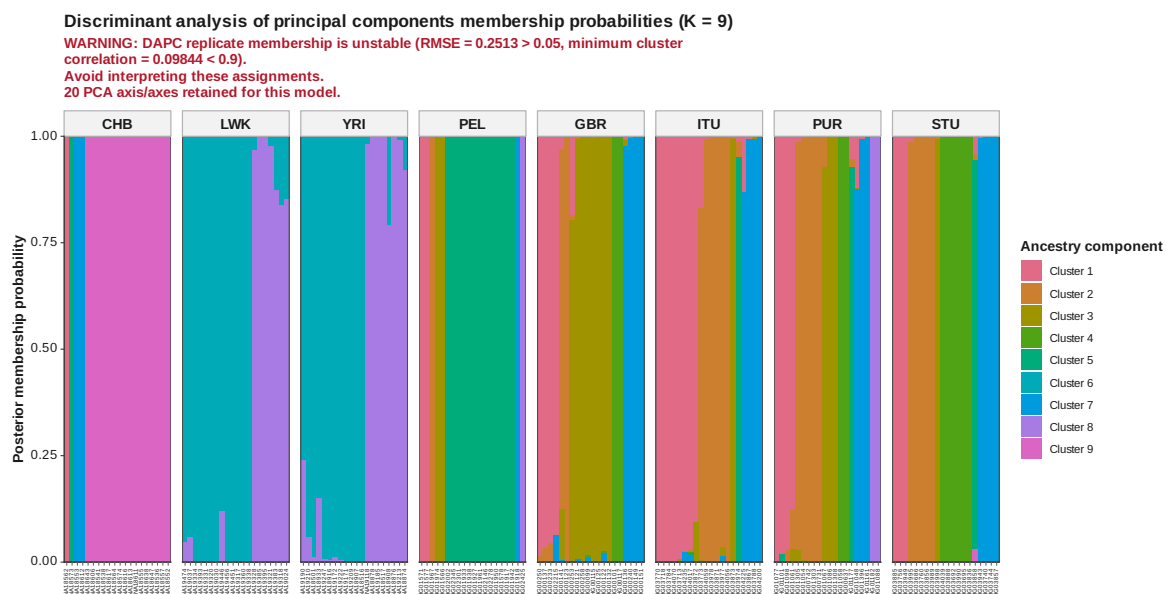


Figure 70: DAPC membership K 9

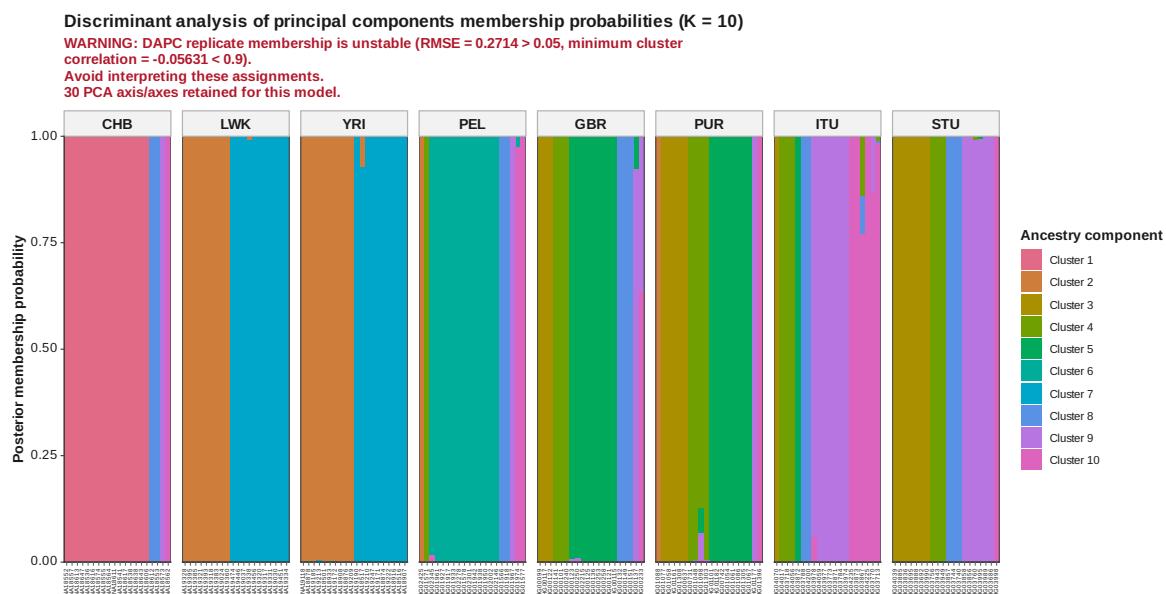
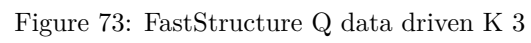
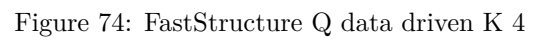


Figure 71: DAPC membership K 10







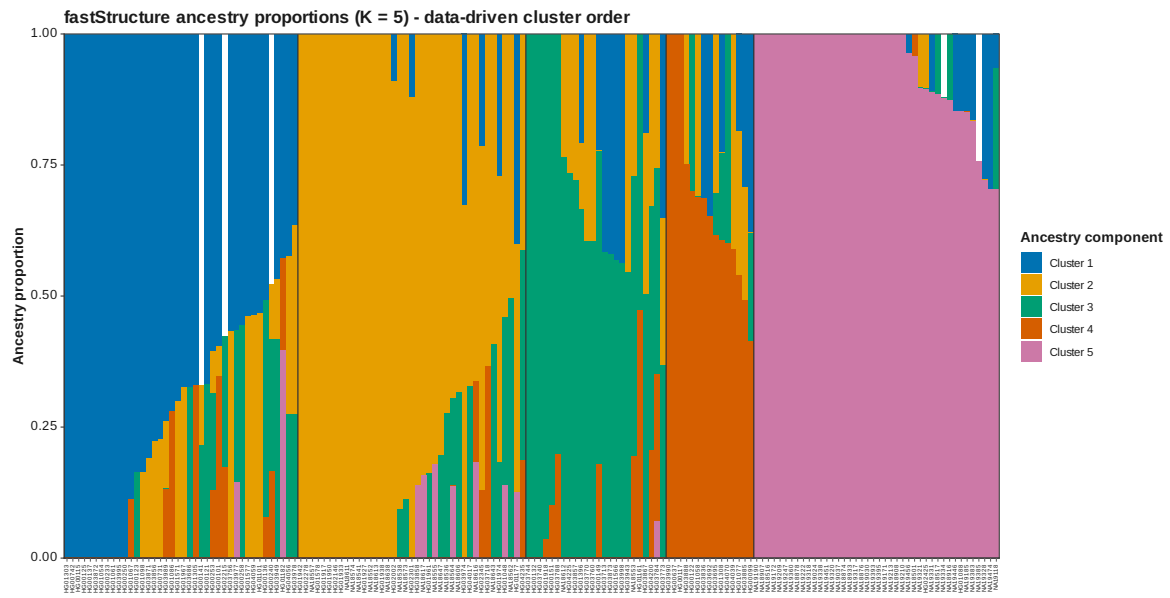


Figure 75: FastStructure Q data driven K 5

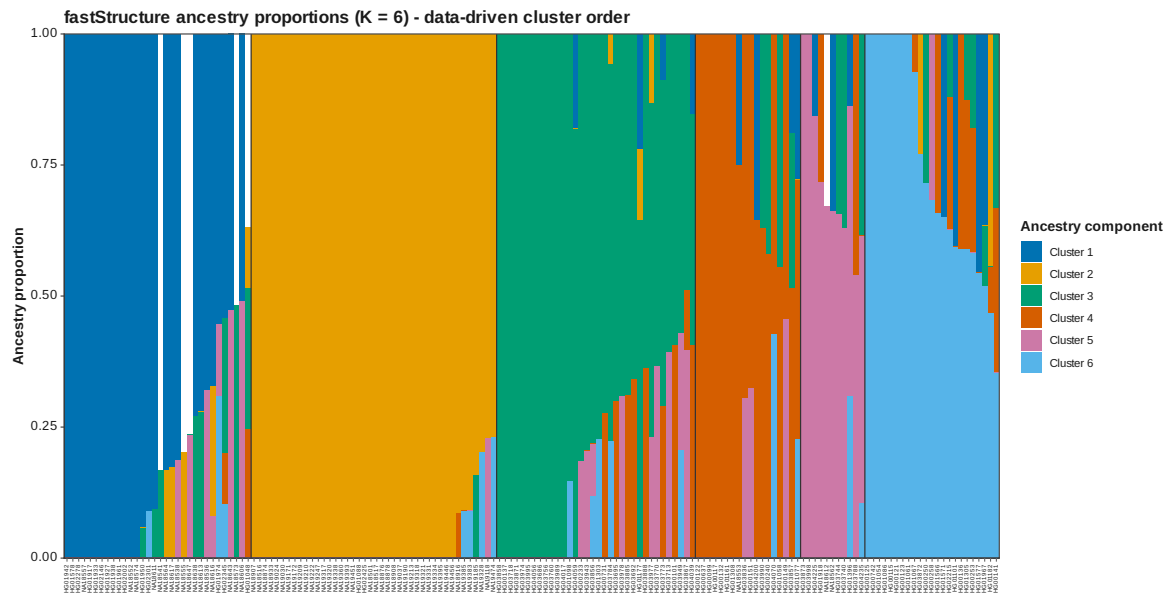


Figure 76: FastStructure Q data driven K 6

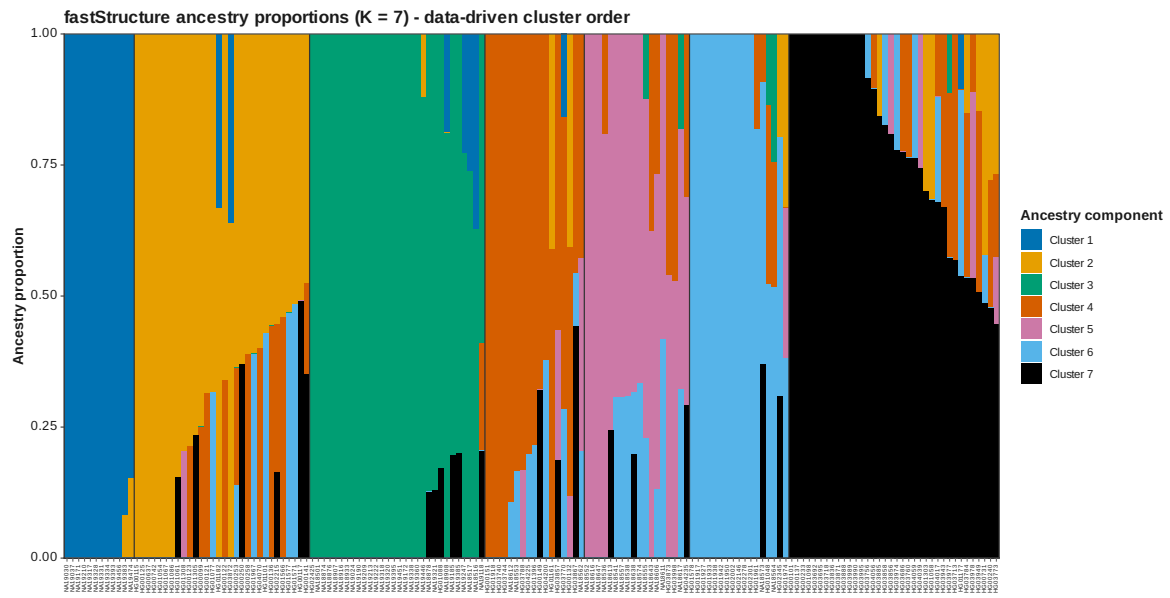


Figure 77: FastStructure Q data driven K 7



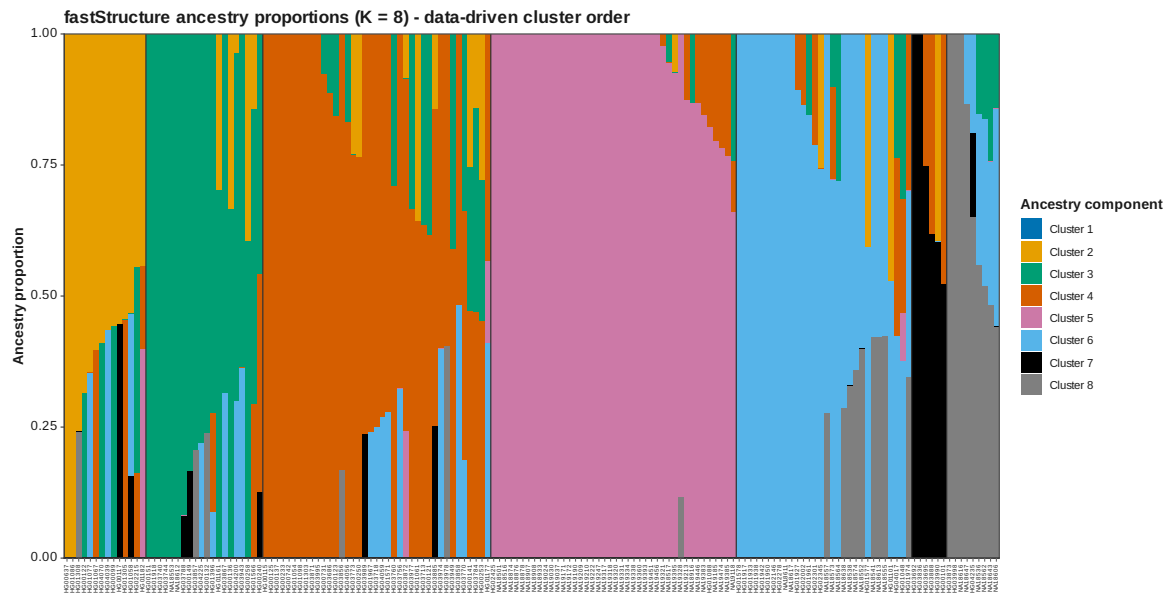


Figure 78: FastStructure Q data driven K 8

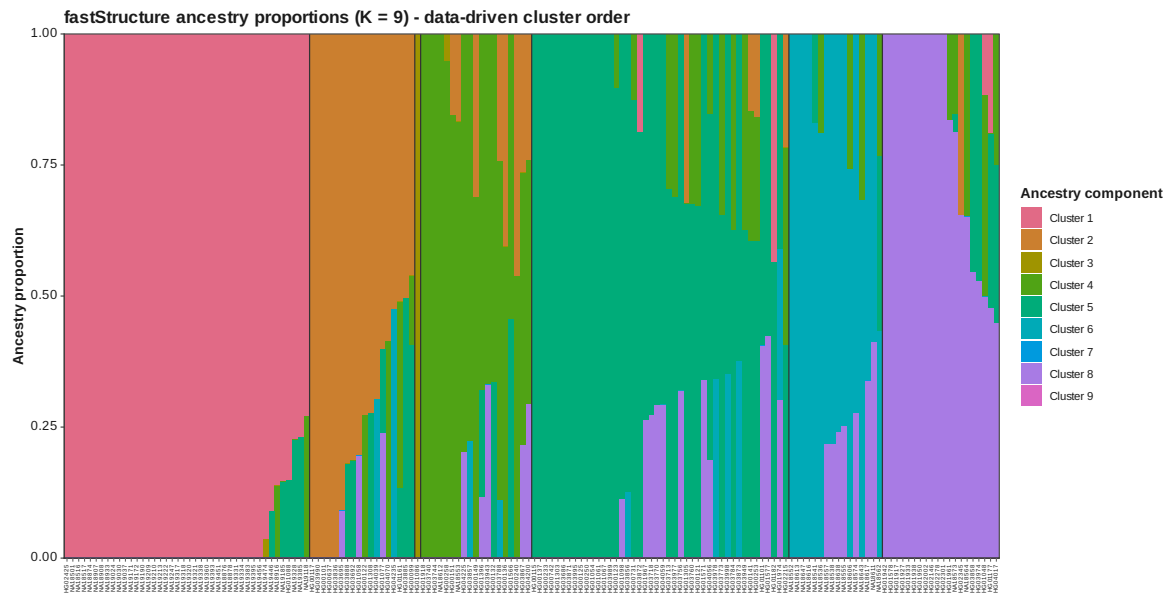


Figure 79: FastStructure Q data driven K 9

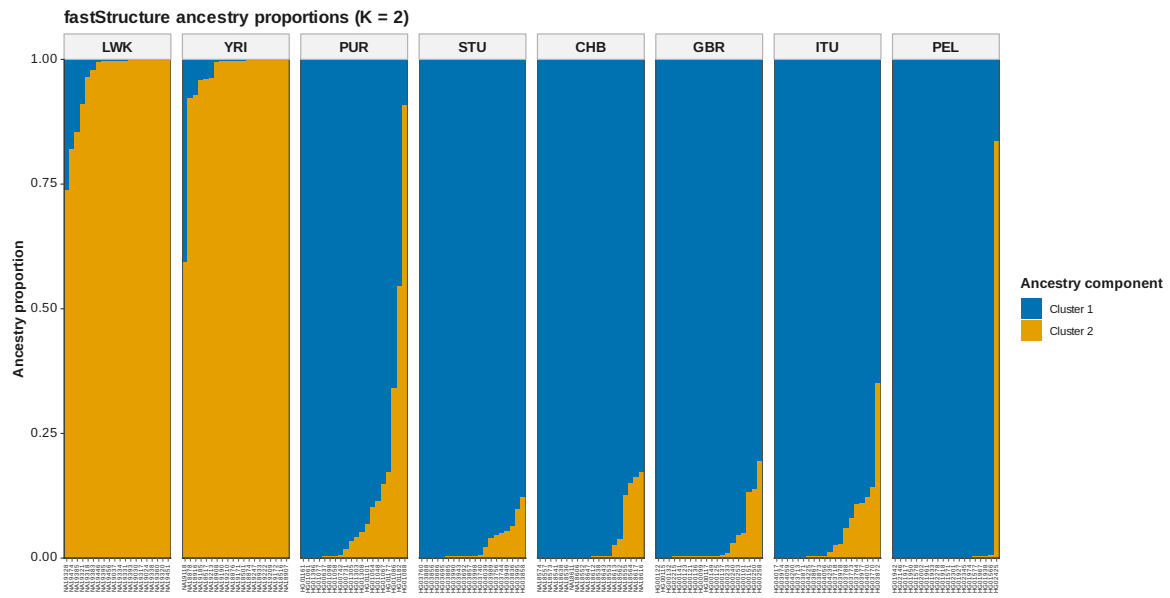


Figure 80: FastStructure Q K 2

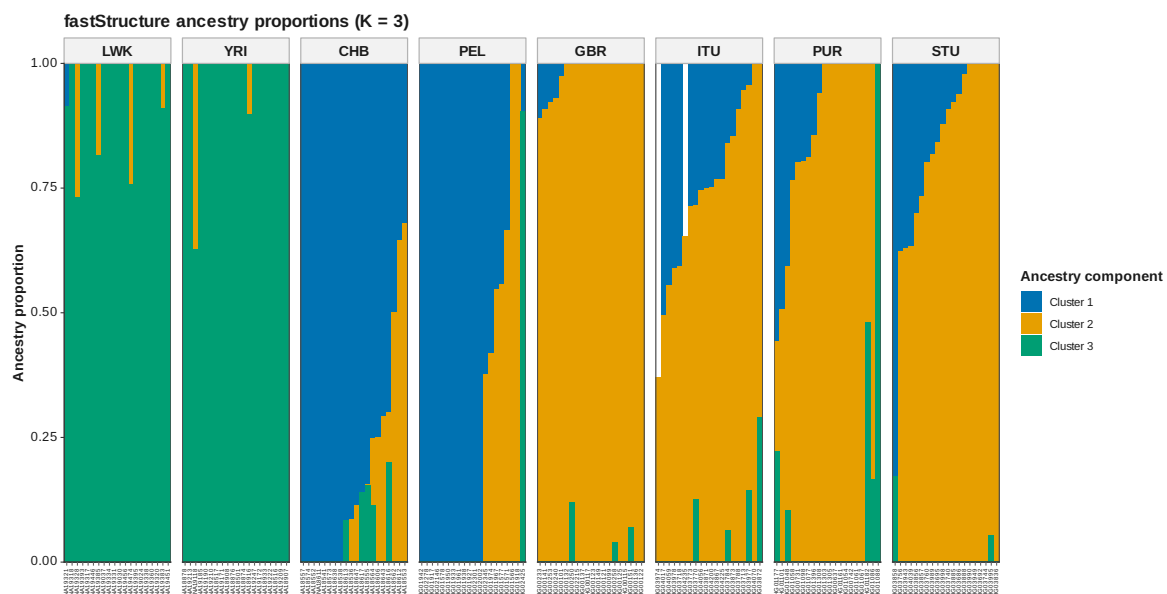


Figure 81: FastStructure Q K 3



Figure 82: FastStructure Q K 4

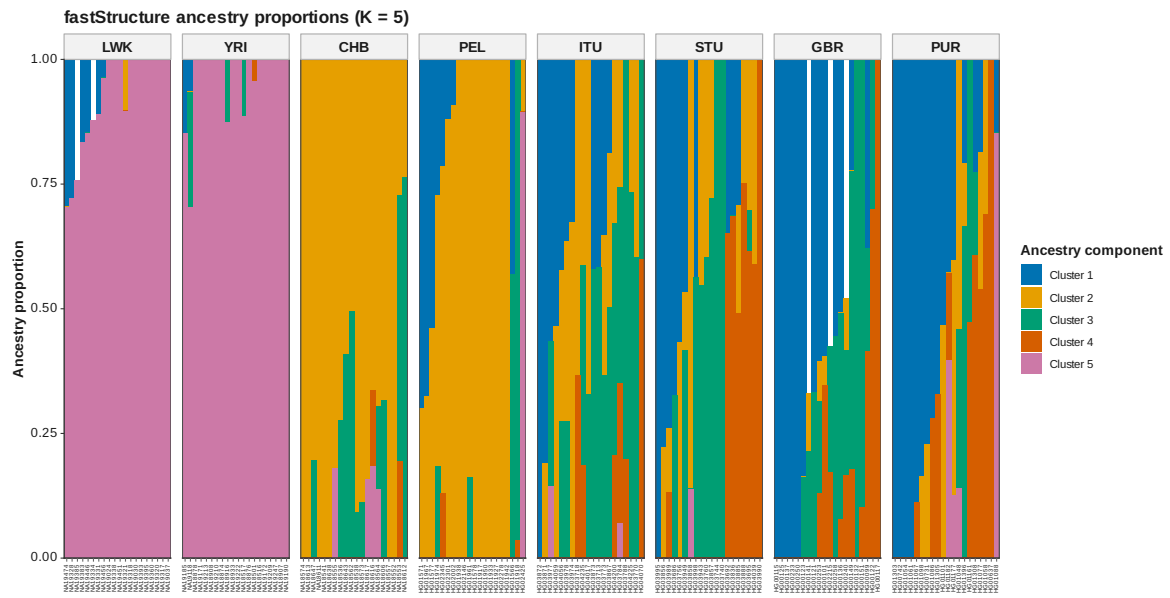


Figure 83: FastStructure Q K 5

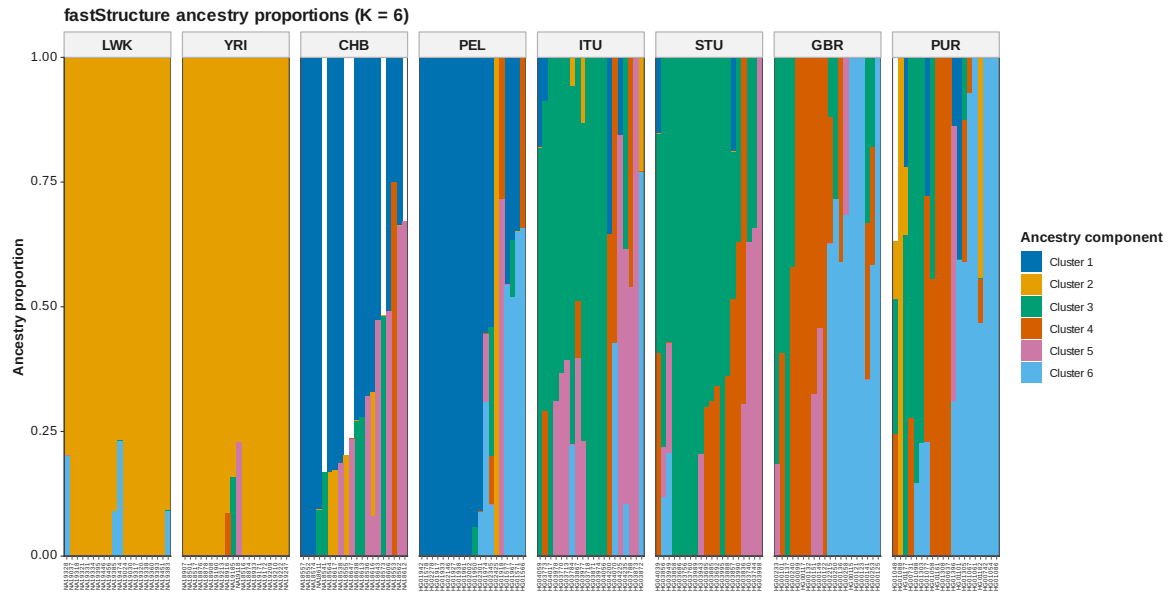


Figure 84: FastStructure Q K 6

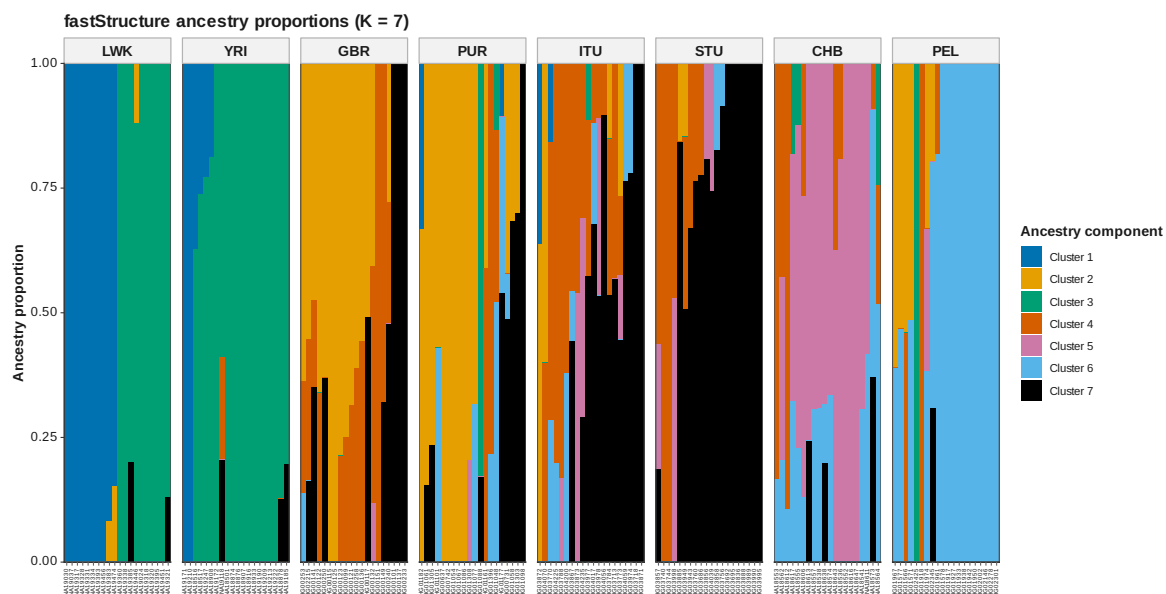


Figure 85: FastStructure Q K 7



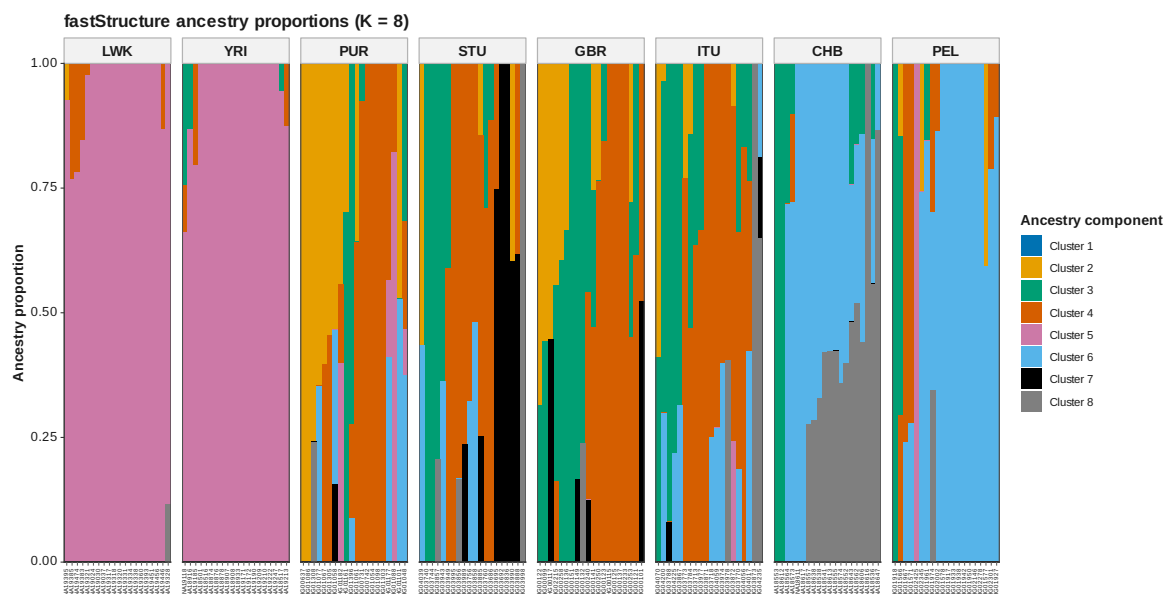


Figure 86: FastStructure Q K 8

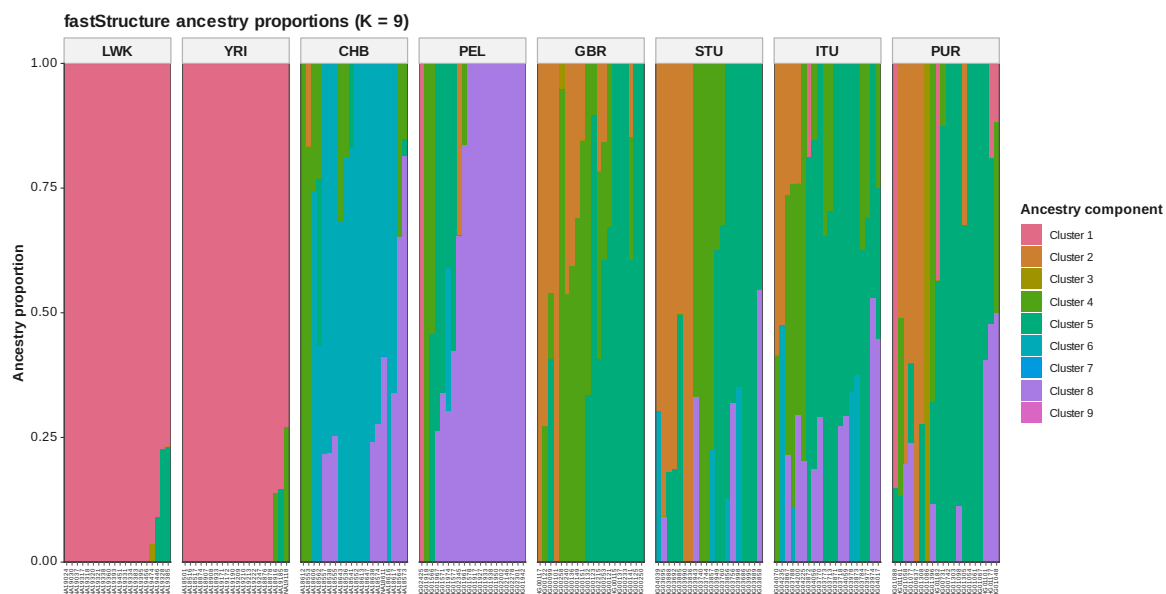
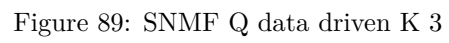


Figure 87: FastStructure Q K 9





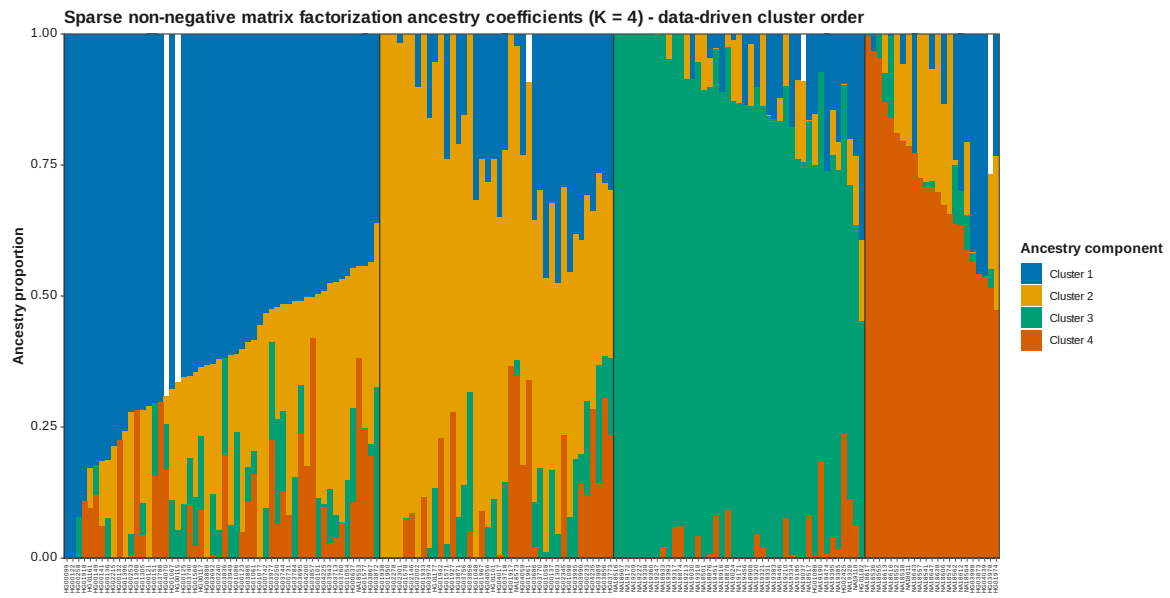


Figure 90: SNMF Q data driven K 4

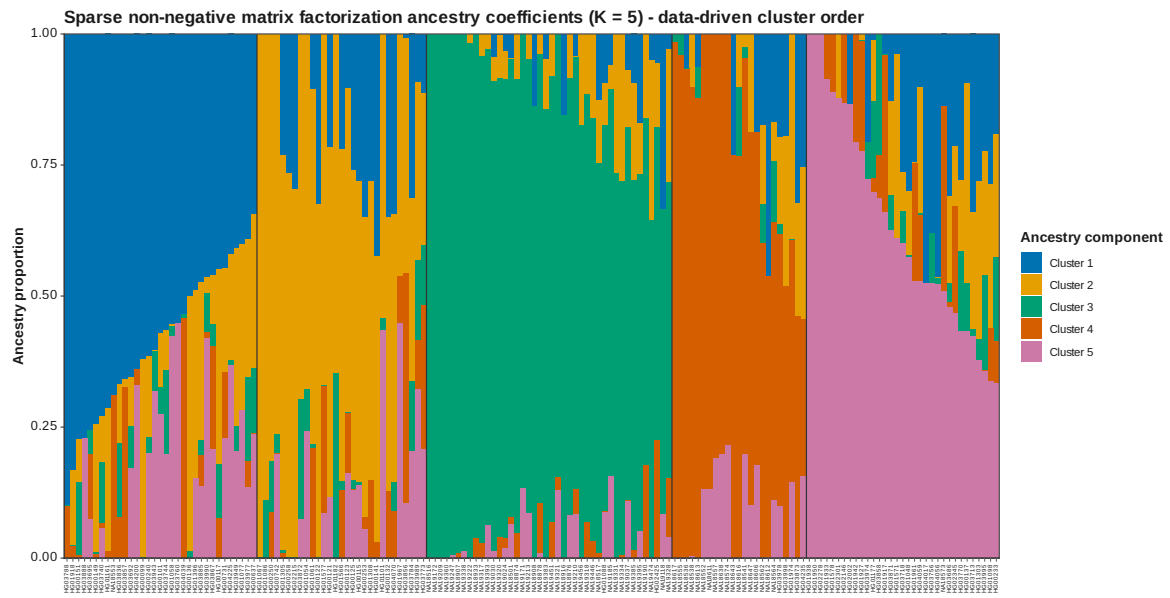


Figure 91: SNMF Q data driven K 5

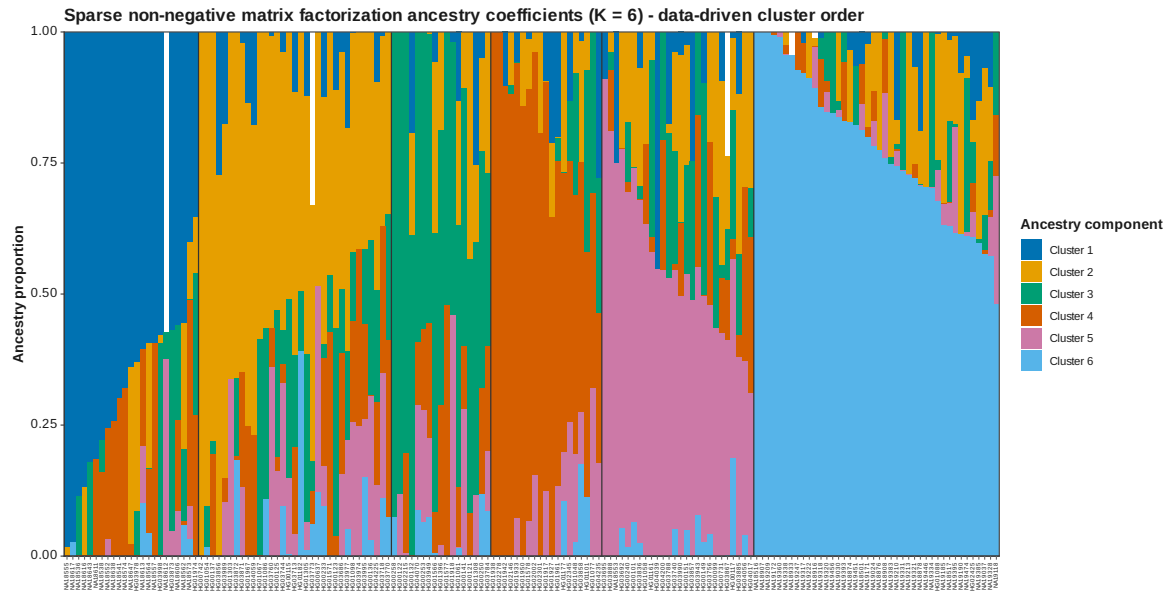


Figure 92: SNMF Q data driven K 6

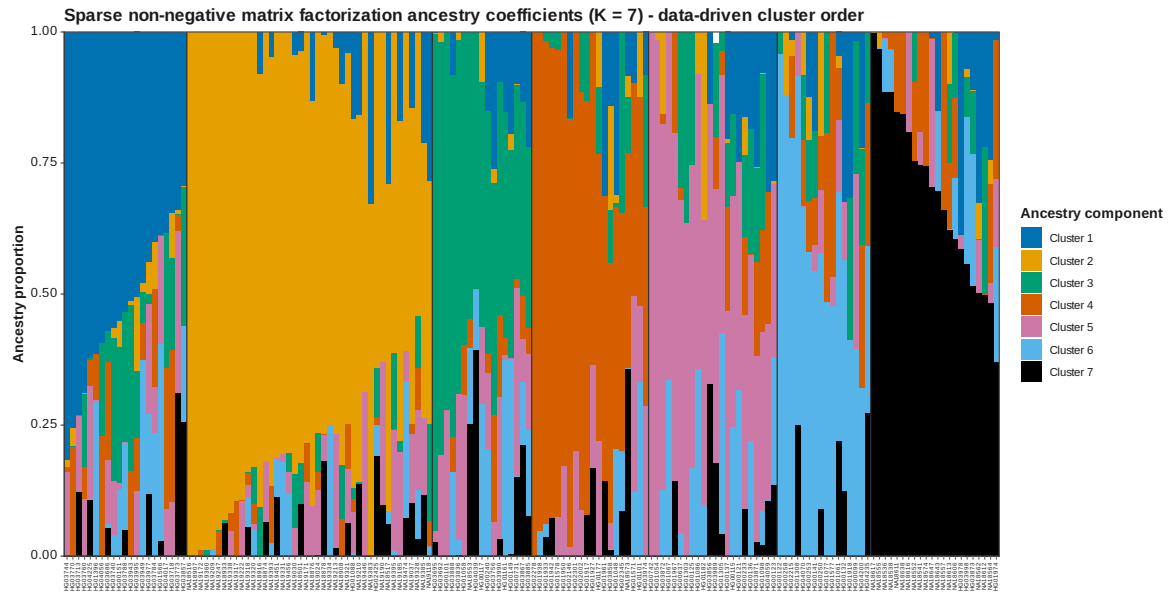


Figure 93: SNMF Q data driven K 7



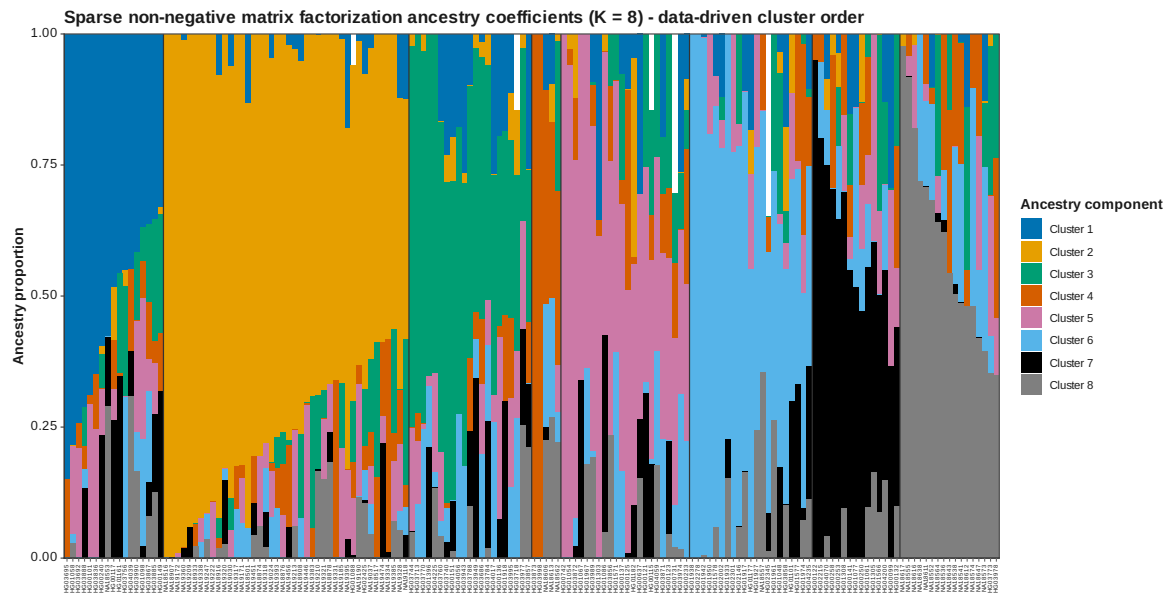


Figure 94: SNMF Q data driven K 8

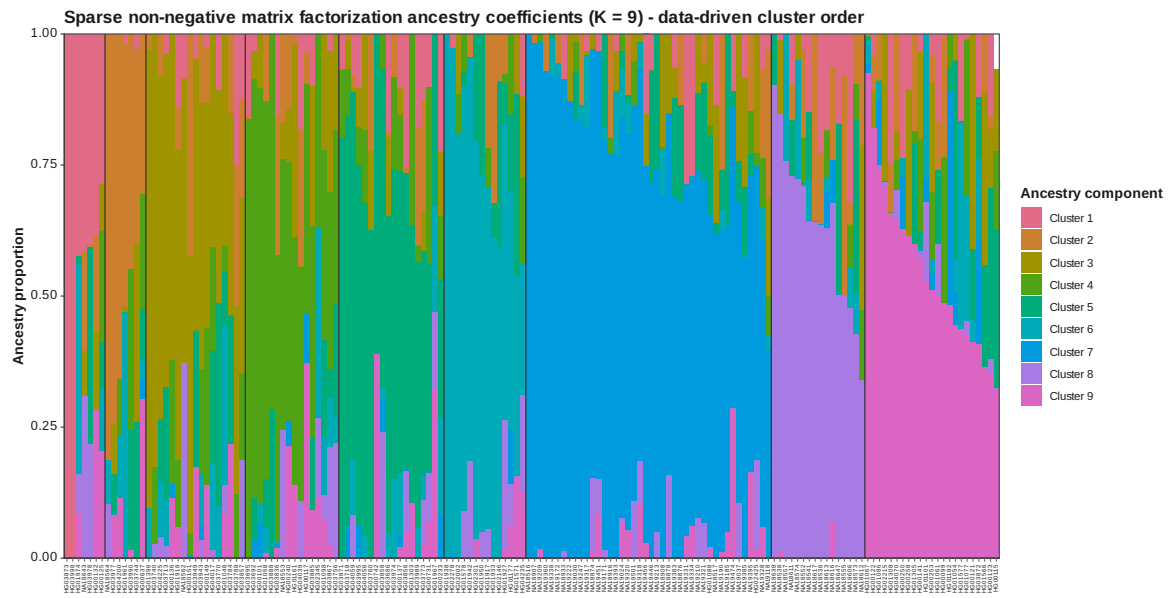


Figure 95: SNMF Q data driven K 9

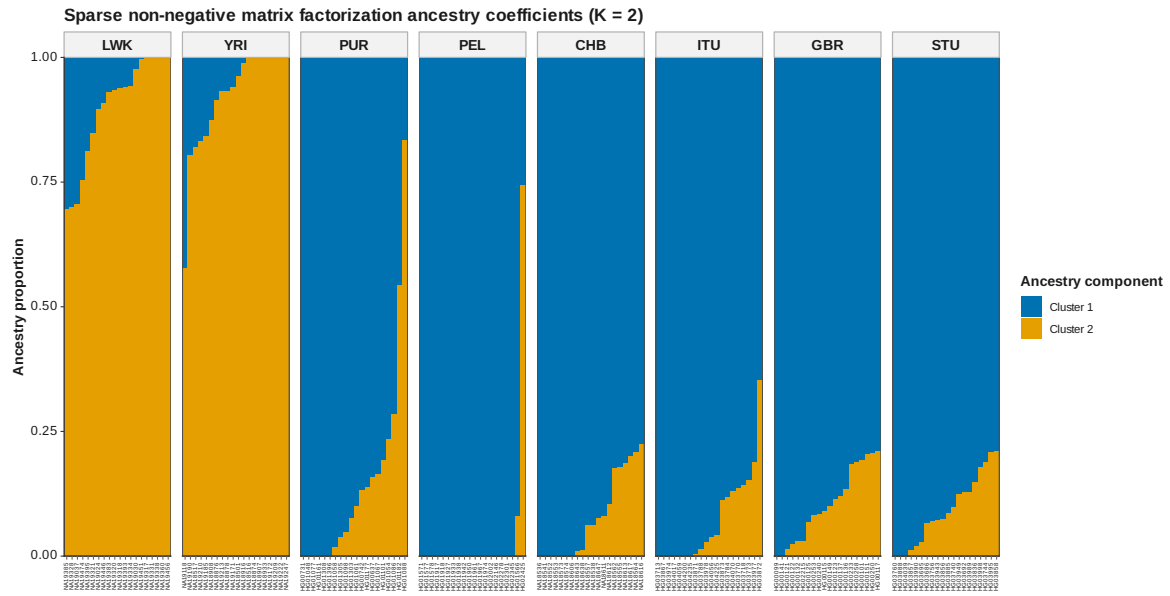


Figure 96: SNMF Q K 2

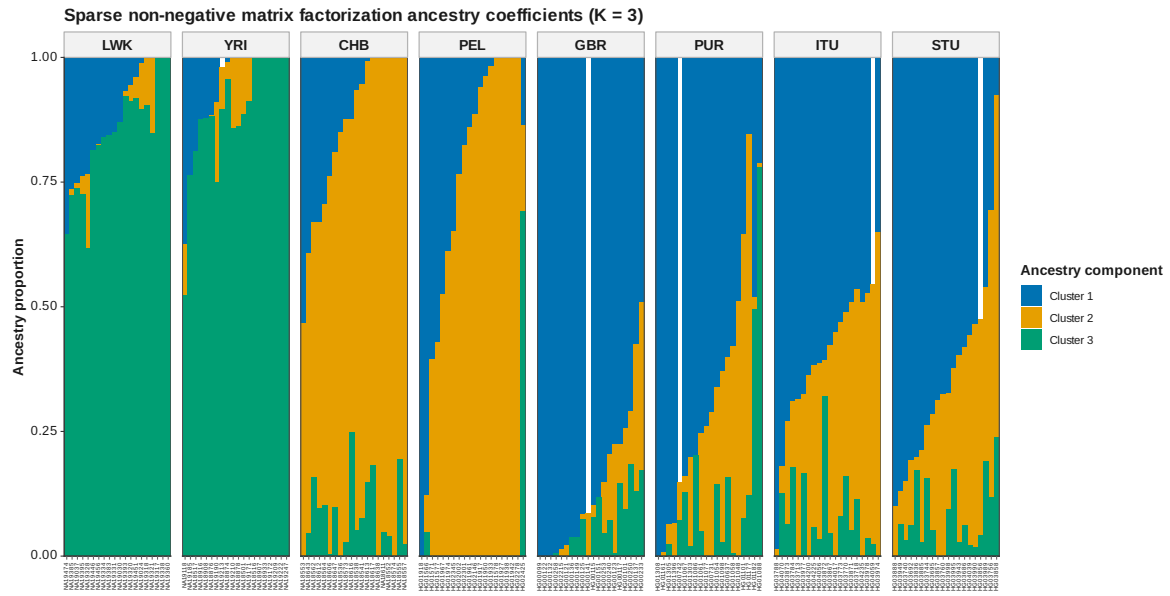


Figure 97: SNMF Q K 3

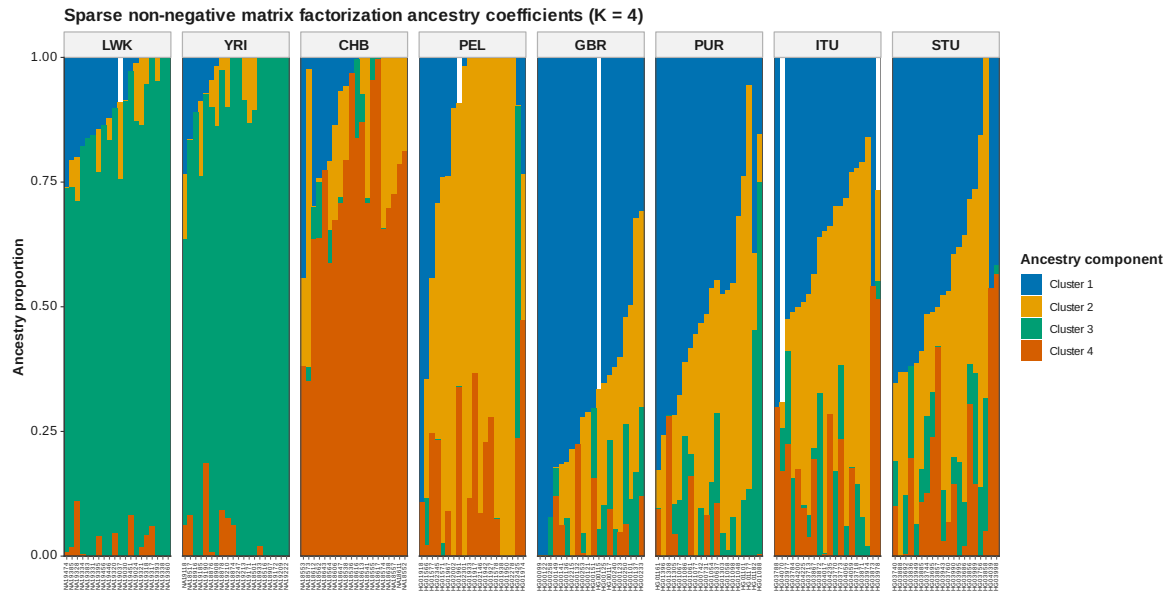


Figure 98: SNMF Q K 4

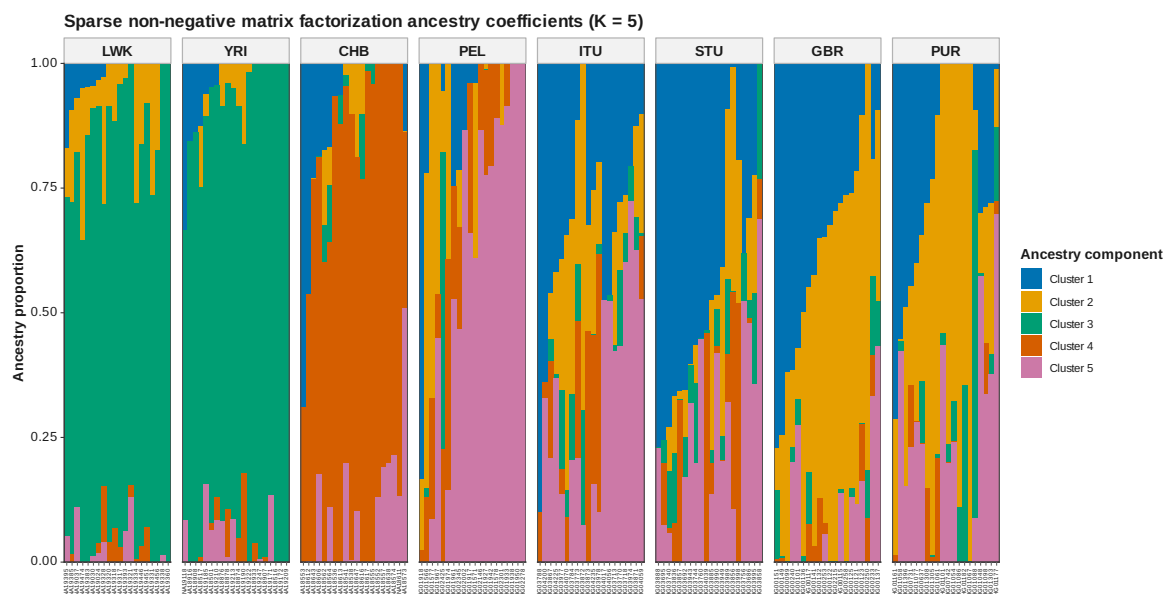


Figure 99: SNMF Q K 5

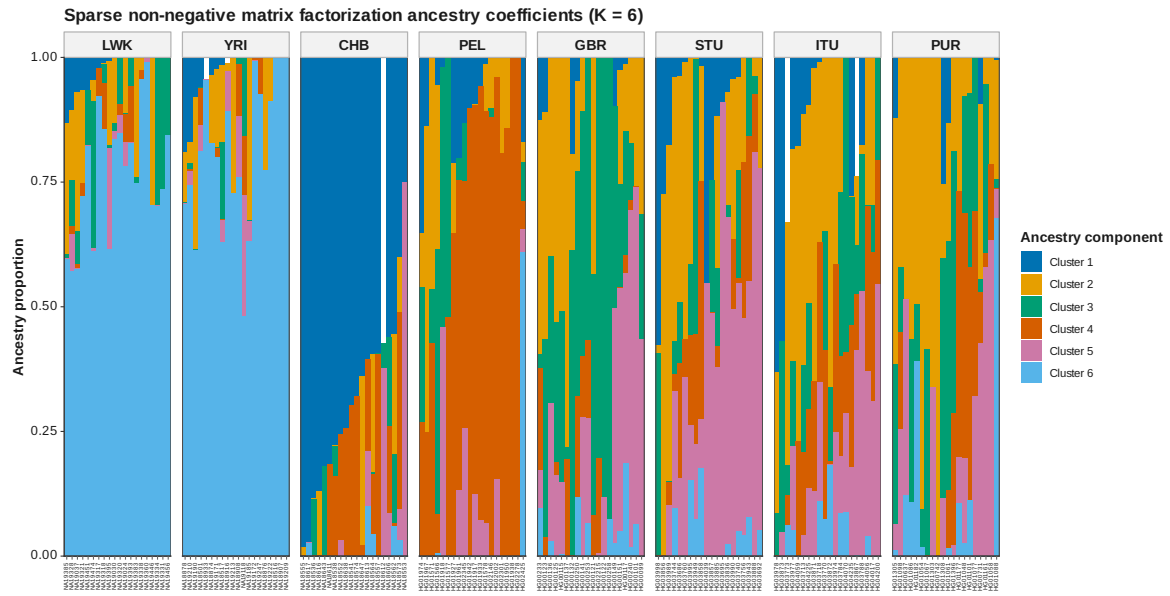


Figure 100: SNMF Q K 6

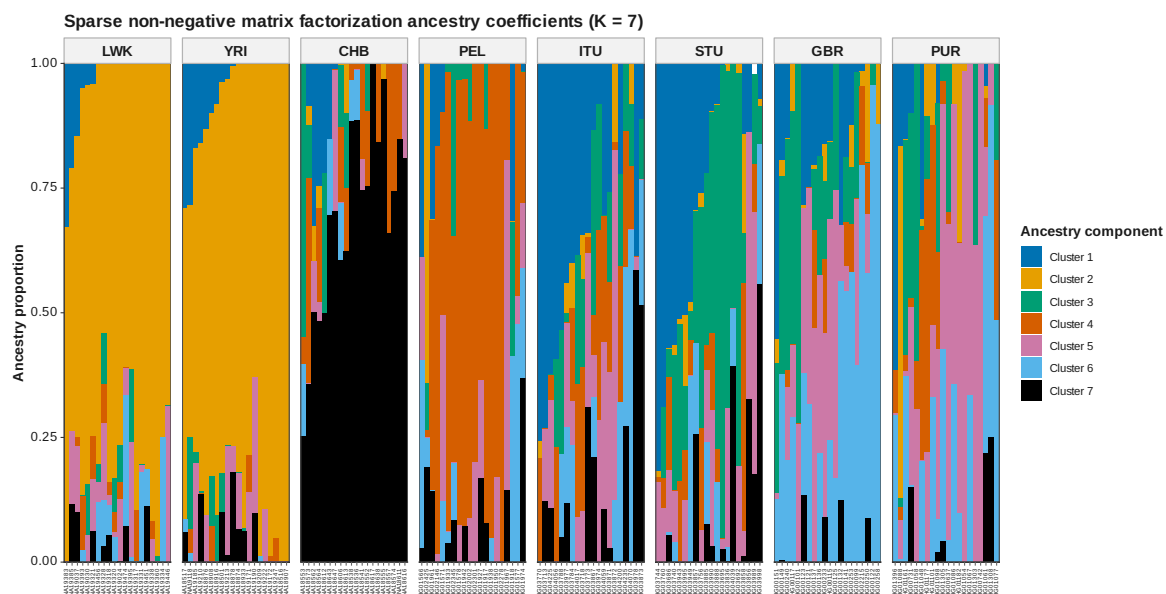


Figure 101: SNMF Q K 7



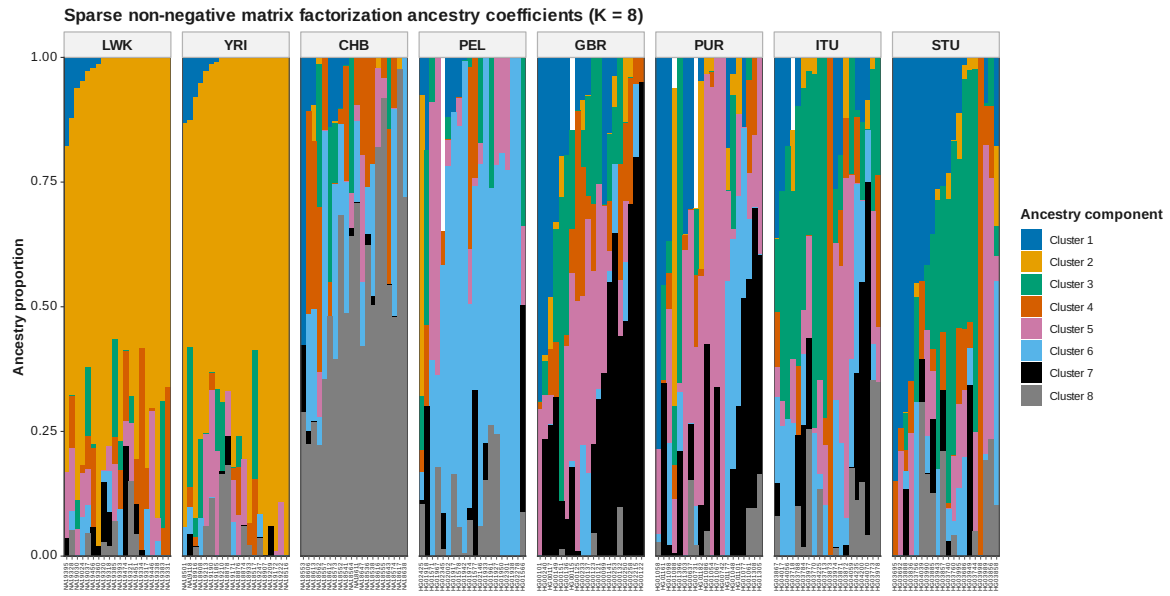


Figure 102: SNMF Q K 8

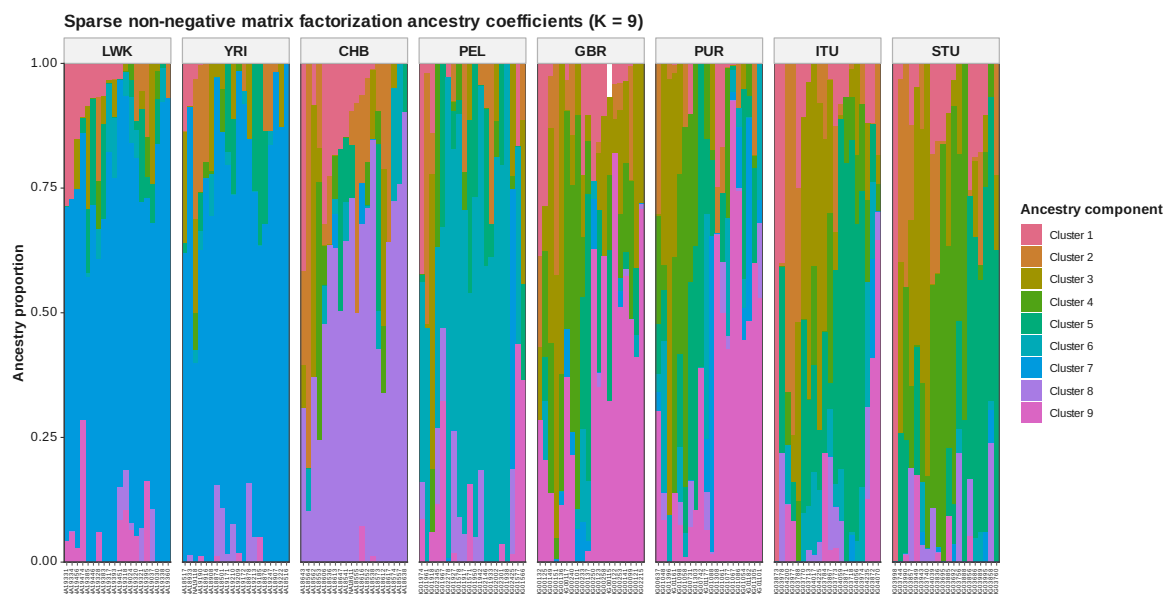


Figure 103: SNMF Q K 9

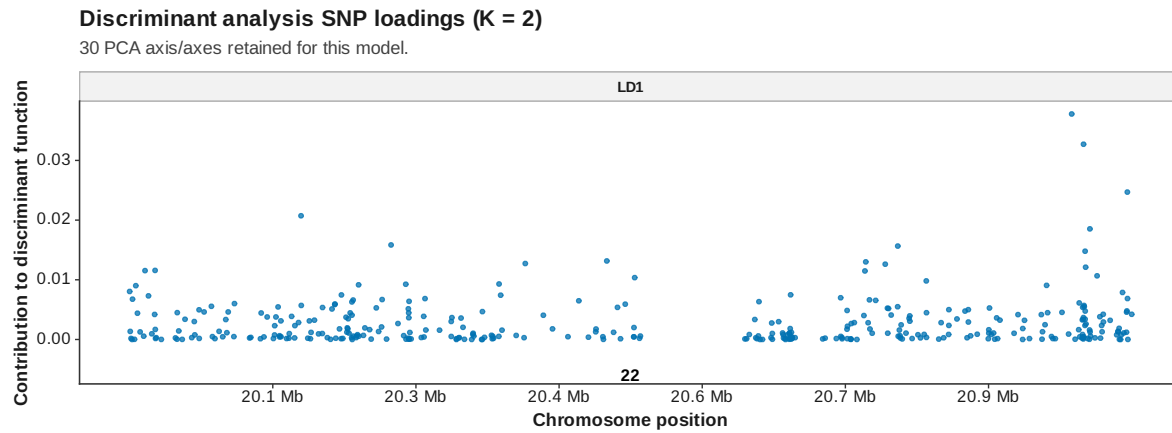


Figure 104: DAPC loadings manhattan K 2

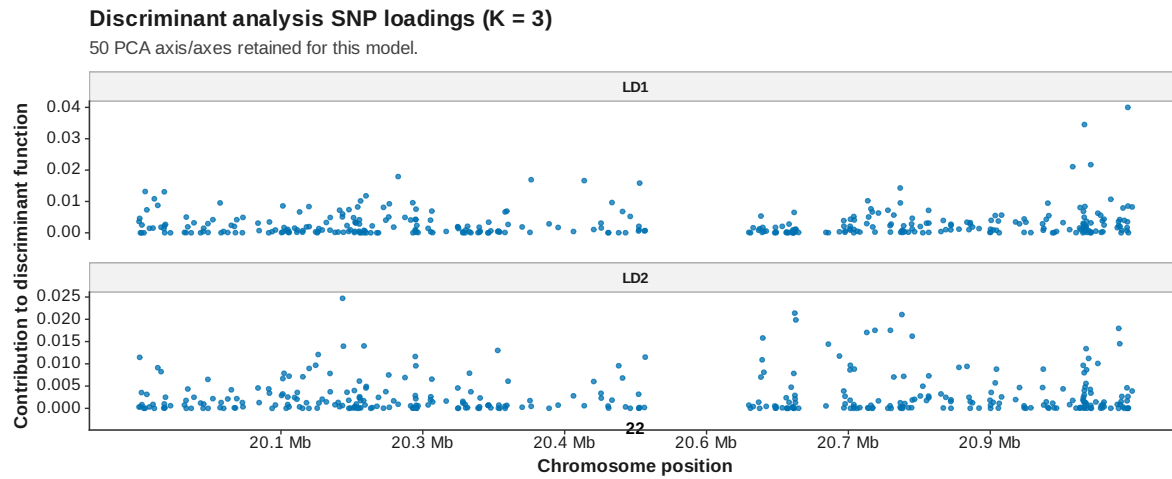


Figure 105: DAPC loadings manhattan K 3

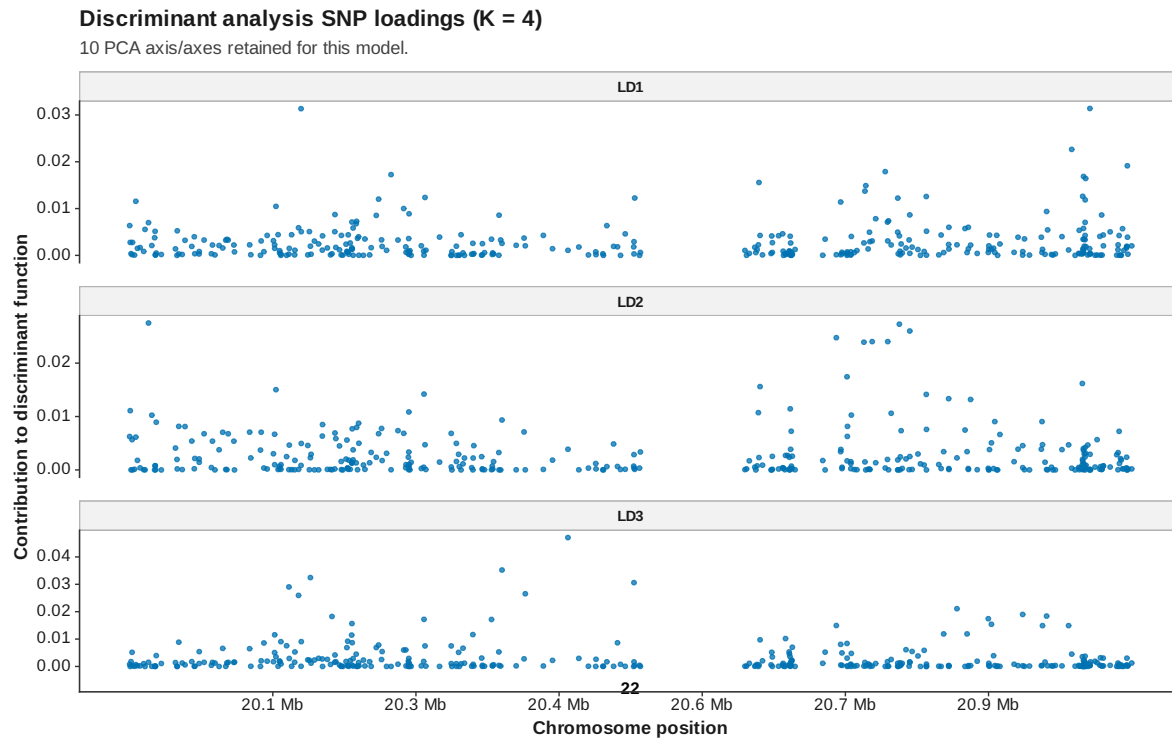


Figure 106: DAPC loadings manhattan K 4



Figure 107: DAPC loadings manhattan K 5

### Discriminant analysis SNP loadings (K = 6)

20 PCA axis/axes retained for this model.

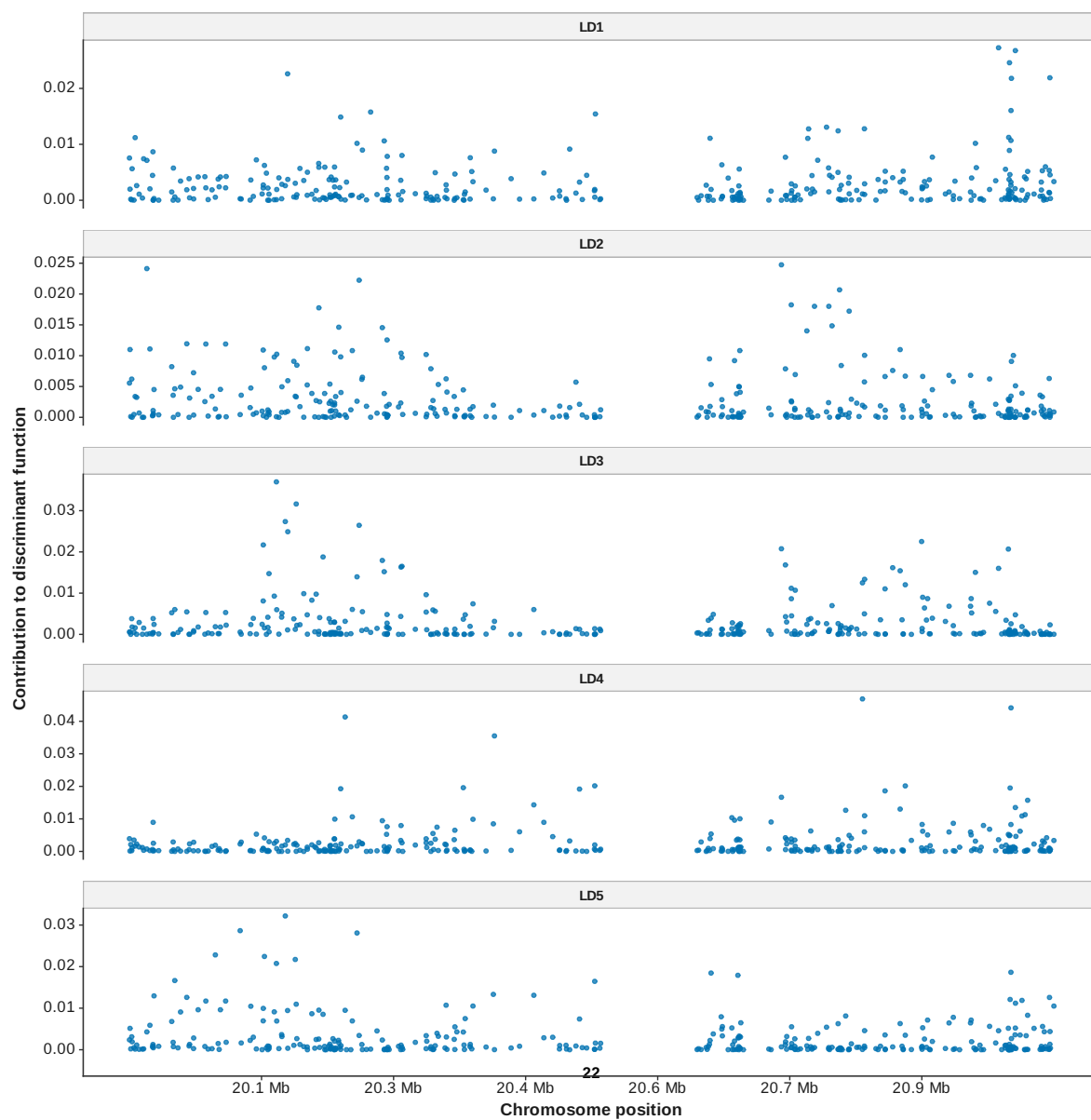


Figure 108: DAPC loadings manhattan K 6

# Discriminant analysis SNP loadings (K = 7)

20 PCA axis/axes retained for this model.

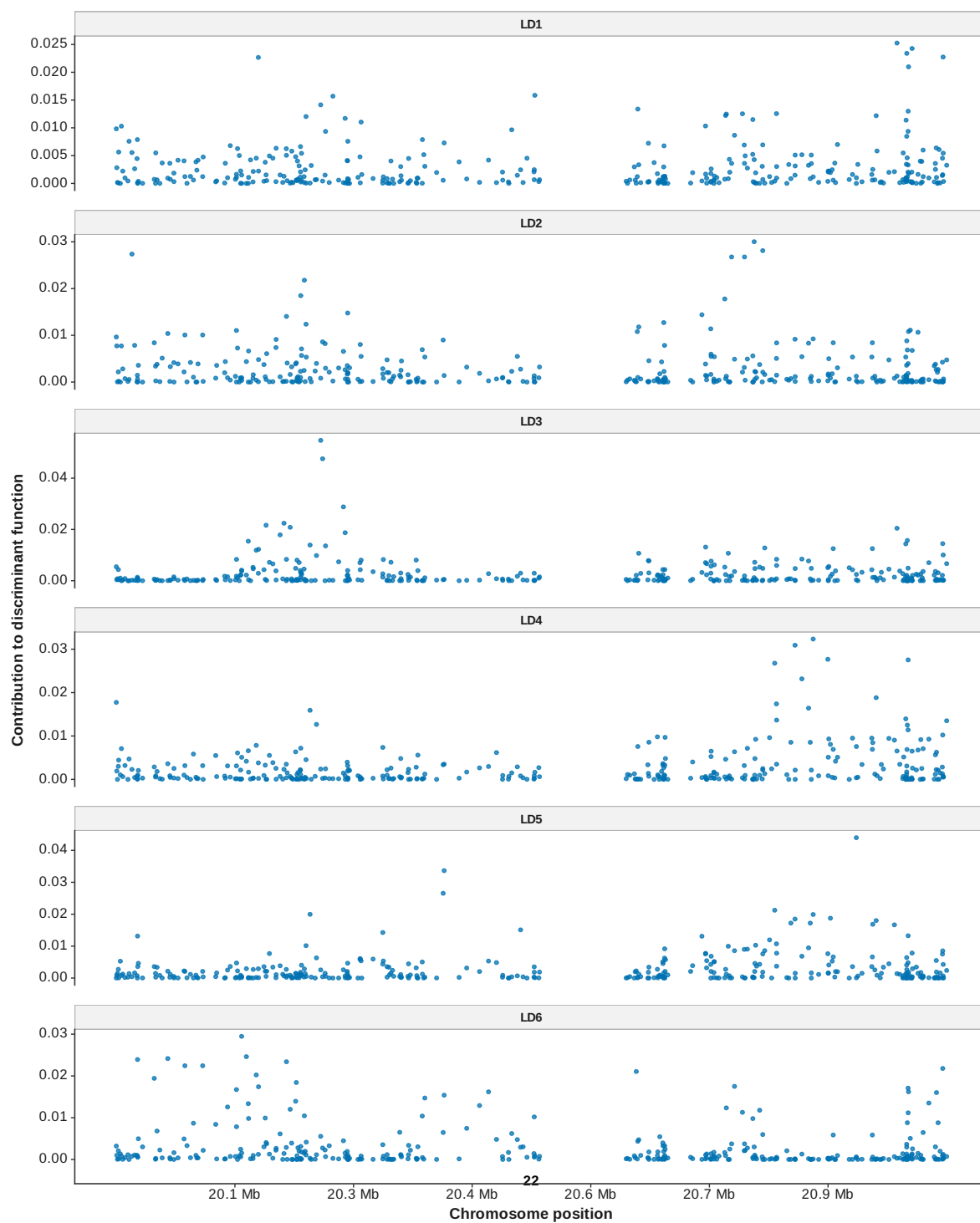


Figure 109: DAPC loadings manhattan K 7





Figure 110: DAPC loadings manhattan K 8  
 121

# Discriminant analysis SNP loadings (K = 9)

20 PCA axis/axes retained for this model.

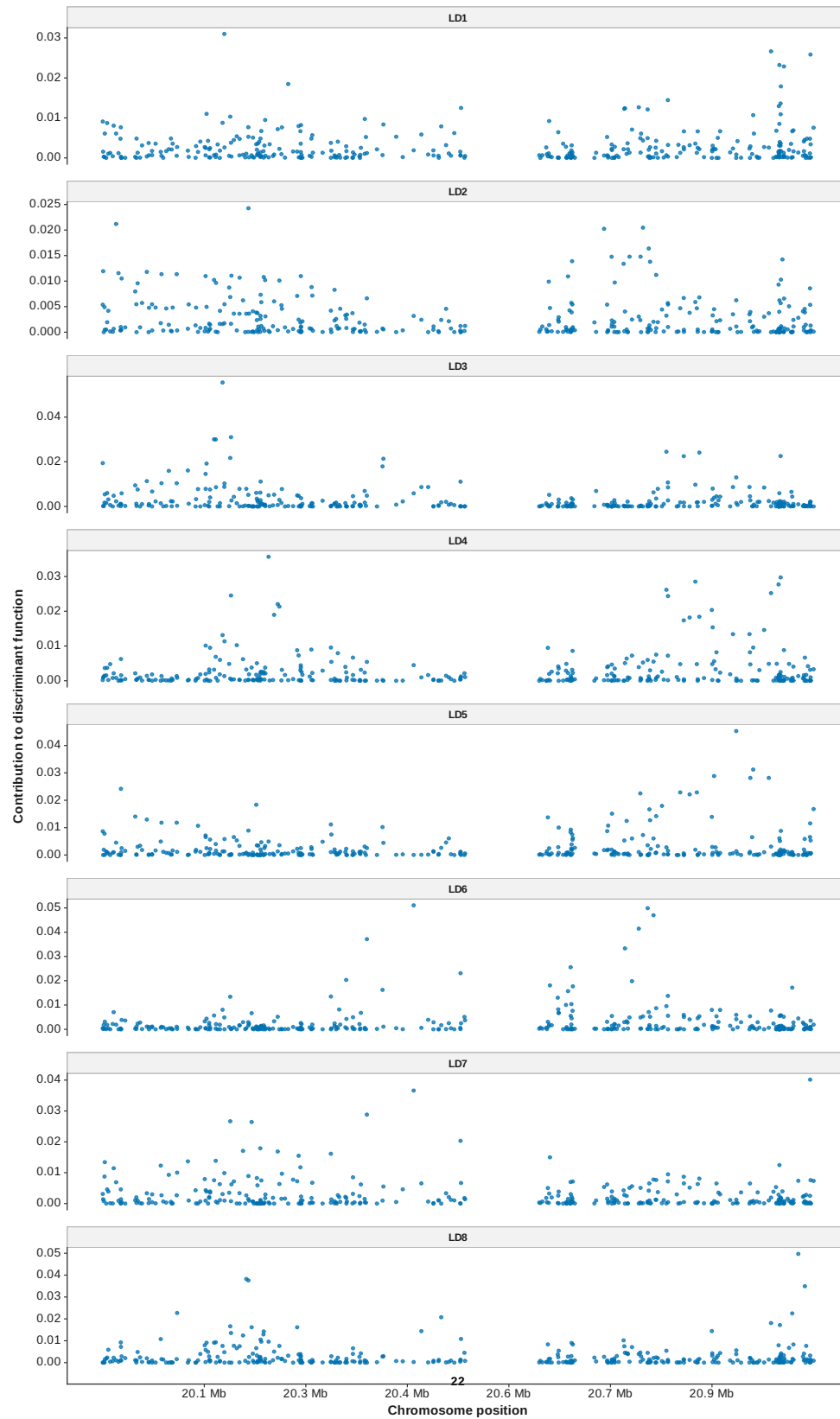


Figure 111: DAPC loadings manhattan K 9



Figure 112: DAPC loadings manhattan K 10

### Discriminant analysis SNP loadings, ranked (K = 2)

30 PCA axis/axes retained for this model.

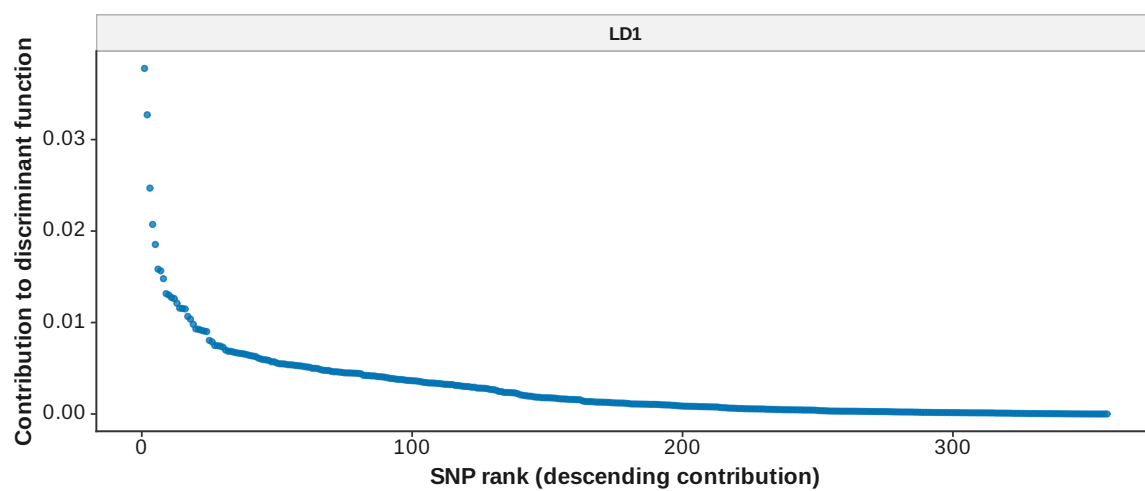


Figure 113: DAPC loadings ranked K 2

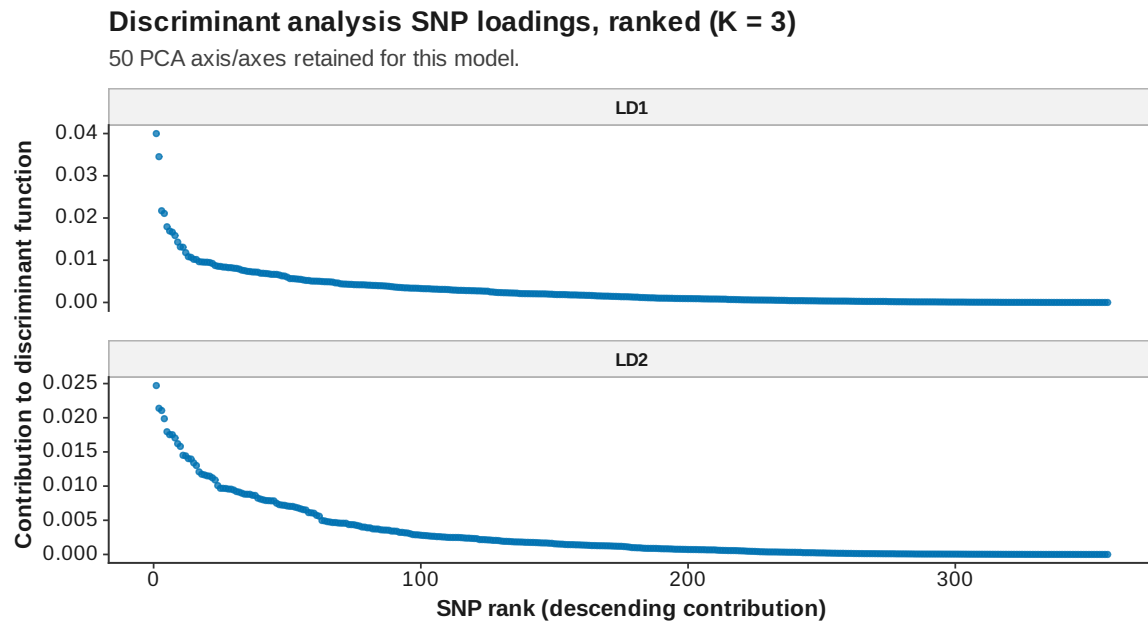


Figure 114: DAPC loadings ranked K 3

### Discriminant analysis SNP loadings, ranked (K = 4)

10 PCA axis/axes retained for this model.

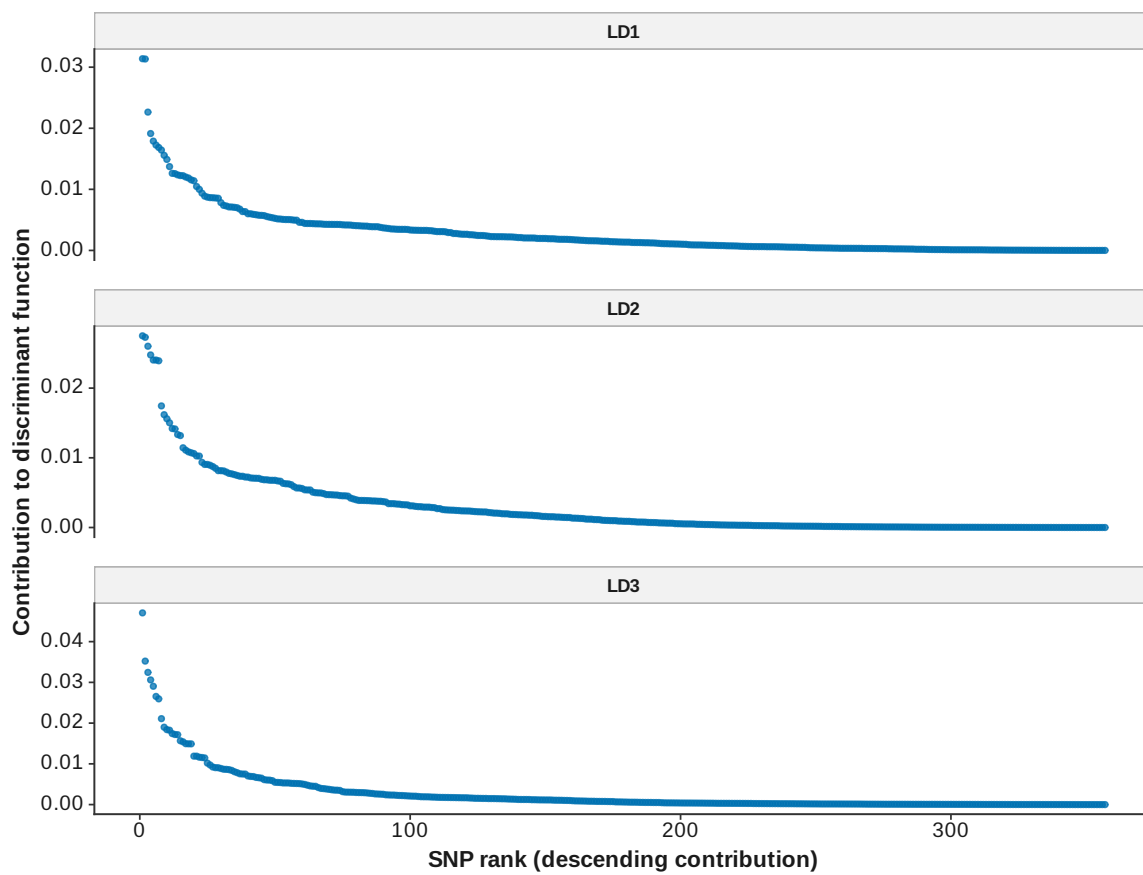


Figure 115: DAPC loadings ranked K 4

### Discriminant analysis SNP loadings, ranked (K = 5)

30 PCA axis/axes retained for this model.

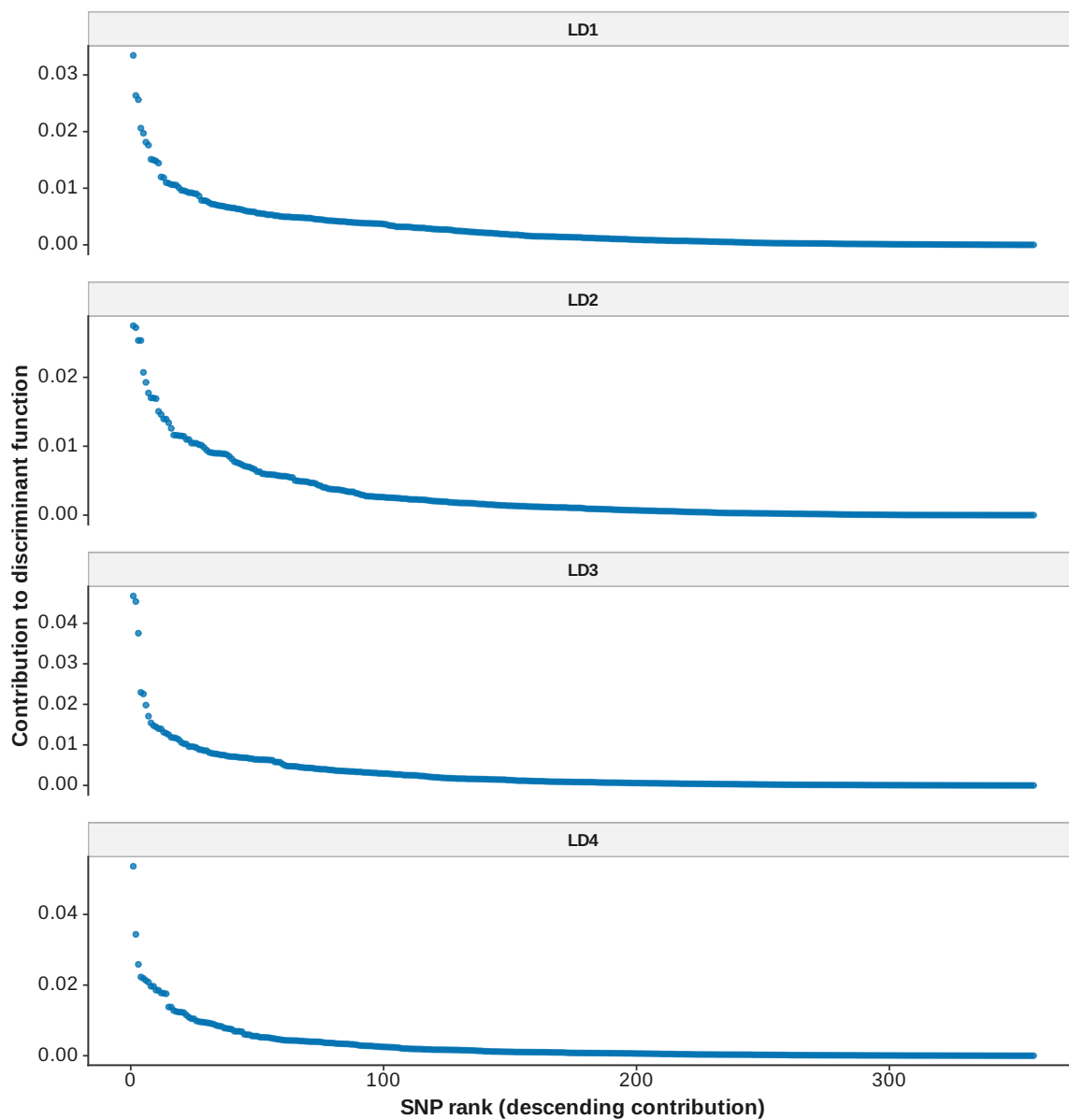


Figure 116: DAPC loadings ranked K 5

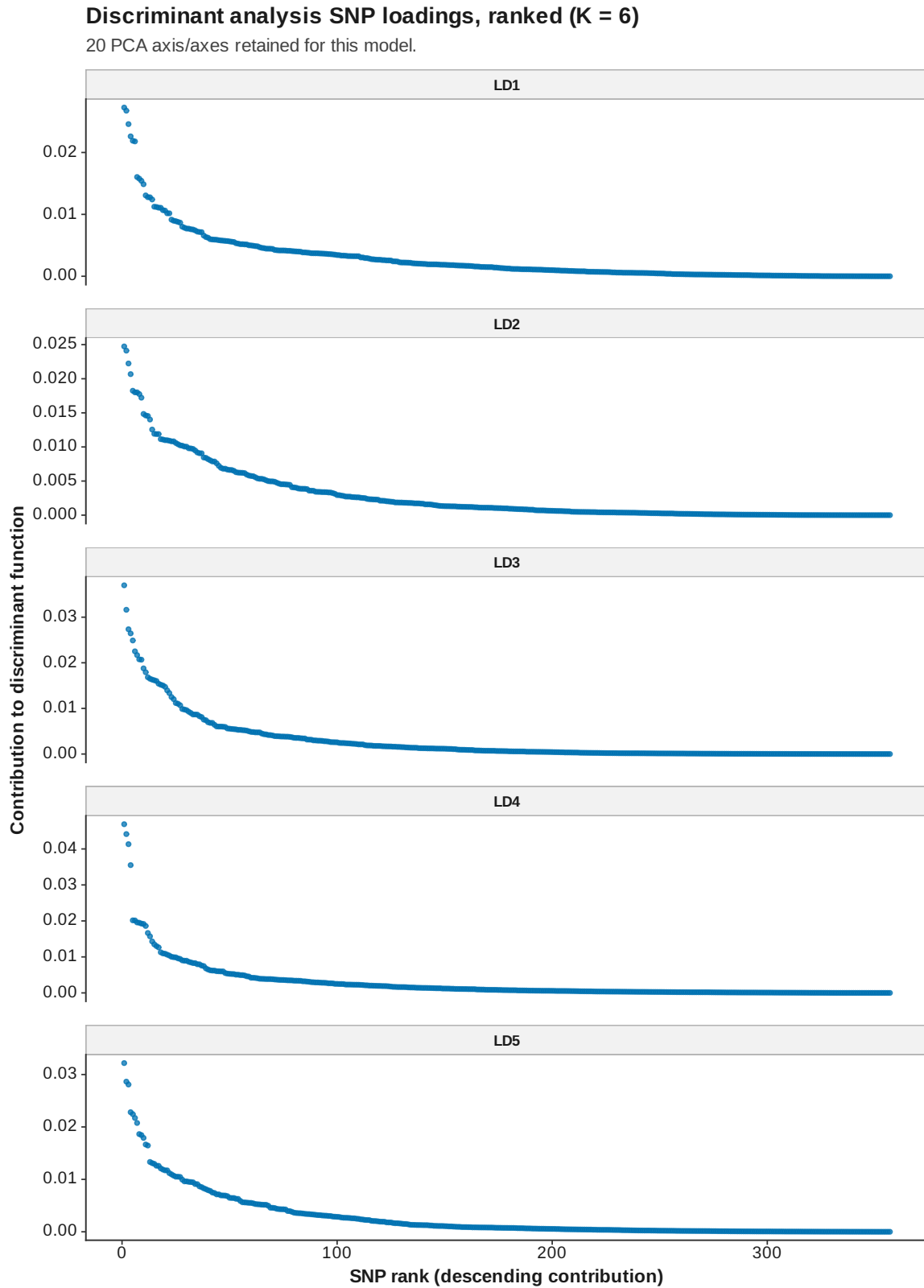


Figure 117: DAPC loadings ranked K 6



### Discriminant analysis SNP loadings, ranked (K = 7)

20 PCA axis/axes retained for this model.

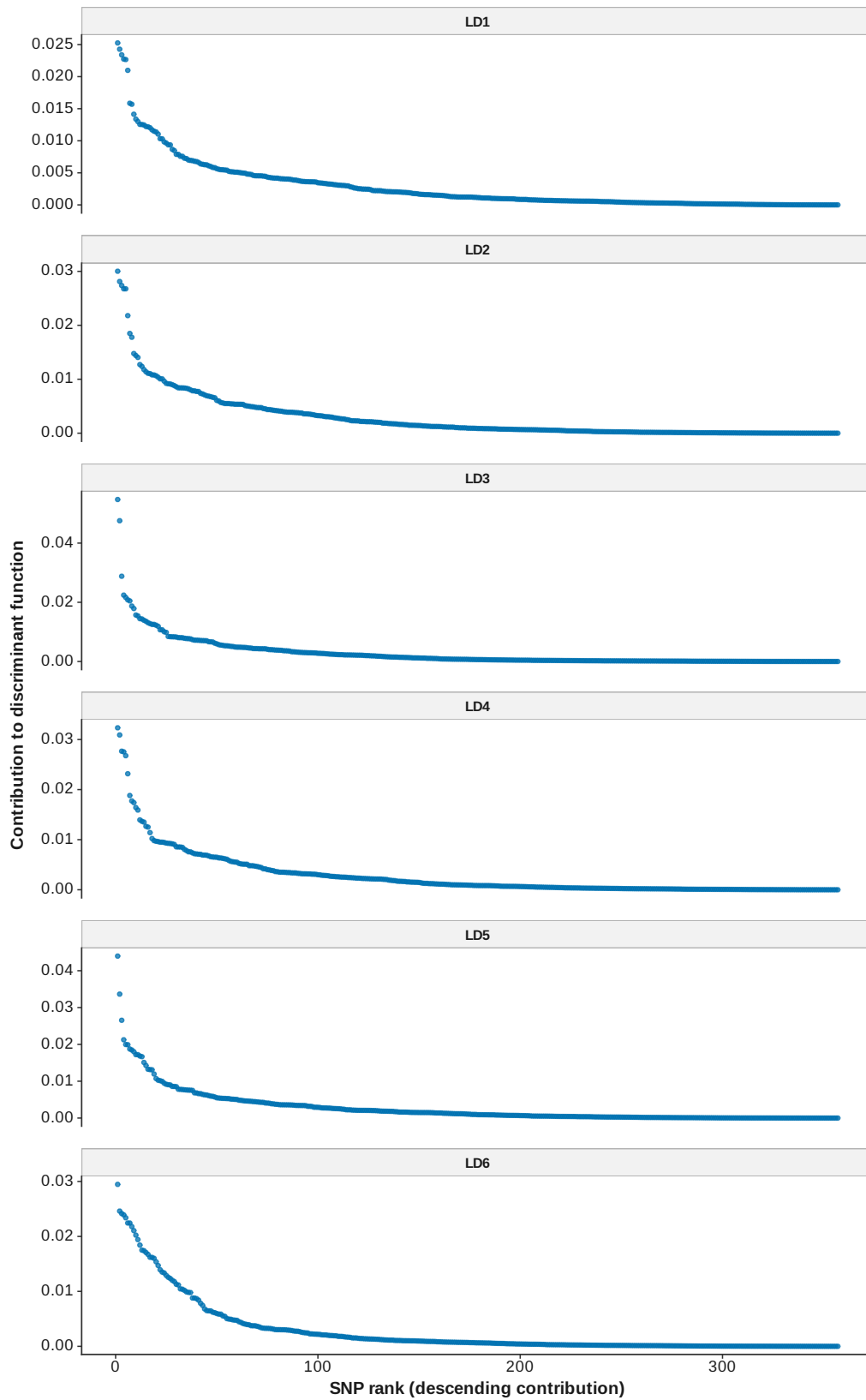


Figure 118: DAPC loadings ranked K 7

### Discriminant analysis SNP loadings, ranked (K = 8)

10 PCA axis/axes retained for this model.

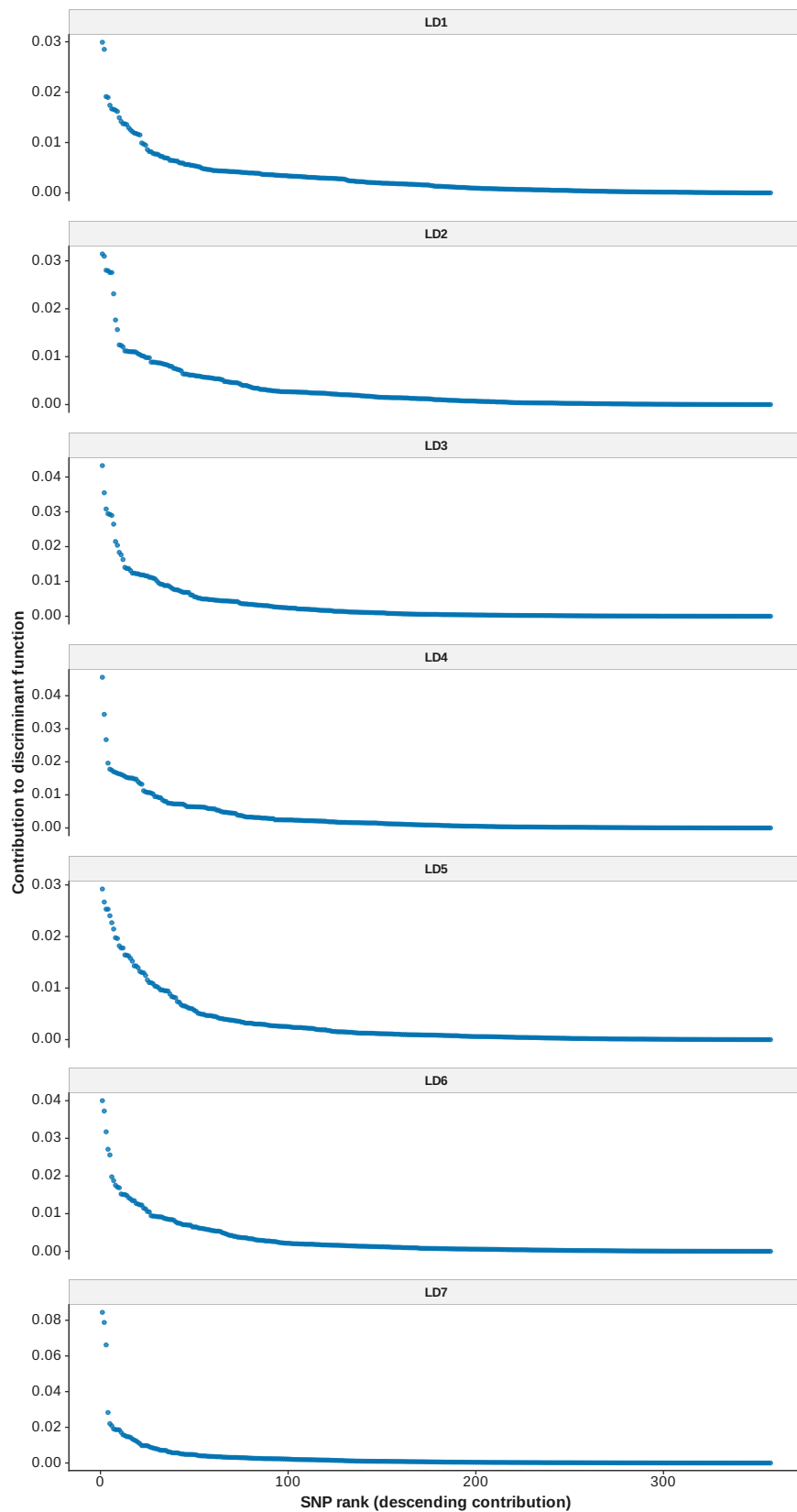


Figure 119: DAPC loadings ranked K 8

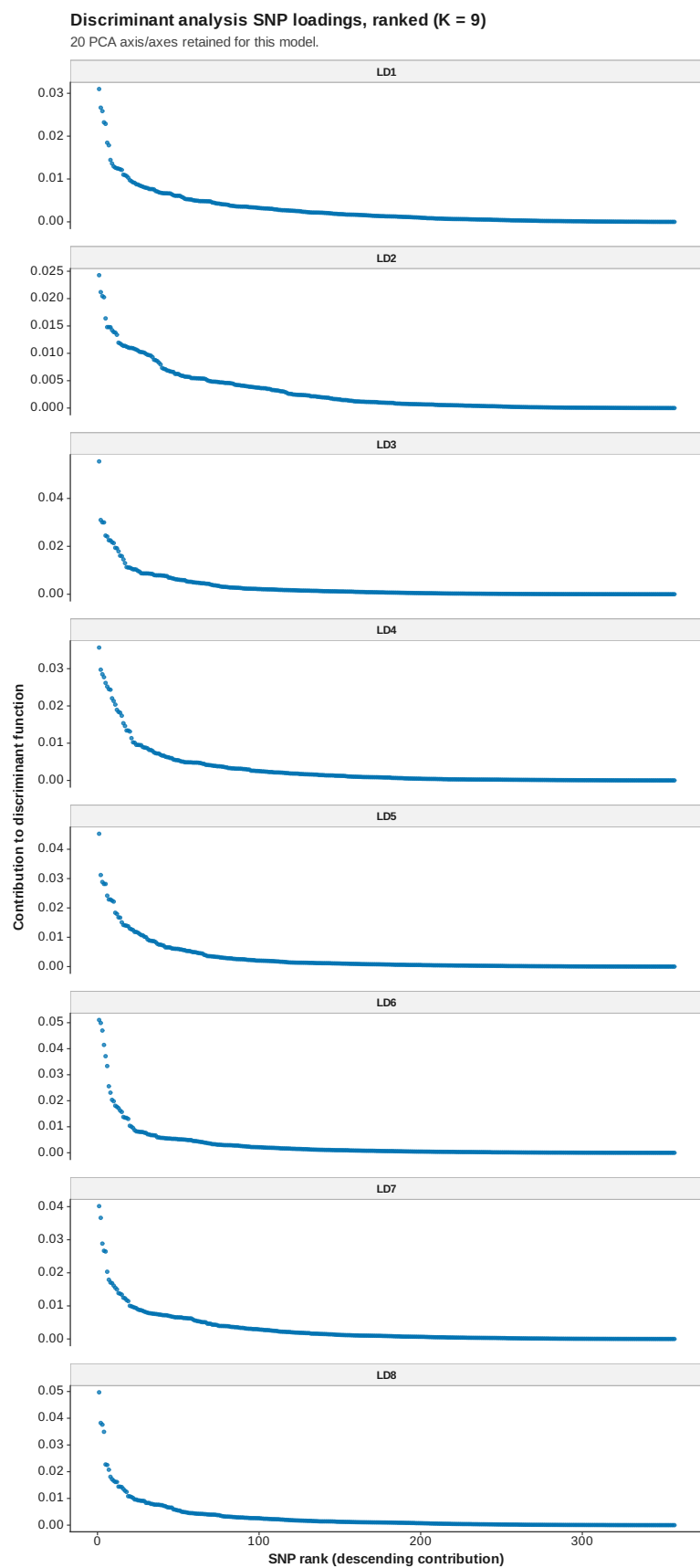


Figure 120: DAPC loadings ranked K 9

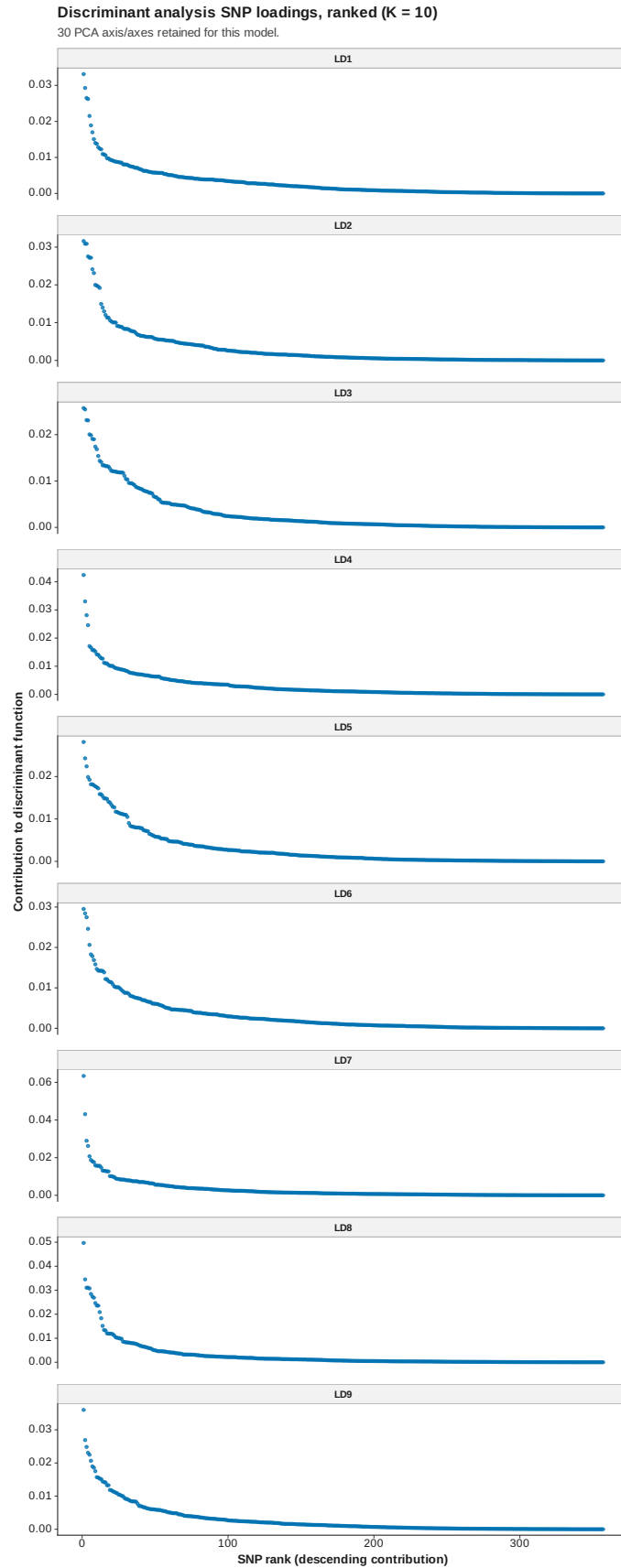


Figure 121: DAPC loadings ranked K 10

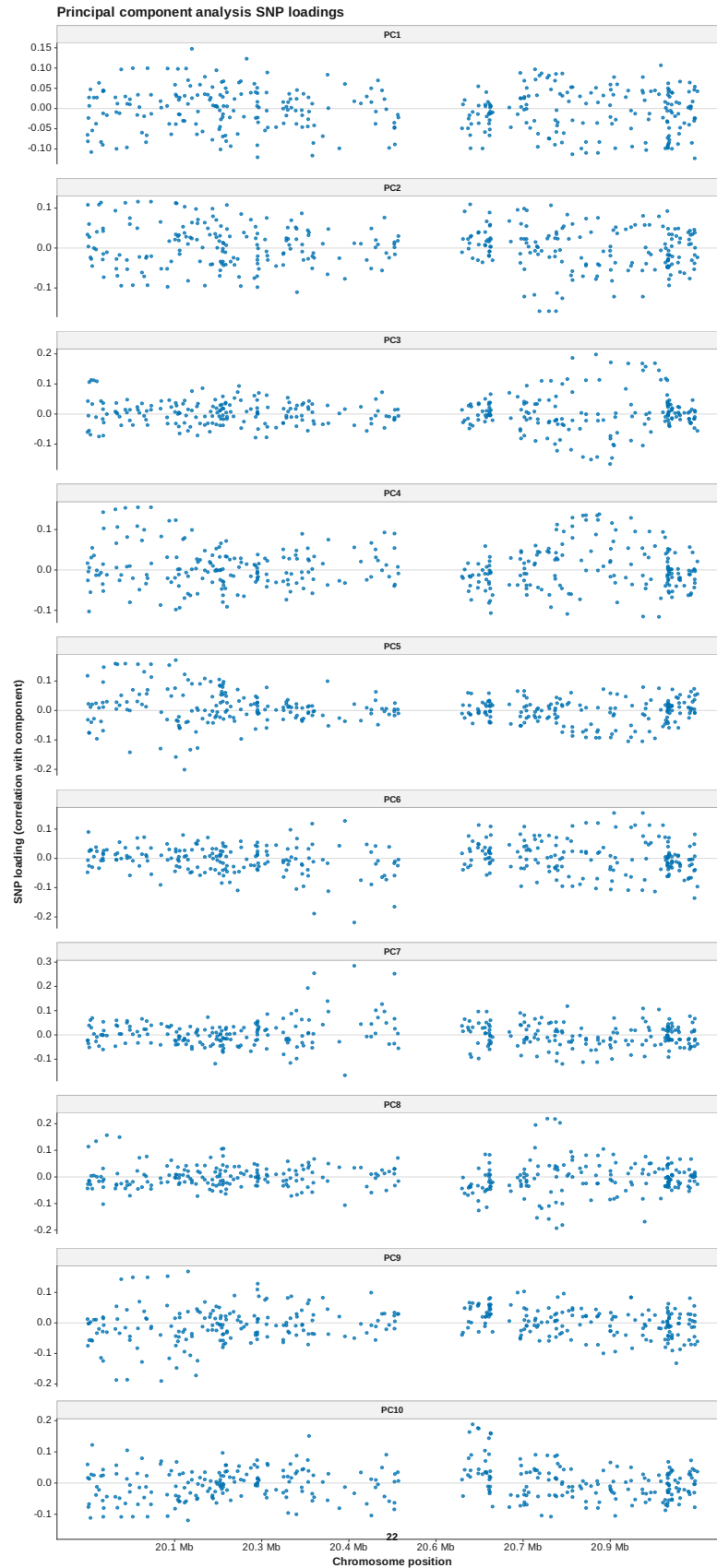


Figure 122: PCA loadings manhattan

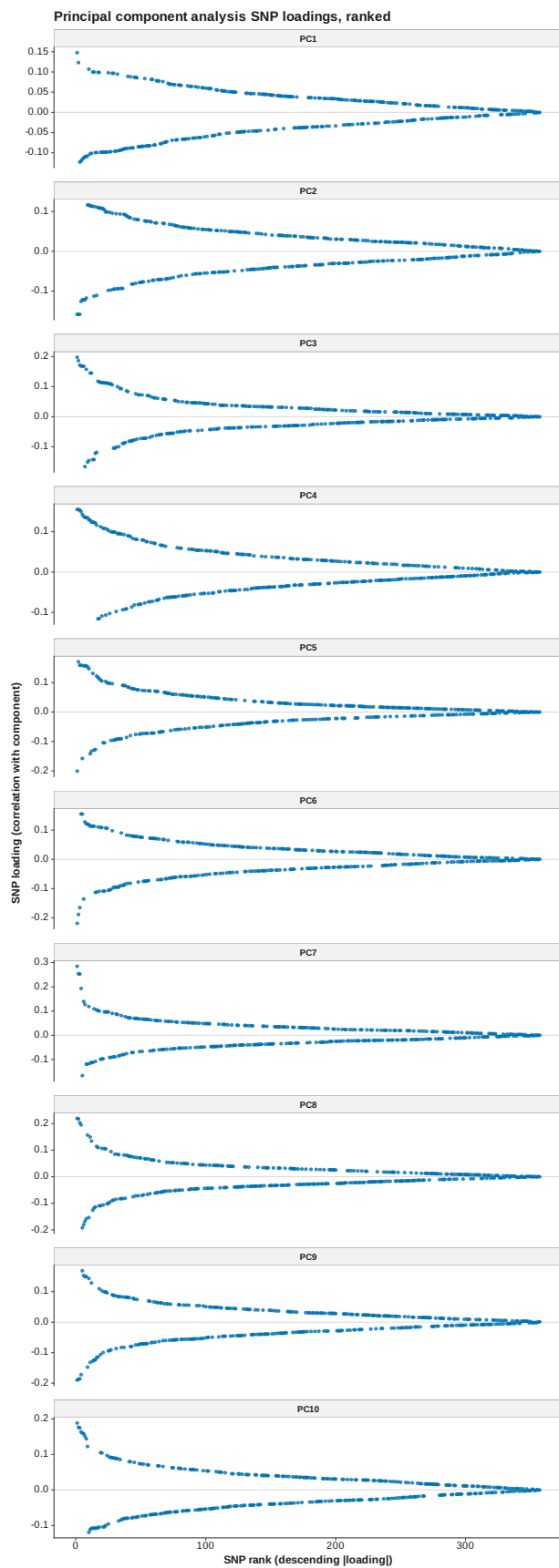


Figure 123: PCA loadings ranked

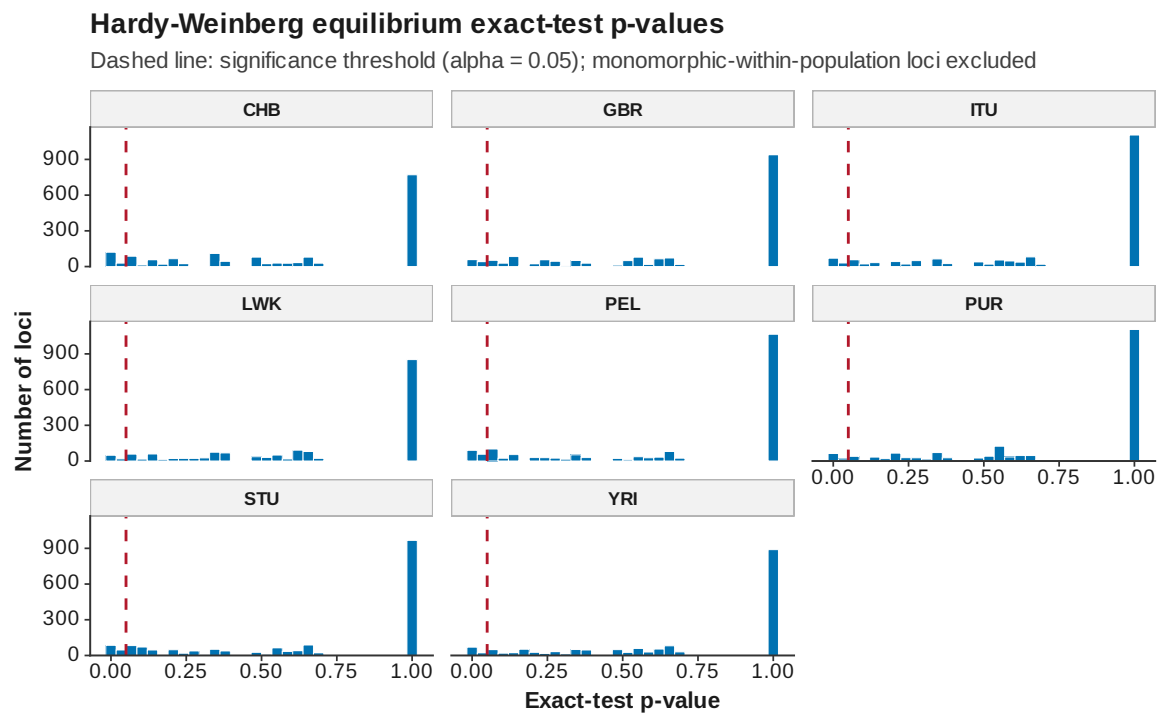


Figure 124: HWE pvalues

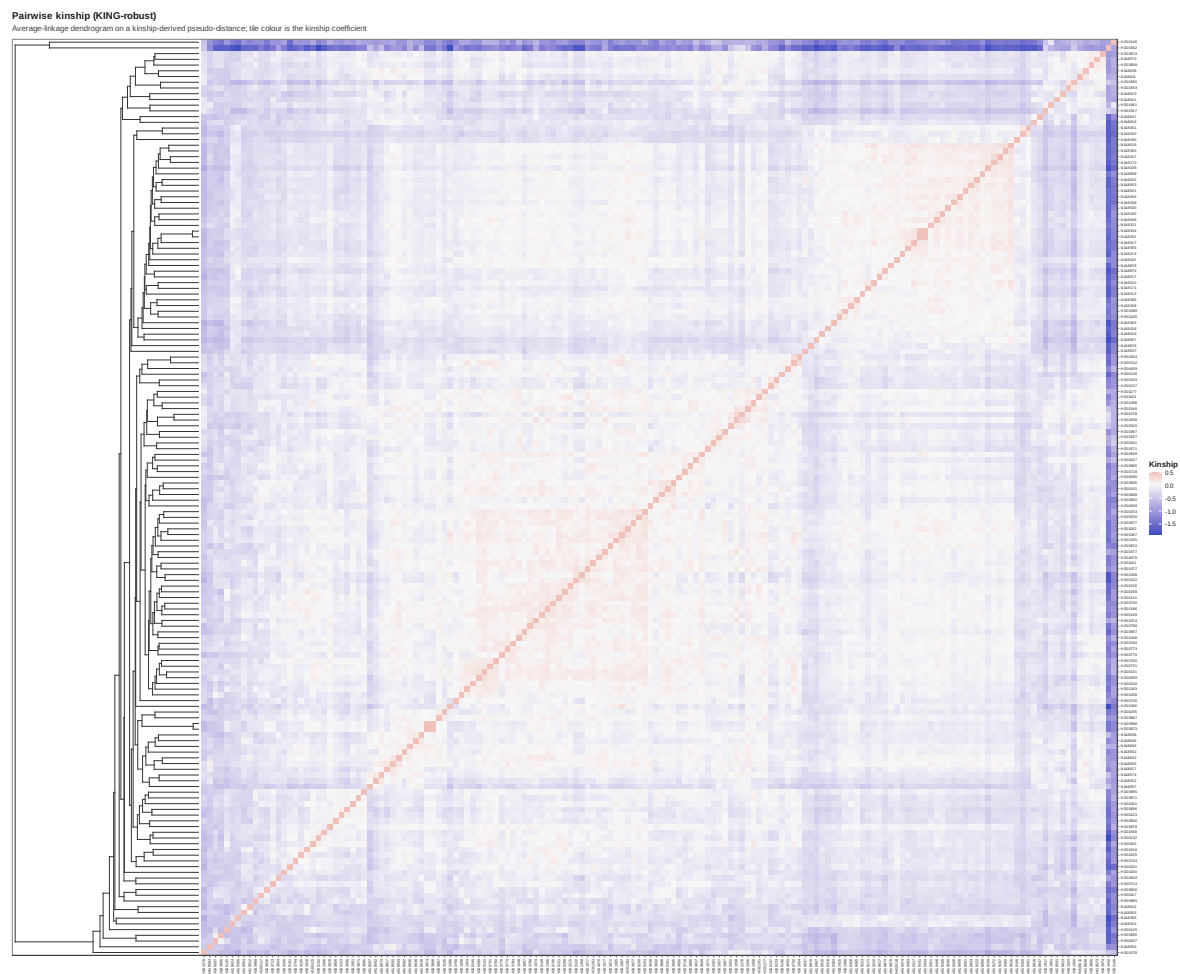


Figure 125: Kinship heatmap



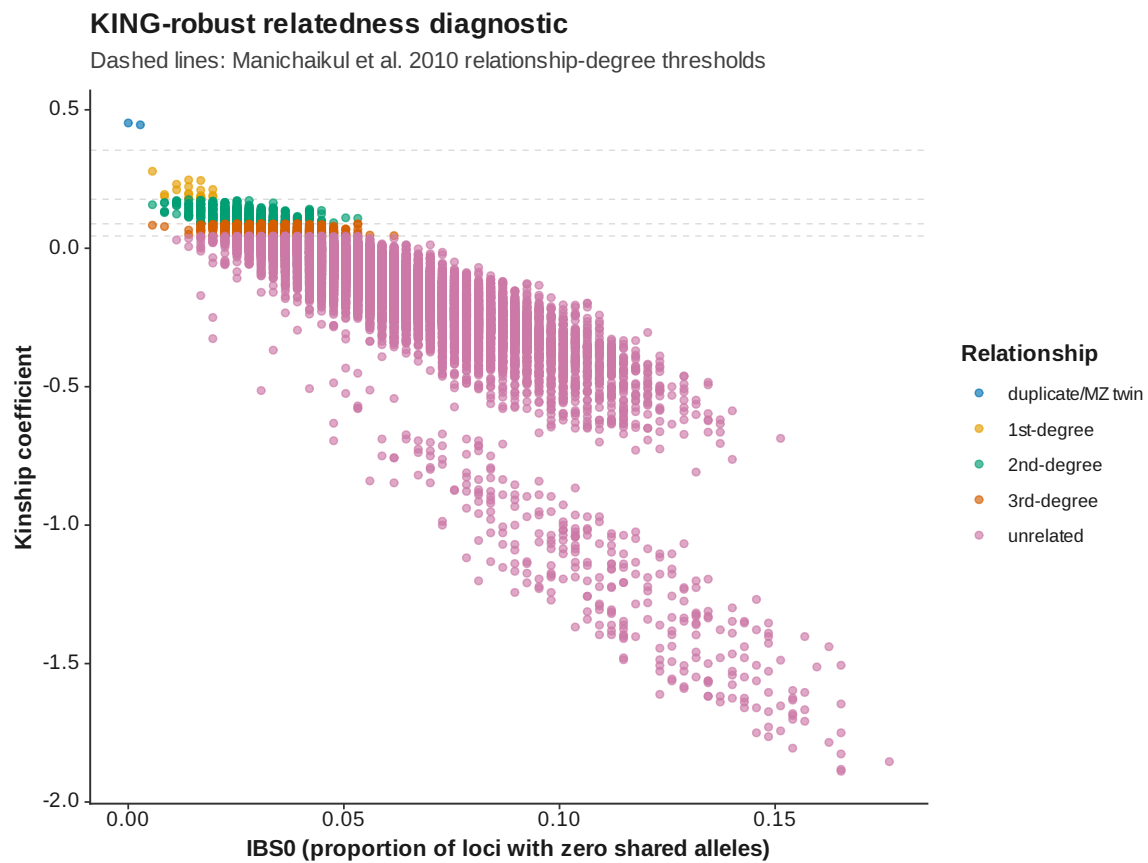


Figure 126: Kinship IBS 0 vs kinship

### Runs of homozygosity: length distribution

Every detected run across all samples

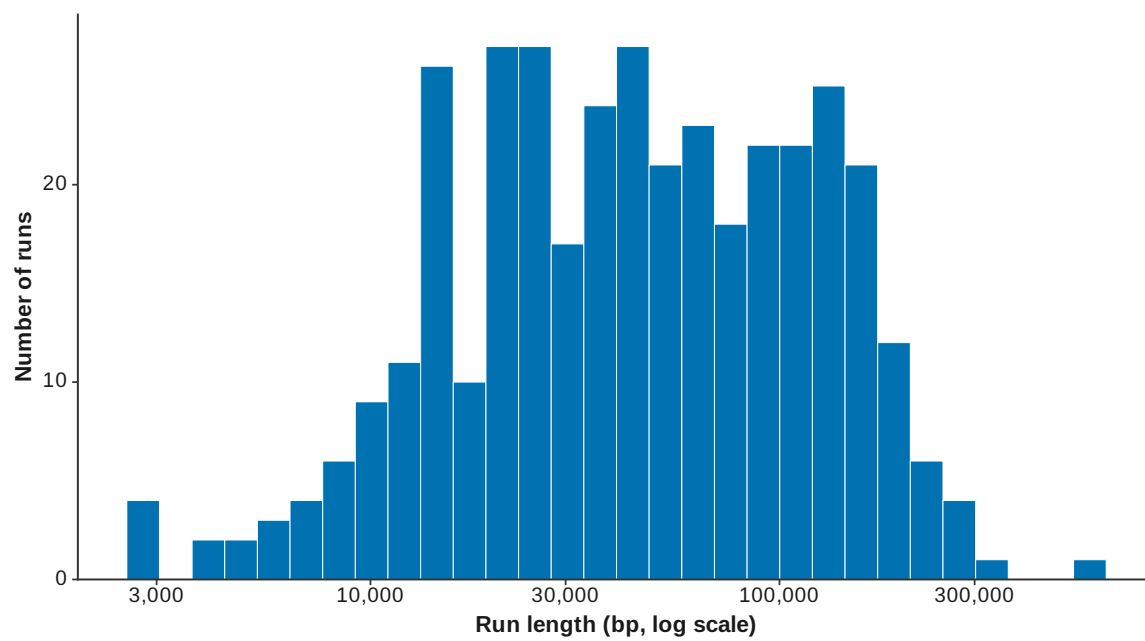


Figure 127: ROH length distribution

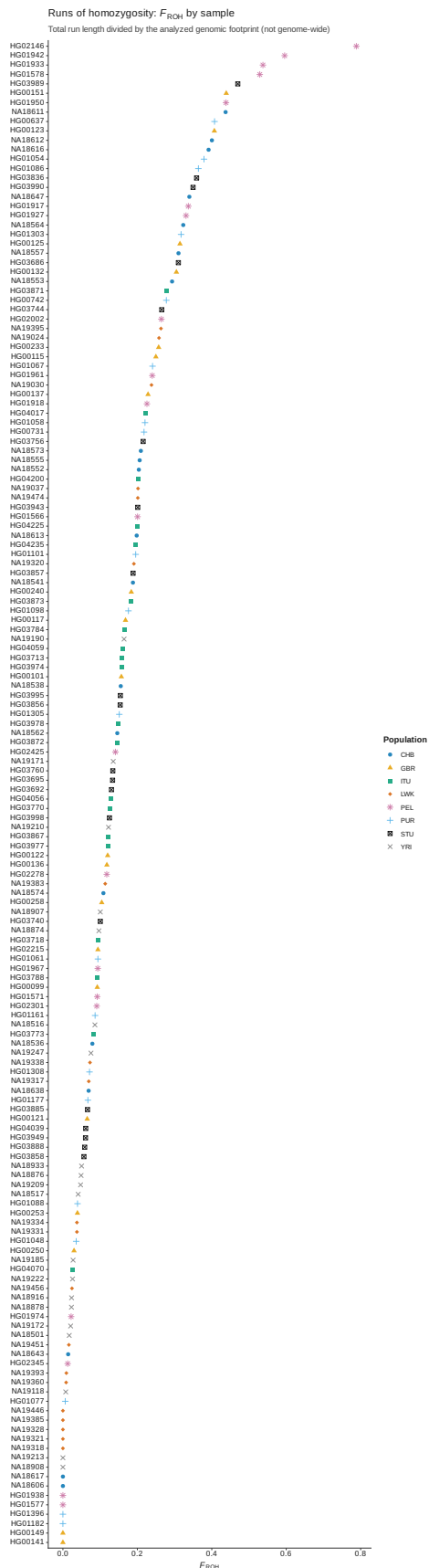


Figure 128: ROH FROH by sample

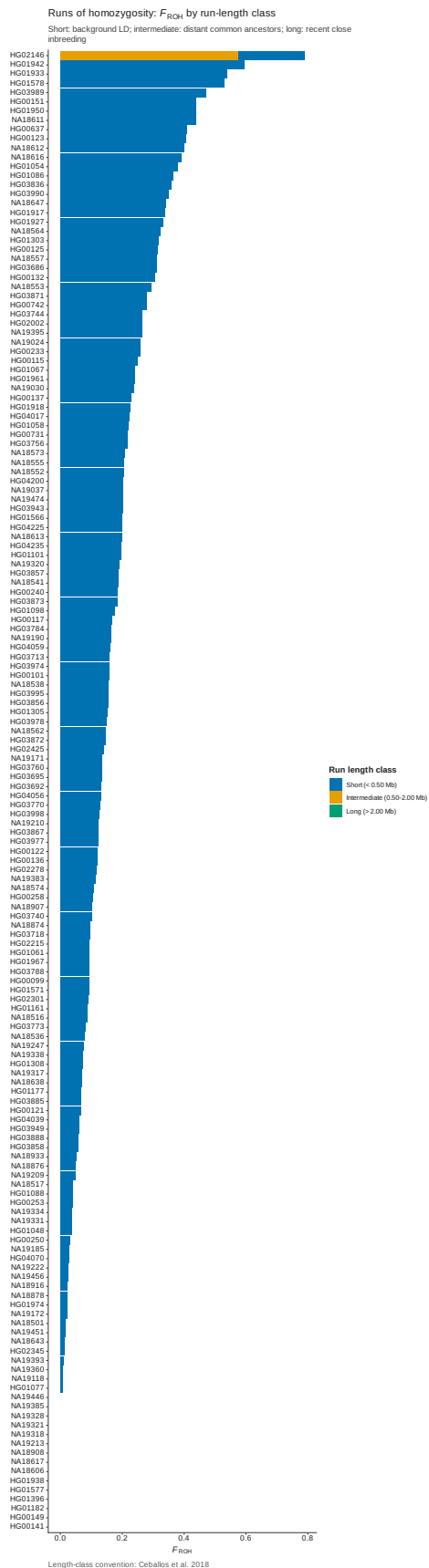


Figure 129: ROH FROH by length class

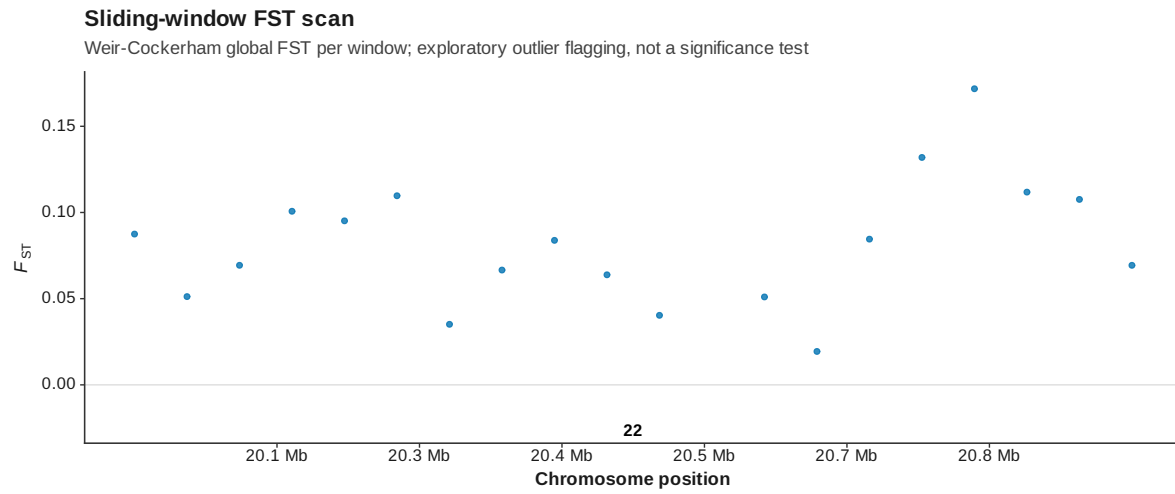


Figure 130: Genome scan FST manhattan

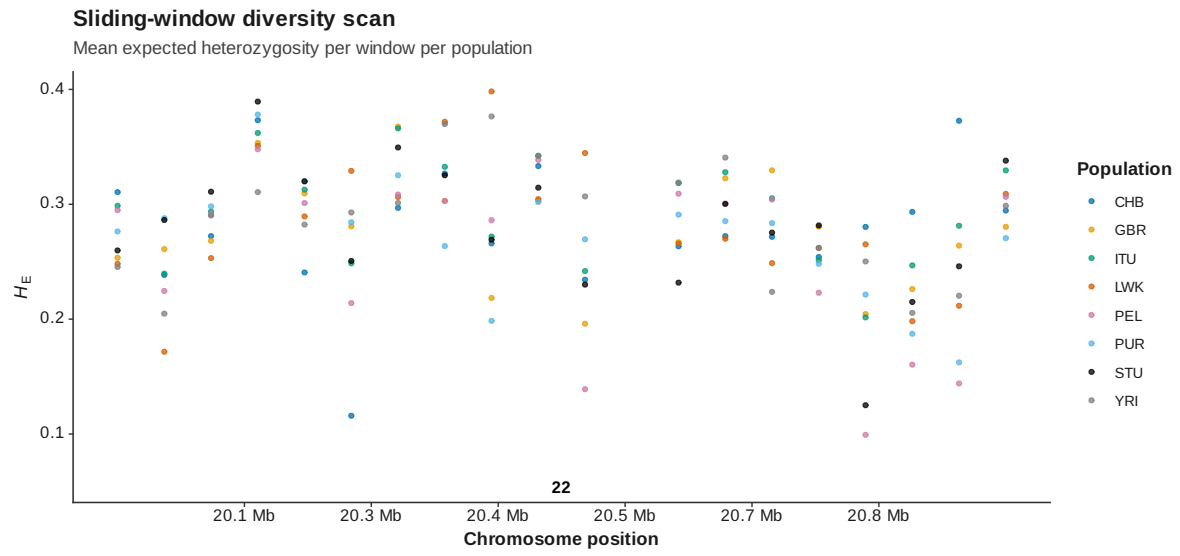


Figure 131: Genome scan diversity manhattan

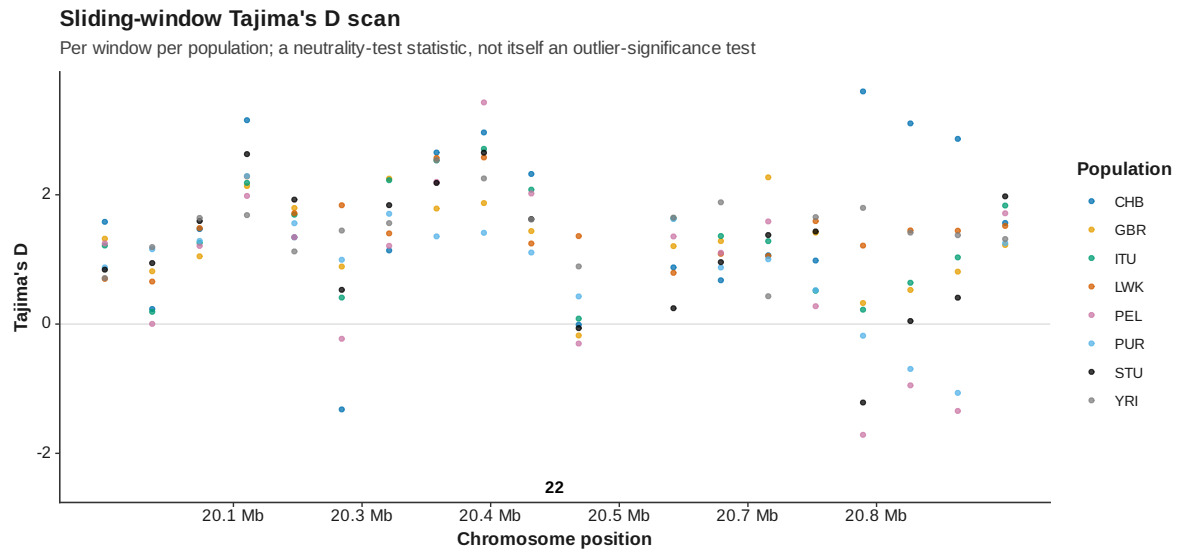


Figure 132: Genome scan tajima d manhattan

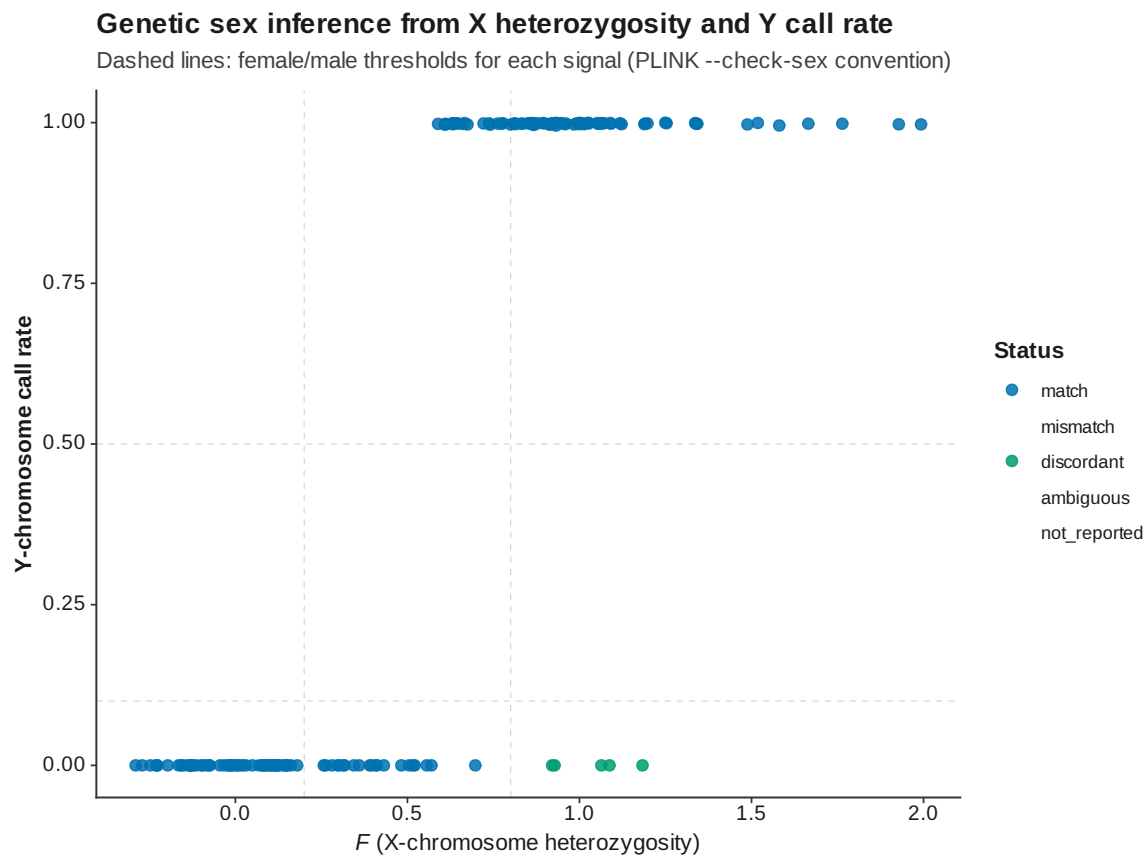


Figure 133: Sex check F by sample



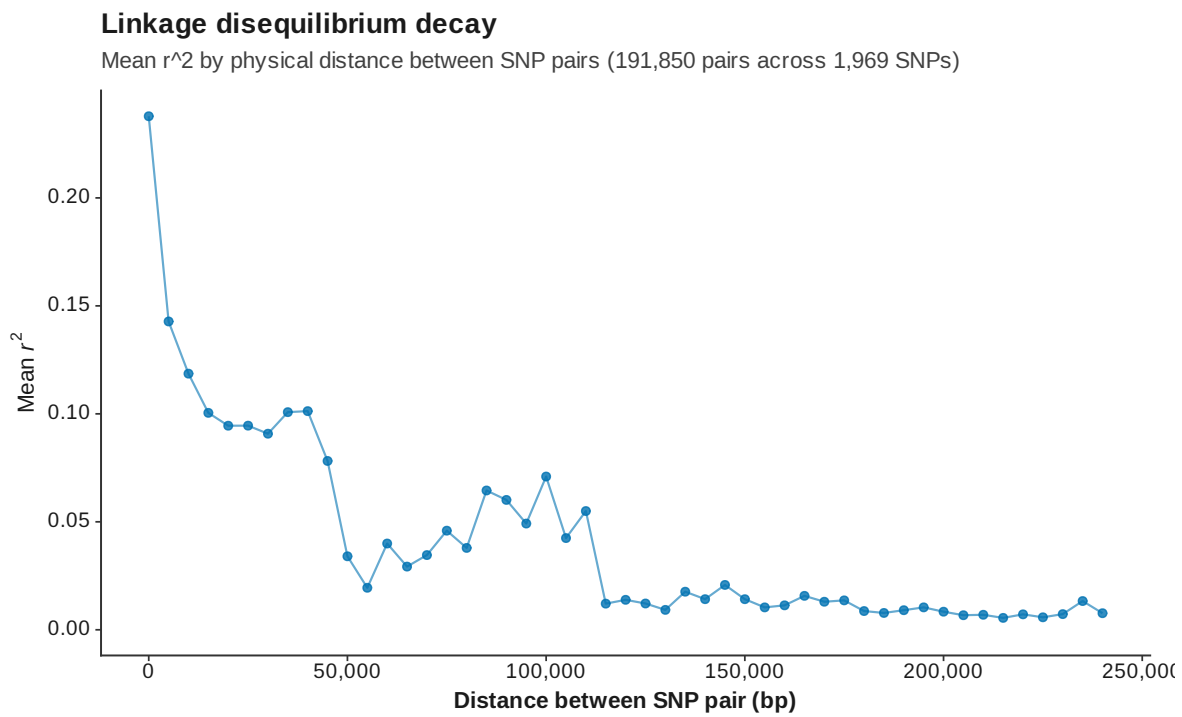


Figure 134: LD decay

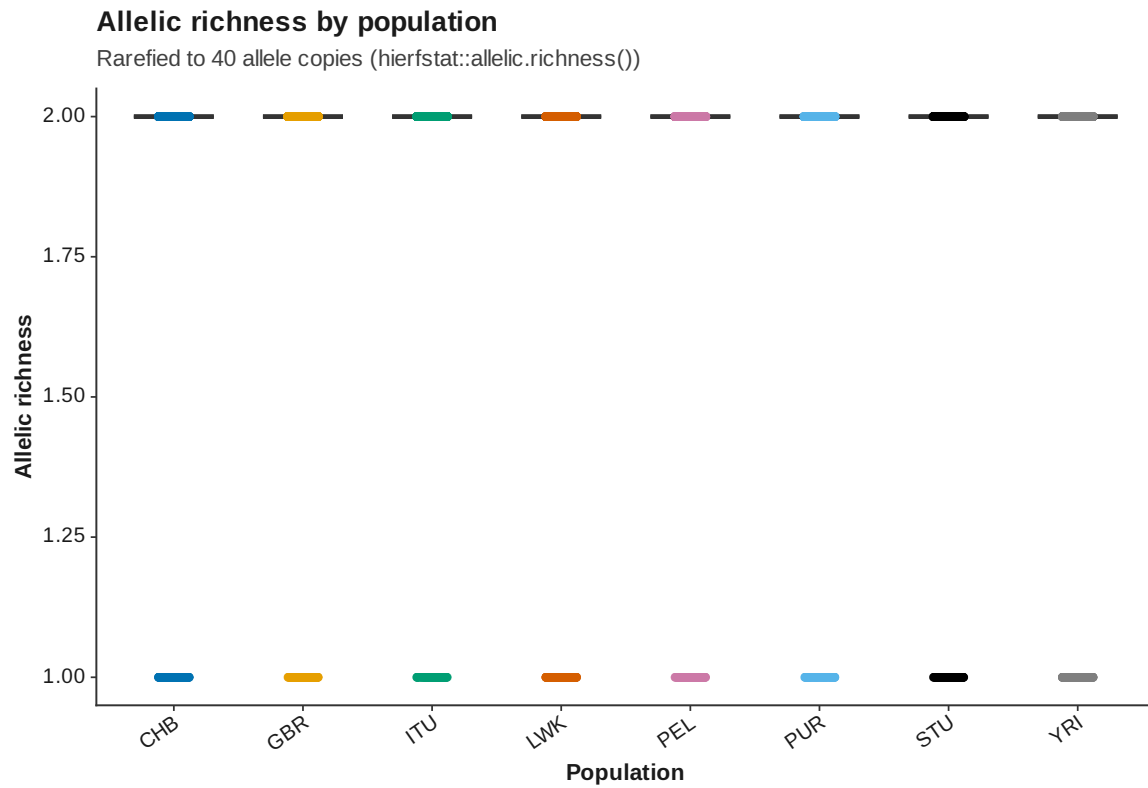


Figure 135: Allelic richness

## Population assignment test

106 / 160 samples (66.2%) assign to their recorded population

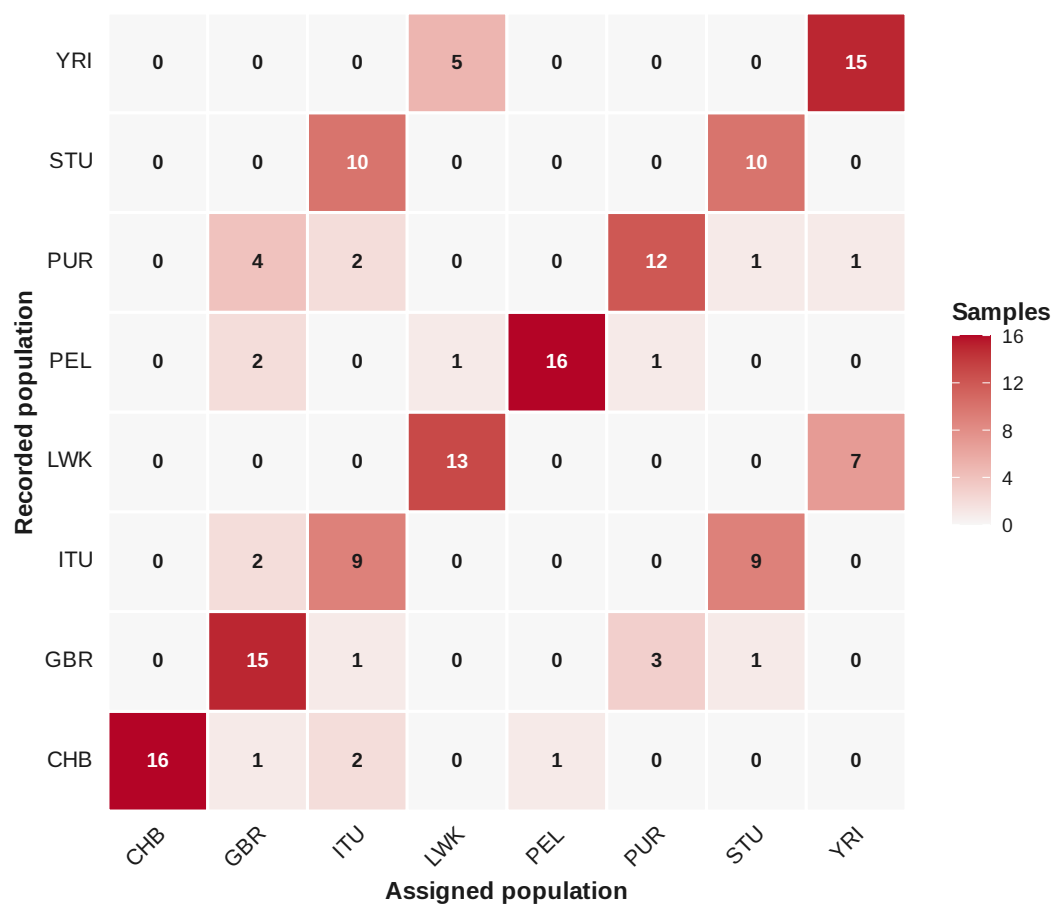
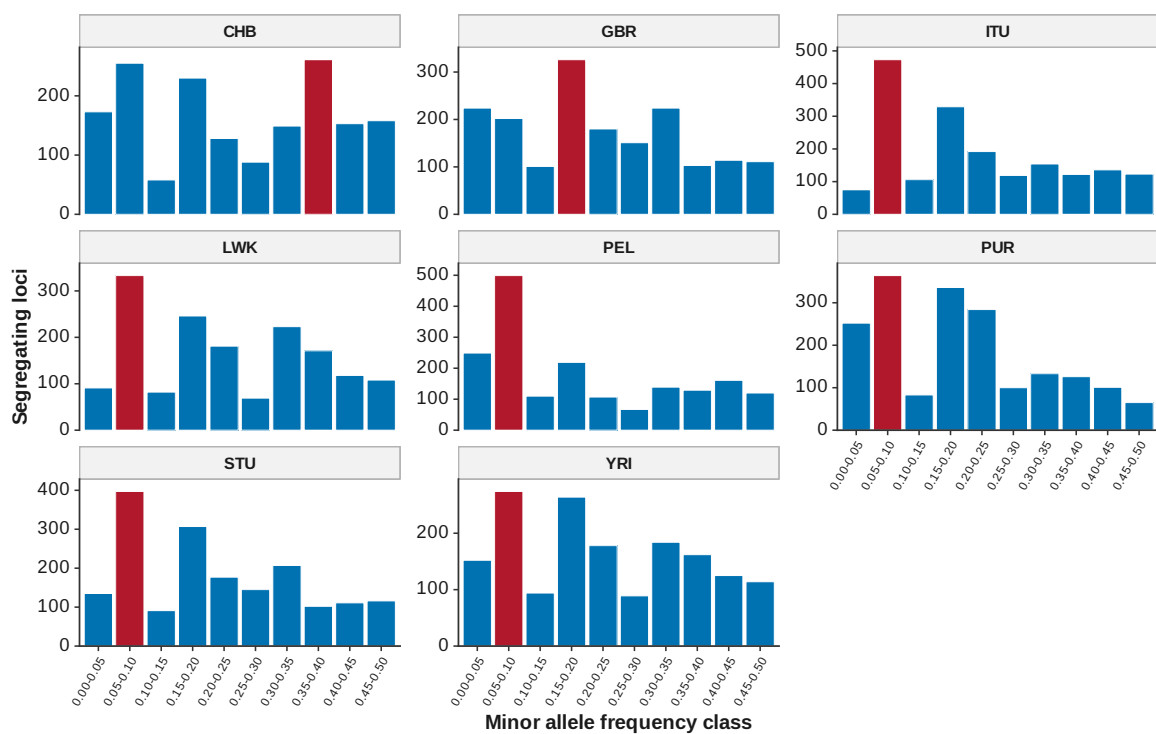


Figure 136: Population assignment

### Site frequency spectrum (folded) and mode-shift bottleneck screen

Highlighted bar: each population's modal minor-allele-frequency class



A mode away from the lowest class is a possible recent-bottleneck signature (Luikart and Comuet 1998)

Figure 137: Site frequency spectrum

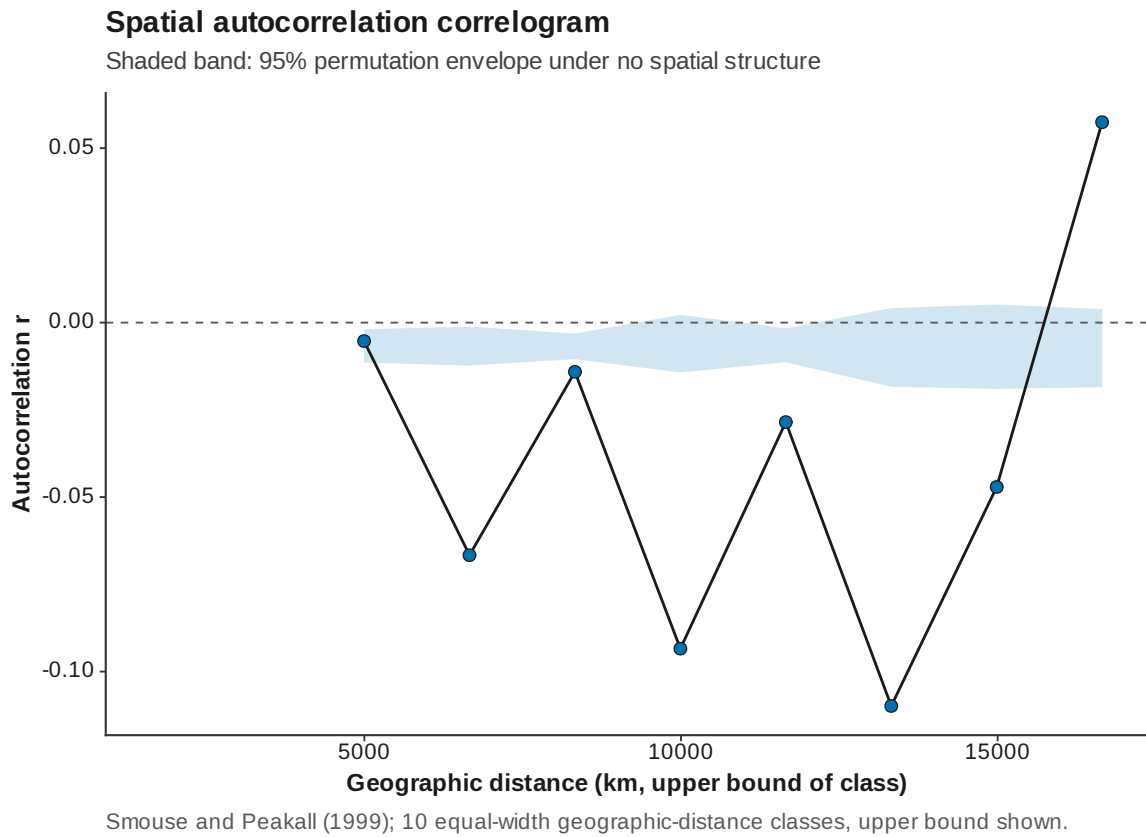


Figure 138: Spatial autocorrelation

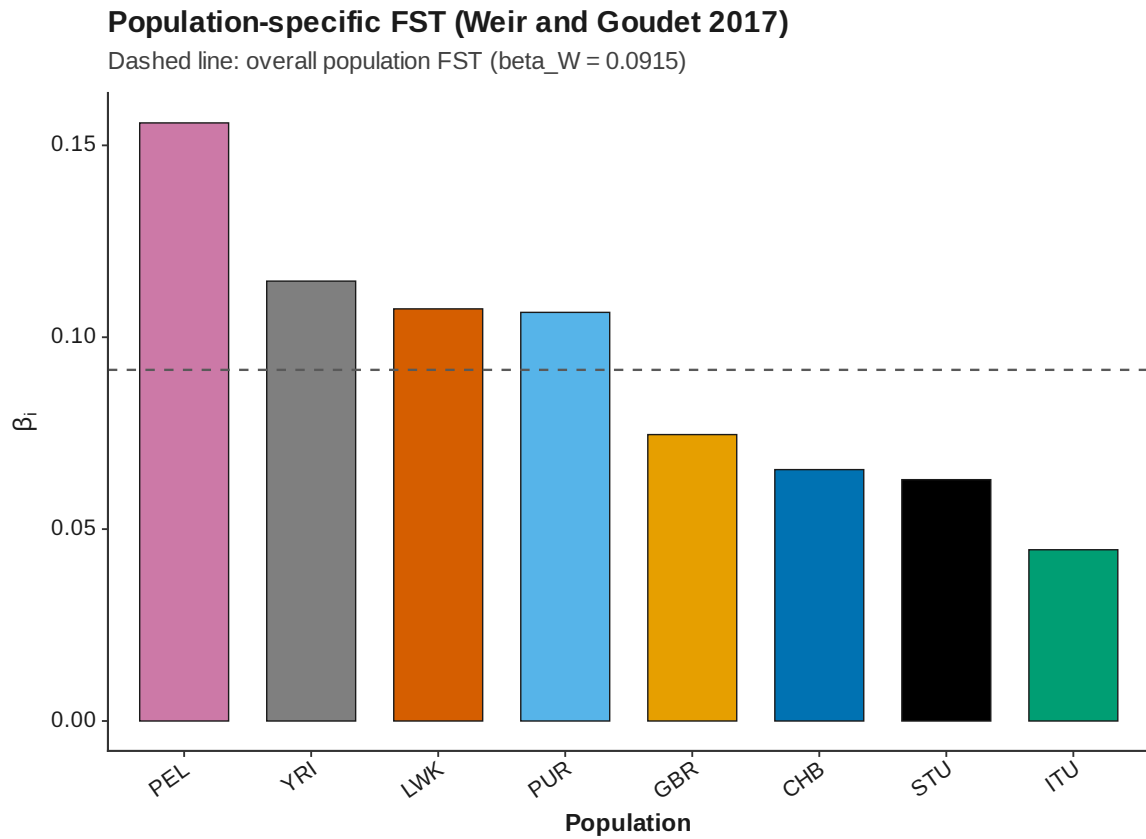
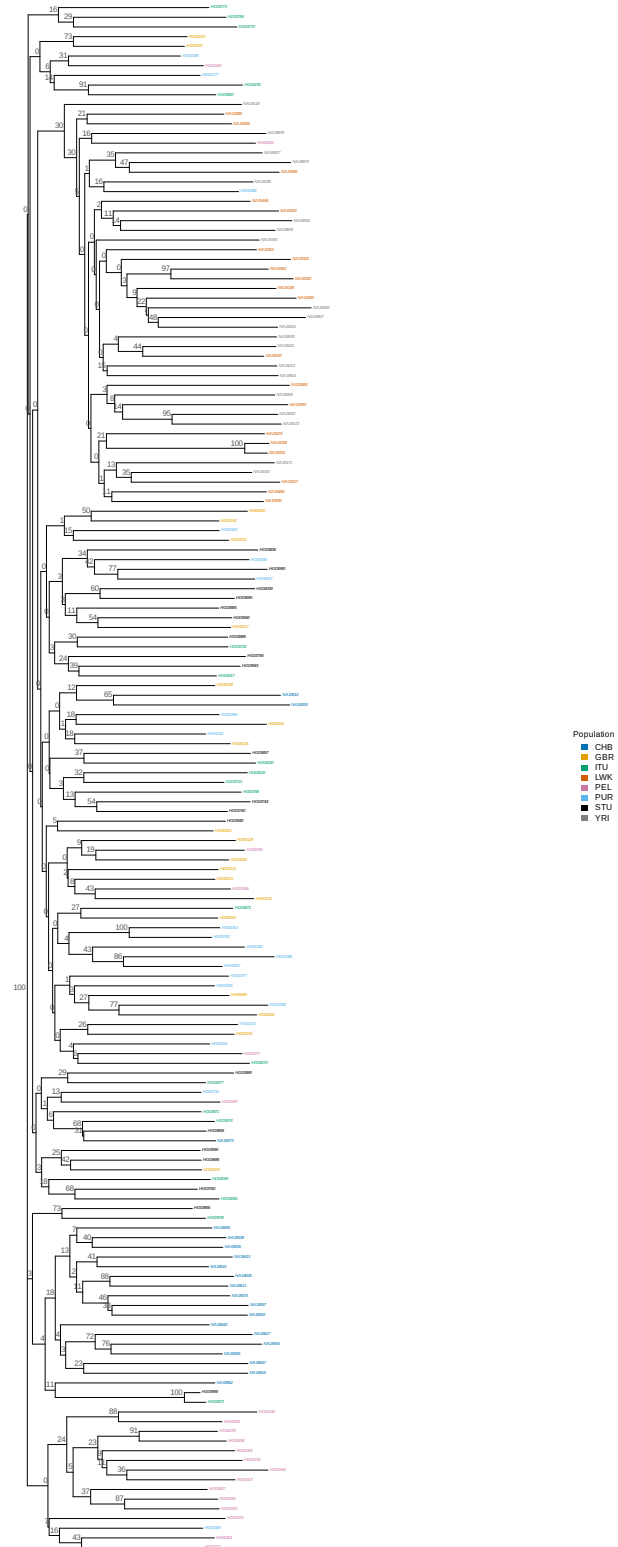


Figure 139: Population specific fst

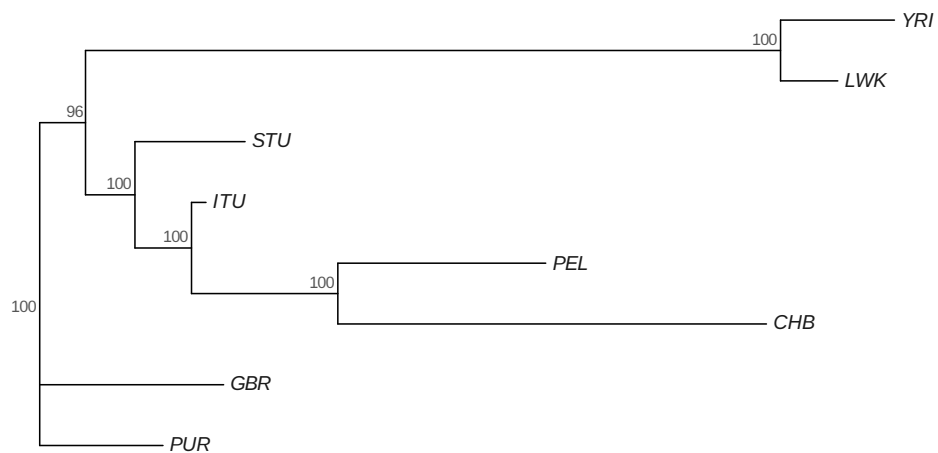
Neighbor-joining tree (identity-by-state)



Node labels: bootstrap support (% of 100 locus-resampled replicates)

Figure 140: IBS tree  
151

Neighbor-joining tree (Nei's genetic distance)



Node labels: bootstrap support (% of 100 locus-resampled replicates)

Figure 141: Population tree



Maximum-likelihood tree (GTR+Gamma, ascertainment-bias corrected)

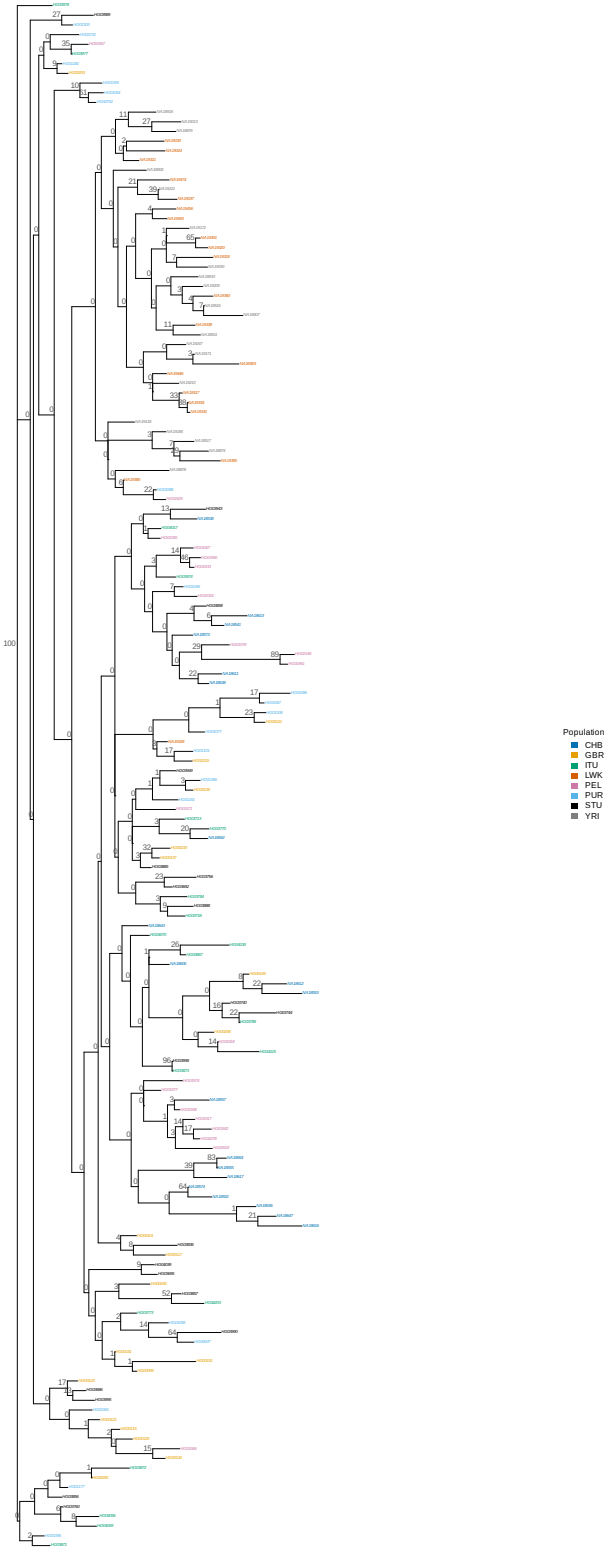


Figure 142: ML tree

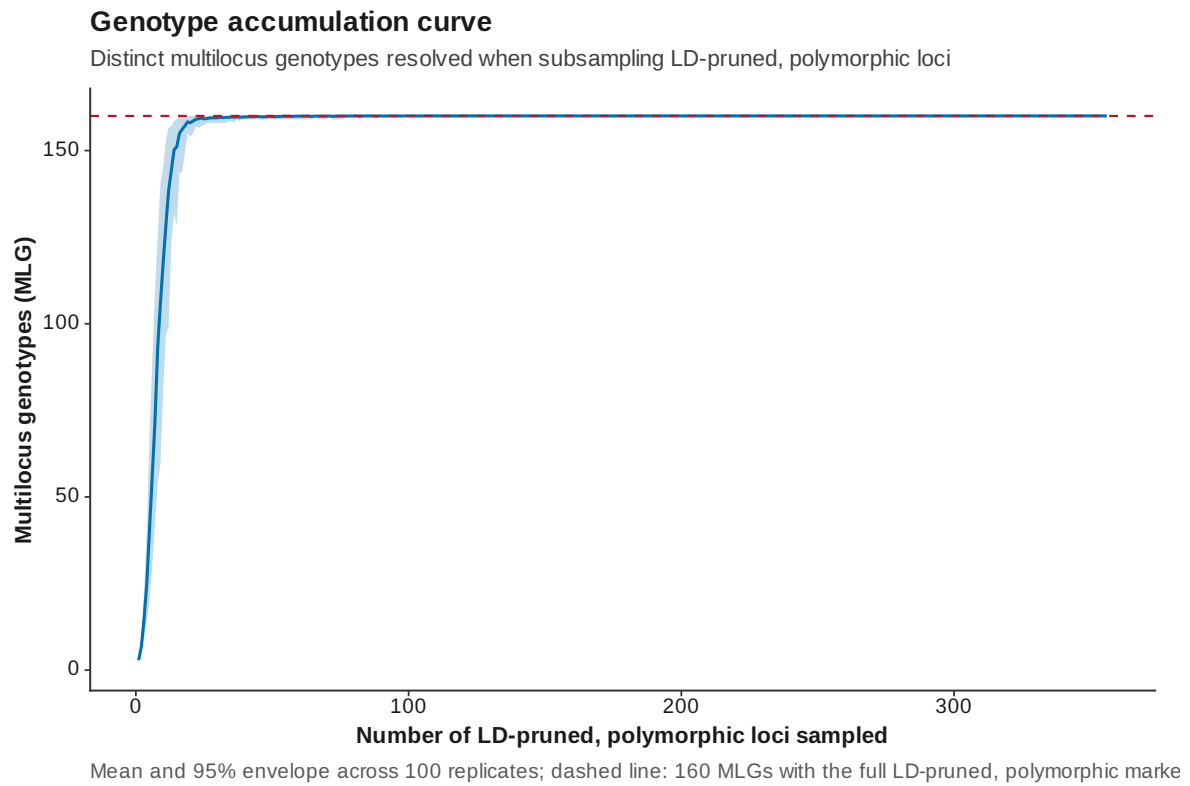


Figure 143: Genotype accumulation curve

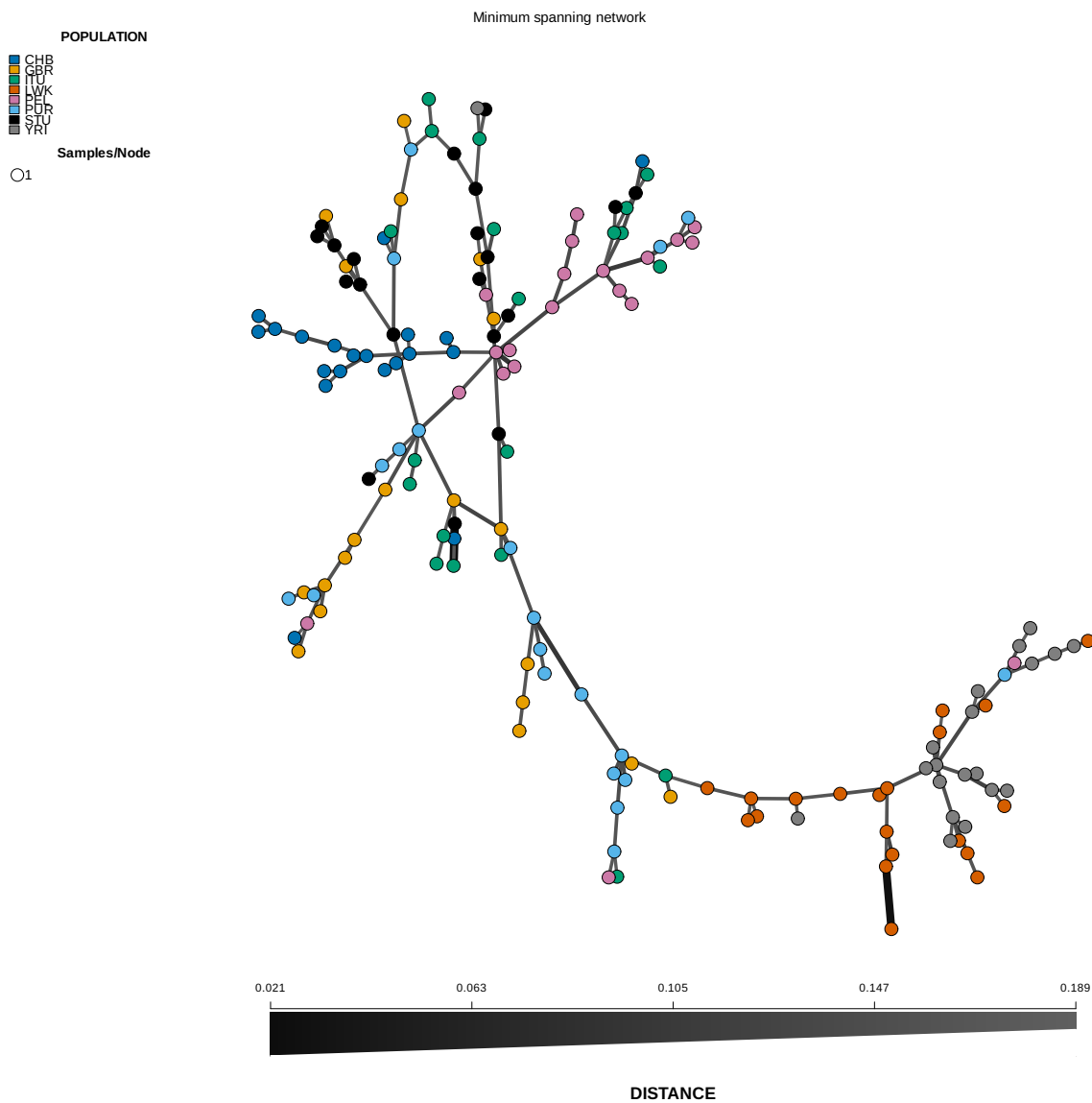


Figure 144: MSN network

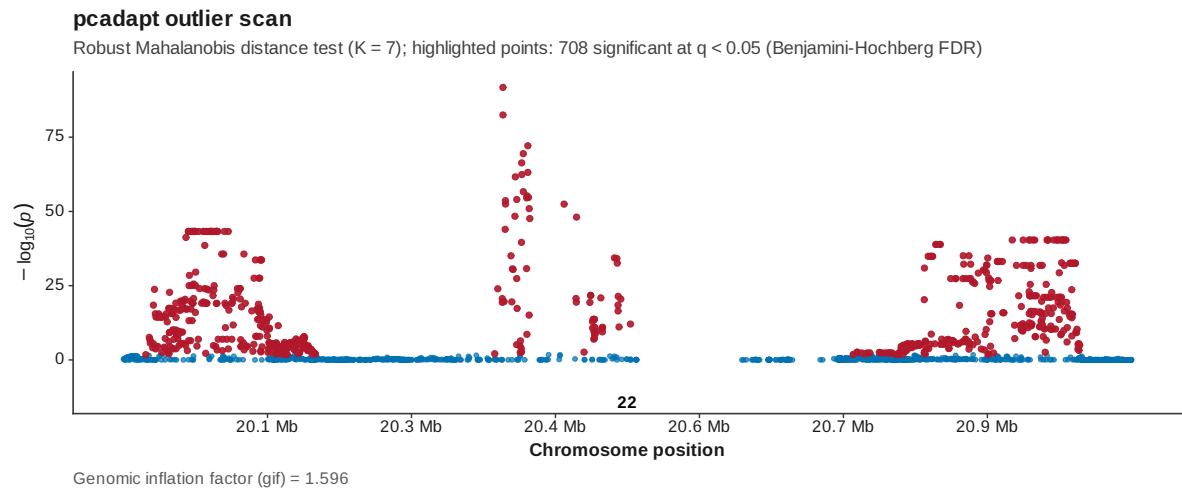
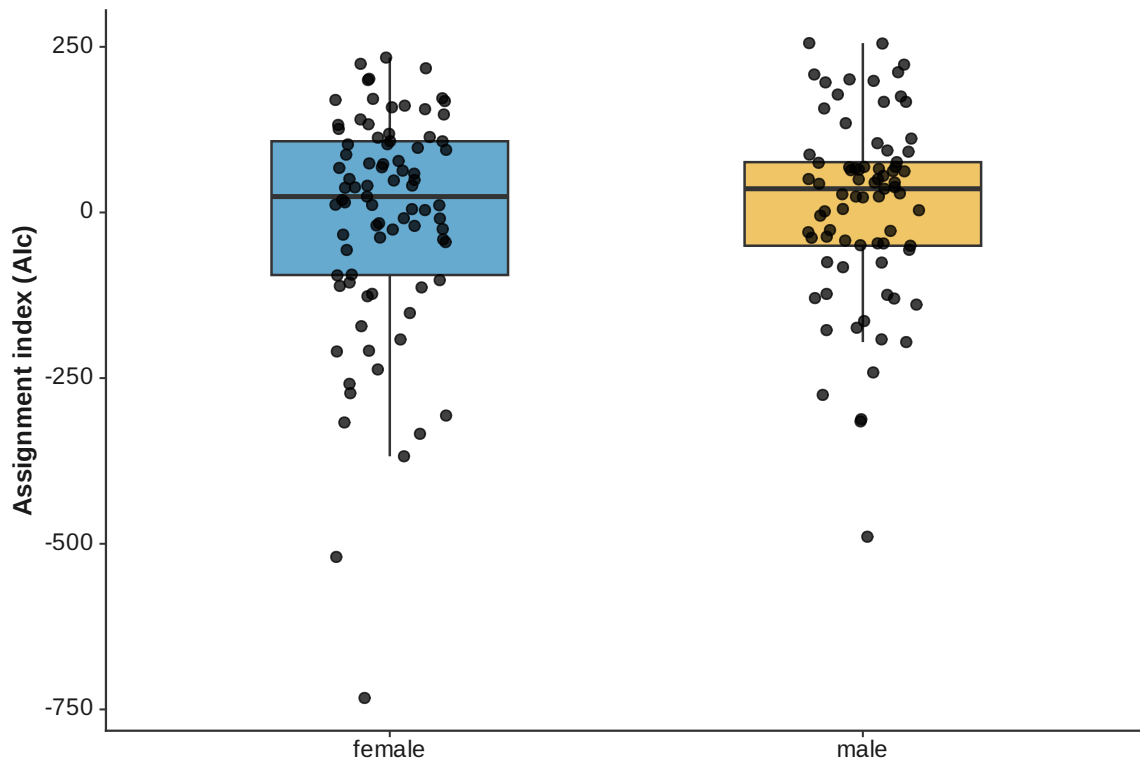


Figure 145: Pcadapt manhattan

### Sex-biased dispersal test

mAlc test: statistic = -0.648,  $p = 0.5177$



Assignment index (AIC, Favre et al. 1997) by recorded sex;  $n = 83$  female, 77 male

Figure 146: Sexbias AIC by sex

## 25 Reproducibility

This report was generated from the serialized pipeline result object. Exact input hashes, settings, package versions, and stage runtimes are stored alongside the report.

### 25.1 Population-structure reproducibility

| K  | assignment_accuracy | replicate_max_rmse |
|----|---------------------|--------------------|
| 2  | 0.2500              | 0.0000             |
| 3  | 0.3562              | 0.0000             |
| 4  | 0.4000              | 0.1514             |
| 5  | 0.4562              | 0.2692             |
| 6  | 0.5000              | 0.2732             |
| 7  | 0.5000              | 0.1490             |
| 8  | 0.5250              | 0.2186             |
| 9  | 0.4812              | 0.2513             |
| 10 | 0.4812              | 0.2714             |

### 25.2 fastStructure

| K | exit_status | marginal_likelihood |
|---|-------------|---------------------|
| 2 | 0           | -0.8538             |
| 3 | 0           | -0.8492             |
| 4 | 0           | -0.8545             |
| 5 | 0           | -0.8604             |
| 6 | 0           | -0.8673             |
| 7 | 0           | -0.8724             |
| 8 | 0           | -0.8724             |
| 9 | 0           | -0.8710             |

### 25.3 Sparse non-negative matrix factorization

| K | run | cross_entropy |
|---|-----|---------------|
| 2 | 1   | 0.658188      |
| 2 | 2   | 0.641827      |
| 2 | 3   | 0.632785      |
| 2 | 4   | 0.665263      |
| 2 | 5   | 0.652600      |
| 3 | 1   | 0.652267      |
| 3 | 2   | 0.621568      |
| 3 | 3   | 0.625406      |
| 3 | 4   | 0.650066      |
| 3 | 5   | 0.636630      |
| 4 | 1   | 0.645990      |
| 4 | 2   | 0.618994      |
| 4 | 3   | 0.620640      |
| 4 | 4   | 0.646500      |
| 4 | 5   | 0.642422      |
| 5 | 1   | 0.642907      |
| 5 | 2   | 0.617357      |

| K | run | cross_entropy |
|---|-----|---------------|
| 5 | 3   | 0.619569      |
| 5 | 4   | 0.643777      |
| 5 | 5   | 0.631210      |
| 6 | 1   | 0.636456      |
| 6 | 2   | 0.616501      |
| 6 | 3   | 0.614301      |
| 6 | 4   | 0.639528      |
| 6 | 5   | 0.628516      |
| 7 | 1   | 0.639879      |
| 7 | 2   | 0.617185      |
| 7 | 3   | 0.609143      |
| 7 | 4   | 0.640496      |
| 7 | 5   | 0.633083      |
| 8 | 1   | 0.637875      |
| 8 | 2   | 0.620175      |
| 8 | 3   | 0.606337      |
| 8 | 4   | 0.643657      |
| 8 | 5   | 0.643623      |
| 9 | 1   | 0.639730      |
| 9 | 2   | 0.623835      |
| 9 | 3   | 0.597450      |
| 9 | 4   | 0.647969      |
| 9 | 5   | 0.632479      |